

Notes

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Chapter 1

Notation

Following is a list of notation and a brief description, grouped by topic.

Tensors

a	zeroth order (scalar)
$\mathbf{A}$	first order (vector) in the reference configuration
A_I	first order (vector) components in the reference configuration
$\mathbf{a}$	first order (vector) in the current configuration
a_i	first order (vector) components in the current configuration
$\mathbf{A}$	second order (matrix) in the reference configuration
$\mathbf{a}$	second order (matrix) in the current configuration
A_{IJ}	second order (matrix) components in the reference configuration
a_{ij}	second order (matrix) components in the current configuration
$\mathbb{A}$	fourth order in the reference configuration
$\mathbb{A}_{IJKL}$	fourth order components in the reference configuration
$\mathfrak{A}$	fourth order in the current configuration
$\mathfrak{A}_{ijkl}$	fourth order components in the current configuration
$\mathbf{E}_I$	basis vector for the reference (material) configuration
$\mathbf{e}_i$	basis vector for the current (spatial) configuration
O	origin of body B and location of the body reference frame $\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3$
o	origin in spatial configuration

Motion

$\mathbf{X}$	position of material particle in reference configuration
$\mathbf{x}$	position of material particle in current configuration
$\mathbf{U}$	displacement vector from $\mathbf{X}$ to $\mathbf{x}$, objective
φ	motion of material particle $\mathbf{X}$
$\varphi_*[\cdot]$	push forward operator
$\varphi_*^{-1}[\cdot]$	pull back operator
$\mathcal{L}[\cdot]$	Lie derivative, $\mathcal{L}[\cdot] \triangleq \mathbf{F} \frac{\partial}{\partial t} (\mathbf{F}^{-1}[\cdot] \mathbf{F}^{-\top}) \mathbf{F}^\top$
$\mathbf{v}$	velocity of material particle in current configuration, non-objective
$\mathbf{a}$	acceleration of material particle in current configuration, non-objective
$\nabla_0(\cdot)$	gradient operator in the reference configuration
$\nabla(\cdot)$	gradient operator in the current configuration
$\nabla_0 \mathbf{U}$	displacement gradient in the reference configuration, $\nabla_0 \mathbf{U} = \mathbf{F} - \mathbf{1}$
$\nabla \mathbf{u}$	displacement gradient in the current configuration, $\nabla \mathbf{u} = \mathbf{1} - \mathbf{F}^{-1}$
$\mathbf{F} \triangleq \nabla_0 \varphi$	deformation gradient, $\mathbf{F} = \mathbf{vR} = \mathbf{RU}$, non-symmetric, two-point tensor, objective (transforms as a vector, not a tensor)
J	Jacobian of the deformation gradient, $J \triangleq \det \mathbf{F}$
$\mathbf{R}$	rotation, objective (transforms as a vector, not a tensor)
$\mathbf{v}$	left-stretch, objective
$\mathbf{U}$	right-stretch, objective

Strain

$\mathbf{C}$	right Cauchy-Green, $\mathbf{C} \triangleq \mathbf{F}^\top \mathbf{F}$, symmetric, positive definite, objective
$\mathbf{b}$	left Cauchy-Green, $\mathbf{b} \triangleq \mathbf{F} \mathbf{F}^\top$, symmetric, positive definite, objective
$\mathbf{E}$	Green-Lagrange, $\mathbf{E} \triangleq \frac{1}{2} (\mathbf{C} - \mathbf{1})$, objective
$\mathbf{e}$	Euler-Almansi, $\mathbf{e} \triangleq \frac{1}{2} (\mathbf{1} - \mathbf{b}^{-1})$, objective
$\mathbf{E}_{(m)}$	Seth-Hill
λ_I	principal stretch ratio

ϵ	infinitesimal
$\dot{\mathbf{F}}$	material velocity gradient
$\nabla \mathbf{v}$	spatial velocity gradient, non-symmetric, non-objective, covariant
$\mathbf{d}$	stretch rate (spatial rate of deformation, rate of stretch), symmetric, objective, covariant, $\mathbf{d} = \frac{1}{2} (\nabla \mathbf{v} + \nabla \mathbf{v}^T)$
$\mathbf{w}$	rotation rate (spin, rate of rotation, vorticity), skew-symmetric, non-objective, covariant, $\mathbf{w} = \frac{1}{2} (\nabla \mathbf{v} - \nabla \mathbf{v}^T)$
$\mathbf{D}$	rotated rate of deformation, $\mathbf{D} = \mathbf{R}^T \mathbf{d} \mathbf{R}$
$\mathbf{\Omega}$	rate of change of the rotation tensor, skew-symmetric, $\mathbf{\Omega} \triangleq \dot{\mathbf{R}} \mathbf{R}^T$

Stress

$\boldsymbol{\sigma}$	Cauchy, symmetric, objective
$\boldsymbol{\tau}$	Kirchhoff, $\boldsymbol{\tau} \triangleq J \boldsymbol{\sigma}$
$\hat{\boldsymbol{\sigma}}$	Co-rotated Cauchy, $\hat{\boldsymbol{\sigma}} \triangleq \mathbf{R}^T \boldsymbol{\sigma} \mathbf{R}$
$\boldsymbol{\Sigma}$	rotated Kirchhoff, $\boldsymbol{\Sigma} \triangleq \mathbf{R}^T \boldsymbol{\tau} \mathbf{R}$
$\mathbf{P}$	Piola-Kirchhoff One, non-symmetric, objective (transforms as a vector, not a tensor), $\mathbf{P} \triangleq J \boldsymbol{\sigma} \mathbf{F}^{-T} = \boldsymbol{\tau} \mathbf{F}^{-T} = \mathbf{F} \mathbf{S}$
$\mathbf{S}$	Piola-Kirchhoff Two, symmetric, objective, $\mathbf{S} \triangleq J \mathbf{F}^{-1} \boldsymbol{\sigma} \mathbf{F}^{-T} = \mathbf{F}^{-1} \boldsymbol{\tau} \mathbf{F}^{-T}$
$\mathbf{B}$	Biot (Jaumann)
$\mathbf{A}$	Almansi
$\dot{\boldsymbol{\sigma}}$	Cauchy stress rate, non-objective
$\overset{\circ}{\boldsymbol{\sigma}}$	Truesdell rate of Cauchy stress, objective

$\square \sigma$	Green-McInnis-Naghdi rate of Cauchy stress, objective
$\nabla \sigma$	Jaumann-Zaremba rate of Cauchy stress, objective

Energy

P	pressure
V	volume
$\mathcal{V}$	specific volume
T	temperature
S	entropy
ρ_0	density in the reference configuration
ρ	density in the current configuration
m	mass
E	total energy
$\mathcal{E}$	total specific energy
U	internal energy
K	kinetic energy (also denotes bulk modulus)
G	potential energy for external forces
$\mathcal{F}$	Helmholtz free energy
$\mathcal{G}$	Gibbs free energy
$\mathcal{H}$	enthalpy

Fluids

$\bar{\rho}_0$	constant density in the reference configuration
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Elasticity

E	elastic modulus
ν	Poisson ratio
K	bulk modulus (also denotes kinetic energy)
G	shear modulus

Plasticity

$\varepsilon^{\mathbf{e}}$ elastic strain

$\varepsilon^{\mathbf{p}}$ plastic strain

$\dot{\varepsilon}$ strain rate

$\dot{\varepsilon}^{\mathbf{p}}$ plastic strain rate

Φ yield function

$\dot{\gamma}$ plastic multiplier

H hardening modulus

Contact

a a

b b

Shock

a a

b b

Chapter 2

Math

2.1 Mathematical Preliminaries

2.1.1 Continuity of Derivatives

Several results will make use of the **Schwarz theorem**, which states that if function $f : \mathbb{R}^2 \mapsto \mathbb{R}$ has continuous second partial derivatives at a given point $\mathbf{p} \in \mathbb{R}^2 = \{p_1, p_2\}^T$,¹ then

$$\frac{\partial^2 f(\mathbf{p})}{\partial x_i \partial x_k} = \frac{\partial^2 f(\mathbf{p})}{\partial x_k \partial x_i} \quad (\text{Commutative}) \quad (2.1)$$

The Schwarz theorem establishes that the result of differentiation of a function of two variables is independent of the order of that differentiation.

2.1.2 Vector Spaces

As defined by Gurtin,²

- LIN= the set of all tensors
- LIN⁺= the set of all tensors $\mathbf{S}$ with $\det \mathbf{S} > 0$
- SYM= set of all symmetric tensors
- SKW= set of all skew tensors
- PSYM= set of all positive definite, symmetric tensors
- ORTH= set of all orthogonal tensors

¹This relationship can be extended to general dimensionality of $f : \mathbb{R}^n \mapsto \mathbb{R}$, but this higher dimensionality is not used in the development that follows.

²Gurtin ME. An introduction to continuum mechanics. Academic press; 1982, Vol. 158, Jan 12.

- ORTH^+ = set of all rotations

As defined by Curnier,³ for all second-order tensors $\mathbf{S} \in \text{LIN}$, the **tracor** $\mathbb{T}$ and **deviator** $\mathbb{D}$ are fourth-order tensors are defined as

$$\mathbb{T} \triangleq \frac{1}{3} \delta_{ij} \delta_{kl} \hat{\mathbf{e}}_i \otimes \hat{\mathbf{e}}_j \otimes \hat{\mathbf{e}}_k \otimes \hat{\mathbf{e}}_l \quad (2.2)$$

such that

$$\mathbb{T} : \mathbf{S} = \mathbf{S}_{\text{VOL}} \triangleq \frac{1}{3} \text{tr}(\mathbf{S}) \mathbf{1}, \quad (2.3)$$

and

$$\mathbb{D} \triangleq \left(\frac{1}{2} \delta_{ik} \delta_{jl} + \frac{1}{2} \delta_{il} \delta_{jk} - \frac{1}{3} \delta_{ij} \delta_{kl} \right) \hat{\mathbf{e}}_i \otimes \hat{\mathbf{e}}_j \otimes \hat{\mathbf{e}}_k \otimes \hat{\mathbf{e}}_l \quad (2.4)$$

such that

$$\mathbb{D} : \mathbf{S} = \mathbf{S}_{\text{DEV}} \triangleq \mathbf{S} - \frac{1}{3} \text{tr}(\mathbf{S}) \mathbf{1}. \quad (2.5)$$

Note, there are two other fourth-order tensors often used, the **symmetrizer** $\mathbb{S}$ and the **skewer** $\mathbb{W}$ defined, respectively, as

$$\mathbb{S} \triangleq \frac{1}{2} (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) \hat{\mathbf{e}}_i \otimes \hat{\mathbf{e}}_j \otimes \hat{\mathbf{e}}_k \otimes \hat{\mathbf{e}}_l, \quad (2.6)$$

and

$$\mathbb{W} \triangleq \frac{1}{2} (\delta_{ik} \delta_{jl} - \delta_{il} \delta_{jk}) \hat{\mathbf{e}}_i \otimes \hat{\mathbf{e}}_j \otimes \hat{\mathbf{e}}_k \otimes \hat{\mathbf{e}}_l. \quad (2.7)$$

2.1.3 Matrix Operations

Symmetry Operators

Let $\mathbf{A}$ be a matrix of dimension $n \times n$, with components a_{ij} , $i = 1 \dots n$, $j = 1 \dots n$. The symmetry operator $\text{SYM}(\cdot)$, on any square square matrix $\mathbf{A}$, is defined

$$\text{SYM}(\mathbf{A}) \triangleq \frac{1}{2} (\mathbf{A} + \mathbf{A}^\top). \quad (2.8)$$

The skew-symmetry operator $\text{SKW}(\cdot)$, on any square square matrix $\mathbf{A}$, is defined

$$\text{SKW}(\mathbf{A}) \triangleq \frac{1}{2} (\mathbf{A} - \mathbf{A}^\top). \quad (2.9)$$

³Curnier A. Computational methods in solid mechanics. Springer Science & Business Media; 2012 Dec 6.

Then *any* matrix $\mathbf{A}$ can be decomposed into symmetric and skew-symmetric parts through an additive decomposition as

$$\mathbf{A} = \underbrace{\frac{1}{2}(\mathbf{A} + \mathbf{A}^\top)}_{\text{SYM}} + \underbrace{\frac{1}{2}(\mathbf{A} - \mathbf{A}^\top)}_{\text{SKW}} = \text{SYM}(\mathbf{A}) + \text{SKW}(\mathbf{A}). \quad (2.10)$$

Let the symmetric, square matrix $\mathbf{M} = \text{SYM}(\mathbf{A})$ and the skew-symmetric, square matrix $\mathbf{W} = \text{SKW}(\mathbf{A})$. Then the matrix $\mathbf{A}$ can be written as

$$\mathbf{A} = \mathbf{M} + \mathbf{W}. \quad (2.11)$$

The symmetric matrix $\mathbf{M}$ has the property

$$\mathbf{M} = \mathbf{M}^\top \iff m_{ij} = m_{ji}. \quad (2.12)$$

Explicitly, for $\mathbf{M}$ as a (3×3) matrix

$$\begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{bmatrix} = \begin{bmatrix} m_{11} & m_{21} & m_{31} \\ m_{12} & m_{22} & m_{32} \\ m_{13} & m_{23} & m_{33} \end{bmatrix}, \quad (2.13)$$

and of the nine components of $\mathbf{M}$, only six are independent,

$$\mathbf{M} = \begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{12} & m_{22} & m_{23} \\ m_{13} & m_{23} & m_{33} \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & m_{13} \\ & m_{22} & m_{23} \\ \text{sym.} & & m_{33} \end{bmatrix}. \quad (2.14)$$

The skew-symmetric matrix $\mathbf{W}$ has the property

$$\mathbf{W} = -\mathbf{W}^\top \iff w_{ij} = -w_{ji}. \quad (2.15)$$

Explicitly, for $\mathbf{W}$ as a (3×3) matrix

$$\mathbf{W} = \begin{bmatrix} 0 & -w_{12} & w_{13} \\ w_{12} & 0 & -w_{23} \\ -w_{13} & w_{23} & 0 \end{bmatrix}, \quad (2.16)$$

and of the nine components of $\mathbf{W}$, only three are independent.

Remark 2.1.1. In some references, skew-symmetric is called *antisymmetric*. Here, we will use *skew-symmetric*, or sometimes just *skew* as an abbreviated form.

Remark 2.1.2. From spectral theory, the non-zero eigenvalues exist in pairs as purely imaginary numbers $0 \pm i\lambda_k$, $\lambda \in \mathbb{R}$, with

- $k = 1 \dots n$ for $\dim(\mathbf{W}) = (n, n)$ and n even and
- $k = 1 \dots (n-1)/2$ for $\dim(\mathbf{W}) = (n, n)$ and n odd.

For even n dimensions, there are no zero eigenvalues. For odd n dimensions, there is exactly one zero eigenvalue. In the explicit example for $\dim(\mathbf{W}) = (3, 3)$,

$$\mathbf{\Lambda} = \text{EIG}(\mathbf{W}) = \begin{bmatrix} 0 & \lambda & 0 \\ -\lambda & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (2.17)$$

Remark 2.1.3. For the skew-symmetric matrix $\mathbf{W}$, the tensor $\mathbf{Z} = \mathbf{W}^2$ is negative semi-definite,

$$\mathbf{v}^\top \mathbf{Z} \mathbf{v} \leq 0 \quad \forall \text{ vectors } \mathbf{v} \in \mathbb{R}^n. \quad (2.18)$$

Result 2.1.1 All $(n \times n)$ skew-symmetric matrices of odd dimension are singular.

Proof.

For skew-symmetric matrices,

$$\mathbf{W} = -\mathbf{W}^\top. \quad (2.19)$$

From the properties of determinants

$$\det(\mathbf{W}) = \det(-\mathbf{W}^\top) = (-1)^n \det(\mathbf{W}), \quad (2.20)$$

which, since n is odd, becomes,

$$\det(\mathbf{W}) = -\det(\mathbf{W}) = 0, \quad (2.21)$$

■

Result 2.1.2 The rank of a skew-symmetric matrix must be even.

Proof.

For skew-symmetric matrices,

$$\mathbf{W} = -\mathbf{W}^T. \quad (2.22)$$

From the properties of determinants

$$\det(\mathbf{W}) = \det(-\mathbf{W}^T) = (-1)^n \det(\mathbf{W}). \quad (2.23)$$

If n is odd, then $\det(\mathbf{W}) = 0$ (see Result 2.1.1). Therefore, the rank of $\mathbf{W}$ must be even. ■

Cross-Product Operator

One special form of a skew-symmetry operator is a **cross-product operator**, defined as

$$[\mathbf{a}]_{\times} \triangleq \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}. \quad (2.24)$$

Thus $\mathbf{a} \times \mathbf{b} = [\mathbf{a}]_{\times} \mathbf{b}$.

2.1.4 Invariants of Second Order Tensors

For a second order tensor $\mathbf{A}$, the three invariants, $I_1(\mathbf{A})$, $I_2(\mathbf{A})$, and $I_3(\mathbf{A})$ of matrix $\mathbf{A}$ satisfy the equation

$$\det(\mathbf{A} - \lambda \mathbf{1}) \stackrel{\text{set}}{=} 0 \quad (2.25)$$

for any λ , and produce the characteristic equation

$$0 = -\lambda^3 + I_1 \lambda^2 - I_2 \lambda + I_3, \quad (2.26)$$

$$\begin{aligned} &= -\lambda^3 + \underbrace{(a_{11} + a_{22} + a_{33})}_{I_1(\mathbf{A}) = \text{tr}(\mathbf{A})} \lambda^2 - \\ &\quad \underbrace{(a_{11}a_{22} + a_{22}a_{33} + a_{33}a_{11} - a_{12}a_{21} - a_{23}a_{32} - a_{31}a_{13})}_{I_2(\mathbf{A}) = \frac{1}{2} [\text{tr}(\mathbf{A})\text{tr}(\mathbf{A}) - \text{tr}(\mathbf{A} \cdot \mathbf{A})]} \lambda + \\ &\quad \underbrace{(a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31})}_{I_3(\mathbf{A}) = \det(\mathbf{A})} \lambda^0. \end{aligned} \quad (2.27)$$

2.1.5 Extrema

Arg Max

The **arguments of the maxima**, also known as **arg max** or **argmax**, at the values of the domain x that maximize the range $f(x)$.

Example 1.

Defined over the domain $x \in [0, 2\pi]$, the function $f(x) = \cos(x)$ has maximum values of 1 at $x = 0$ and $x = 2\pi$. This notion is written with the max as

$$\max_{x \in [0, 2\pi]} f(x) = \max_{x \in [0, 2\pi]} \cos(x) = 1. \quad (2.28)$$

The **arg max** is then

$$\arg \max_{x \in [0, 2\pi]} f(x) = \arg \max_{x \in [0, 2\pi]} \cos(x) = \{0, 2\pi\}. \quad (2.29)$$

More generally, for x defined over the entire real number line,

$$\arg \max_{x \in \mathbb{R}} f(x) = \arg \max_{x \in \mathbb{R}} \cos(x) = \{2\pi k\} \text{ for } k \in \mathbb{J}. \quad (2.30)$$

...

Chapter 3

Formulation

3.1 Fixed Places

Let $\mathfrak{X}$ denote a **fixed place**, which will be understood as a fundamental and elementary answer to the question of “Where?” For now, a fixed place is understood without recourse to concepts of location or position, since location and position require at least *two* fixed places to be defined (the subject fixed place itself and an origin’s fixed place).¹ Let $\mathbb{S}$ denote **space**, defined as the infinite set of fixed places $\mathfrak{X}$.

3.2 Coordinates

Now, we define the concept of location of fixed places through introduction of an origin, bases vectors, which embed both a direction and a magnitude on the space $\mathbb{S}$. With space, an origin, and a basis, **coordinates** arise as a measure of the distance from the origin to the fixed place, understood as the location. Coordinates are measure, given a particular **coordinate system**. The coordinate system considered herein will be the Cartesian coordinate system, as opposed to the polar coordinate system or the spherical coordinate system.

3.2.1 Space-Fixed Coordinates

From the infinite set of fixed places $\mathbb{S}$, we select one fixed place to designate as the **origin** o . (See Fig. 3.1.)

The term **dextral** designates a *right-handed rule*, such that $\mathbf{e}_1 \times \mathbf{e}_2 = \mathbf{e}_3$, $\mathbf{e}_2 \times \mathbf{e}_3 = \mathbf{e}_1$, and $\mathbf{e}_3 \times \mathbf{e}_1 = \mathbf{e}_2$, or in a more compact notation, that $\mathbf{e}_i \times \mathbf{e}_j = \mathbf{e}_k$. The term **orthonormal** indicates that the basis vectors are straight and are mutually perpendicular. The term **unit**

¹With the notions of space coordinates, introduced in the following section, we can think of a fixed place as a location that has an immutable position.

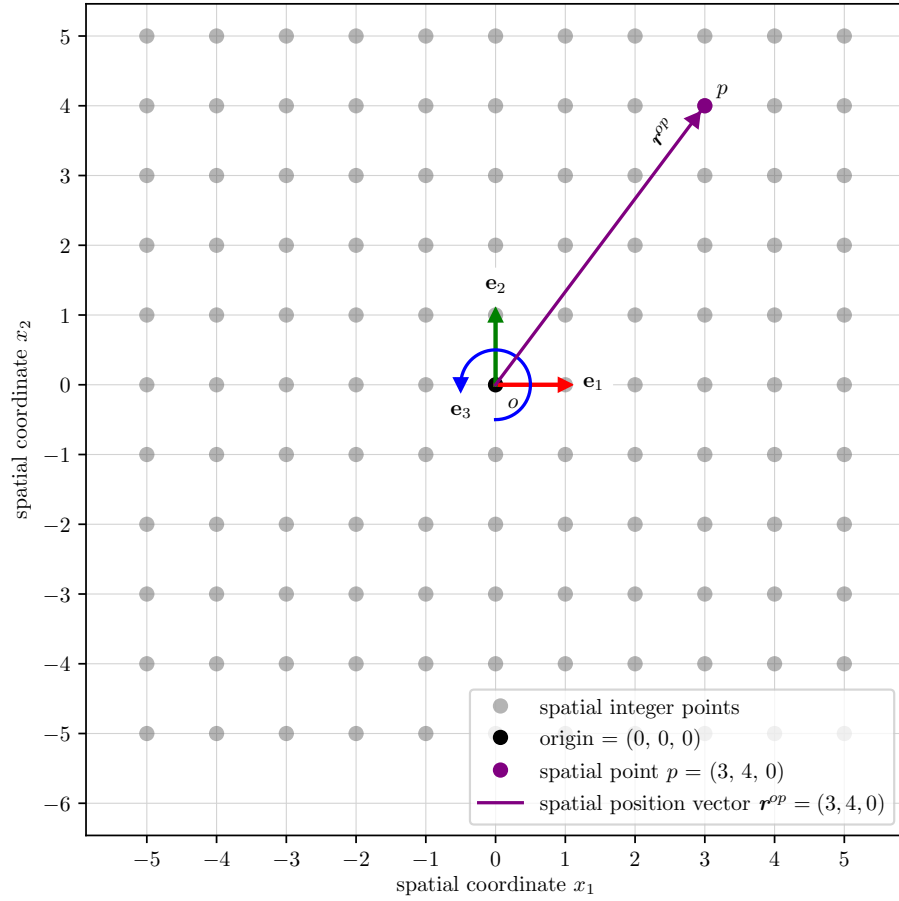


Figure 3.1: Spatial points plotted at integers in gray over the range $\mathbf{x} = (x_1, x_2, x_3) \in \mathbb{R}^3$, such that $x_1 \in [-5, 5]$, $x_2 \in [-5, 5]$, and $x_3 = 0$. The origin o is indicated at $(0, 0, 0)$. The $\mathbf{e}_3$ vector points out of the page. A particular spatial point p , located at $(3, 4, 0)$, is indicated in purple. The position vector $\mathbf{r}^{op}$ locates points p in the spatial coordinate system S , spanned by $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, with origin o . Source: `spatial_points.py`.

indicates the length of each vector is unity. Thus,

$$\| \mathbf{e}_1 \| = \| \mathbf{e}_2 \| = \| \mathbf{e}_3 \| = 1. \quad (3.1)$$

The origin o and the spatial basis vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ together compose the **spatial coordinate system** S , which is a one-to-one correspondence between fixed places $\mathfrak{X}$ in space $\mathbb{S}$ and their **spatial coordinates** $\mathbf{x}$. The spatial coordinates are an ordered set of three real numbers $\{x_1, x_2, x_3\}^T$ measured in a particular coordinate system from the coordinate system's origin to the fixed place $\mathfrak{X}$ in space $\mathbb{S}$.

A place $\mathfrak{X}$ in space $\mathbb{S}$ is *located* the spatial coordinate system S by the particular

spatial coordinate $\mathbf{x}$, written in **explicit form** as

$$\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + x_3\mathbf{e}_3 = x_i\mathbf{e}_i. \quad (3.2)$$

Spatial coordinates locate the position of a place $\mathfrak{X}$ in space $\mathbb{S}$. Thus the coordinate system S is a map from a place $\mathfrak{X}$, which is a member of space $\mathbb{S}$, to a coordinate in real, three-dimensional space. Mathematically, we write, for $\mathfrak{X} \in \mathbb{S}$ and $\mathbf{x} \in \mathbb{R}^3$,

$$S : \mathfrak{X} \mapsto \mathbf{x}. \quad (3.3)$$

Since, at the present moment, we have only introduced one coordinate system (and thus only one origin and one set of basis vectors), we need not designate the basis vectors explicitly, and may write in **implicit form**,

$$\mathbf{x} = \{x_1, x_2, x_3\}^\top, \quad (3.4)$$

without ambiguity. For simplicity, we sometimes will write the vector $\mathbf{x}$ as a triple,

$$\mathbf{x} = (x_1, x_2, x_3), \quad (3.5)$$

without the braces and the transpose, which are formalisms to explicitly indicate a vector. This abbreviated form (3.4), however, will cause ambiguity later, when two or more coordinate systems are present to describe places in space. In this latter case, the explicit basis vectors or the explicit coordinate system must be stated to avoid confusion. To explicitly state the coordinate system in which the coordinates are measures, we indicate the basis vector, such as in $x_i\mathbf{e}_i$.

3.2.2 Body-Fixed Coordinates

Next, envision a body as a collection of particles existing in space. Denote the set of particles as $B \in \mathbb{R}^3$. From either one of the particles in body B , or from a particles that moves with the body,² we selected one reference point as the **body origin** O . (See Fig. 3.2.)

At the body origin O , we construct three dextral, orthonormal be unit vectors $\mathbf{E}_1$, $\mathbf{E}_2$, $\mathbf{E}_3$, fixed in the body, called **material basis vectors**. The origin O and the material basis vectors $\mathbf{E}_1$, $\mathbf{E}_2$, $\mathbf{E}_3$ together compose the **material coordinate system** M , which is a one-to-one correspondence between places in both B and their material coordinates $\mathbf{X}$. The material coordinates are an ordered set of three real numbers $\{X_1, X_2, X_3\}^\top$ measured in the body's coordinate system. A particle in body B is *located* in the material coordinate system M by the particular material coordinate $\mathbf{X}$, written in **explicit form** as

$$\mathbf{X} = X_1\mathbf{E}_1 + X_2\mathbf{E}_2 + X_3\mathbf{E}_3 = X_I\mathbf{E}_I. \quad (3.6)$$

²A particle not on the body but moving with the body will be referred to as a *carrier frame*, defined later.

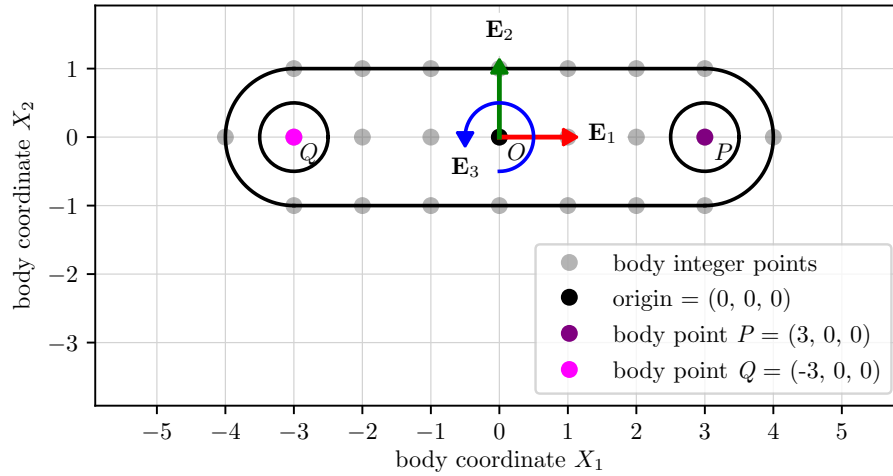


Figure 3.2: Body points plotted at integers in gray over the domain $B \subset \mathbb{R}^3$, bounded by the solid black lines. The origin O is indicated at $(0, 0, 0)$. The $\mathbf{E}_3$ vector points out of the page. Two particular material points P and Q are located at $(3, 0, 0)$ and $(-3, 0, 0)$, respectively. The position vector $\mathbf{R}^{OP}$ locates point P in the body coordinate system B , spanned by $\mathbf{E}_1$, $\mathbf{E}_2$, $\mathbf{E}_3$, with origin O . Likewise, position vector $\mathbf{R}^{OQ}$ locates point Q . Source: `rigid_body.py`.

Thus, material coordinates locate a particle in a body B .

If we do not designate the basis vectors explicitly with capital letters, we may write in **implicit form**,

$$\mathbf{X} = \{X_1, X_2, X_3\}^\top, \quad (3.7)$$

without ambiguity as measures with respect to the body coordinate M . For simplicity, we sometimes will write the vector $\mathbf{X}$ as a triple,

$$\mathbf{X} = (X_1, X_2, X_3), \quad (3.8)$$

without the braces and the transpose, which are formalisms to explicitly indicate a vector.

We have now introduced two coordinate systems: the space-fixed coordinate system and the body-fixed coordinate system. To speak of a particle measured with space coordinates, we will need a way to transform body coordinates into space coordinates. This will be achieved through a coordinate transformation. To lay the groundwork for this transformation, we first define the concept of the **reference configuration**.

3.3 Configurations

3.3.1 Reference Configuration

3.3.2 Current Configuration

Chapter 4

General Motion

4.1 Motion

Let the arbitrary time interval be defined as $\mathbb{T} \triangleq [t_0, t_f]$, from initial to final time, inclusive.¹ Let the **motion**, a one parameter family of configurations, $\varphi : \mathbb{B} \times \mathbb{T} \mapsto \mathbb{R}^3$ map the material particle $\mathbf{X}$ (the reference configuration) into the current configuration $\mathbf{x}$,

$$\mathbf{x} = \varphi(\mathbf{X}, t) = \varphi_t(\mathbf{X}). \quad (4.1)$$

A motion φ evaluated at a particular time $t \in \mathbb{T}$ is referred to as a **current configuration** or **placement**. For any placement at time t , there is a **displacement** field $\mathbf{u} : \mathbb{B} \times \mathbb{T} \mapsto \mathbb{R}^3$,

$$\mathbf{u}(\mathbf{X}, t) \triangleq \varphi(\mathbf{X}, t) - \mathbf{X} \iff u_i \triangleq \varphi_i - \delta_{iJ} X_J. \quad (4.2)$$

Thus, the current configuration φ is simply a function of the original placement $\mathbf{X}$, plus a displacement $\mathbf{u}$, which is **function** of placement $\mathbf{X}$ and time t ,

$$\varphi(\mathbf{X}, t) = \mathbf{X} + \mathbf{u}(\mathbf{X}, t) \iff \varphi_i = \delta_{iJ} X_J + u_i. \quad (4.3)$$

The initial condition $\mathbf{u}_0$ is found from the initial placement φ_0 and the reference configuration $\mathbf{X}$,

$$\mathbf{u}_0(\mathbf{X}) = \mathbf{u}(\mathbf{X}, 0) = \varphi(\mathbf{X}, 0) - \mathbf{X}. \quad (4.4)$$

4.2 Deformation Gradient

To each configuration φ , we define a **deformation gradient** $\mathbf{F} : \mathbb{B} \times \mathbb{T} \mapsto \mathbb{M}_+^3$,

$$\mathbf{F} \triangleq \nabla_0 \varphi(\mathbf{X}, t) \iff F_{iJ} \triangleq \frac{\partial \varphi_i}{\partial X_J}. \quad (4.5)$$

¹Note that t_0 , while typically zero, may be any real number less than t_f .

Lower case indices represent components of tensors in the current configuration. Upper case indices represent components of tensors in the reference configuration. Real, square matrices of dimension three and positive determinants are denoted $\mathbb{M}_+^3$. Gradient operations with and without a subscript “0” are gradients taken in the reference and current configurations, respectively:

$$\nabla_0(\cdot) = \partial(\cdot)/\partial X_J, \quad \text{reference configuration gradient, and} \quad (4.6)$$

$$\nabla(\cdot) = \partial(\cdot)/\partial x_j, \quad \text{current configuration gradient.} \quad (4.7)$$

Alternative notations are $\text{GRAD}(\cdot)$ and $\text{grad}(\cdot)$, respectively.

4.3 Jacobian of the Deformation Gradient

The **Jacobian** of the deformation gradient,

$$J \triangleq \det \mathbf{F}, \quad (4.8)$$

describes the (generally non-uniform) volumetric expansion or contraction of the motion from the reference configuration $\mathbf{X}$. All configurations must be admissible in the sense that the Jacobian of the deformation must be positive ($J > 0$). This requirement keeps the deformations from mapping the body to a single, infinitesimally small point ($J = 0$) or turning the body inside-out ($J < 0$).

Isochoric motions preserve body’s total volume. A Jacobian of unity ($J = 1$) describes an isochoric motion. Table 4.1 describes the categories of motions (expansion, volume-preserving, contraction, and inadmissible) by Jacobian measure.

Table 4.1: Jacobian measure to categorize deformations.

$J < 0$ inadmissible	$J = 0$ inadmissible	$0 < J < 1$ contraction	$J = 1$ isochoric	$J > 1$ expansion
body has turned inside-out	body has shrunk to zero volume	body’s total volume has decreased	body’s total volume is preserved	body’s total volume has increased

Four important isochoric deformations are (1) pure translation, (2) pure rotation, (3) isochoric stretch, and (4) isochoric shear. Section 5 examines these deformations in detail.

4.4 Displacement Gradient

From the displacement field in (4.2), the relationship between the **displacement gradient** and the deformation gradient is given by

$$\nabla_0 \mathbf{u} = \nabla_0 \boldsymbol{\varphi} - \mathbf{1} = \mathbf{F} - \mathbf{1} \iff \frac{\partial u_i}{\partial X_J} = \frac{\partial \varphi_i}{\partial X_J} - \delta_{iJ}. \quad (4.9)$$

4.5 Right Cauchy-Green Deformation

The **right Cauchy-Green deformation** arises from the inner product of two differential fiber elements in the reference configuration, $d\mathbf{X}_1$ and $\mathbf{X}_2$, by the deformation gradient $\mathbf{F}$, to obtain the inner product of the same differential fibers in the current configuration, $d\mathbf{x}_1$ and $d\mathbf{x}_2$,

$$d\mathbf{x}_1 \cdot d\mathbf{x}_2 = d\mathbf{x}_1^\top d\mathbf{x}_2 = (\mathbf{F} d\mathbf{X}_1)^\top \mathbf{F} d\mathbf{X}_2 = d\mathbf{X}_1^\top \mathbf{F}^\top \mathbf{F} d\mathbf{X}_2 = d\mathbf{X}_1^\top \mathbf{C} d\mathbf{X}_2, \quad (4.10)$$

where

$$\mathbf{C} \triangleq \mathbf{F}^\top \mathbf{F} \iff C_{IJ} \triangleq F_{Ii} F_{iJ}. \quad (4.11)$$

The right Cauchy-Green deformation tensor $\mathbf{C}$: (1) is defined in the reference configuration, (2) is symmetric and postive-definite, (3) gets its name from the location of the deformation gradient $\mathbf{F}$ in the definition, which is to the right, (4) is a metric that maps fiber lengths from the reference configuration to the current configuration, and (5) is second-order in reference displacement gradients, as shown in (4.13):

$$\mathbf{C} \stackrel{(4.5)}{=} (\nabla_0 \boldsymbol{\varphi})^\top \nabla_0 \boldsymbol{\varphi} \stackrel{(4.9)}{=} (\mathbf{1} + \nabla_0 \mathbf{u})^\top (\mathbf{1} + \nabla_0 \mathbf{u}), \quad (4.12)$$

$$= \mathbf{1} + \underbrace{\nabla_0 \mathbf{u} + (\nabla_0 \mathbf{u})^\top}_{\text{1st order}} + \underbrace{(\nabla_0 \mathbf{u})^\top \nabla_0 \mathbf{u}}_{\text{2nd order}}. \quad (4.13)$$

This result can be expected since, by definition, $\mathbf{C}$ is second-order in the deformation gradient $\nabla_0 \boldsymbol{\varphi}$, and the relationship between the deformation gradient $\nabla_0 \boldsymbol{\varphi}$ and displacement gradient $\nabla_0 \mathbf{u}$ is linear.

4.6 Left Cauchy-Green Deformation

The **left Cauchy-Green deformation** arises from similar multiplication as with the right Cauchy-Green deformation, but with the stretching going in the reverse, from current configuration back to reference configuration,

$$d\mathbf{X}_1 \cdot d\mathbf{X}_2 = d\mathbf{X}_1^\top d\mathbf{X}_2 = (\mathbf{F}^{-1} d\mathbf{x}_1)^\top \mathbf{F}^{-1} d\mathbf{x}_2 = d\mathbf{x}_1^\top \mathbf{F}^{-\top} \mathbf{F}^{-1} d\mathbf{x}_2 = d\mathbf{x}_1^\top \mathbf{b}^{-1} d\mathbf{x}_2, \quad (4.14)$$

where

$$\mathbf{b} \triangleq \mathbf{F}\mathbf{F}^T \iff b_{ij} \triangleq F_{iJ} F_{Jj}. \quad (4.15)$$

The left Cauchy-Green deformation tensor $\mathbf{b}$: (1) is defined in the current configuration, (2) is symmetric and postive-definite, (3) gets it name from the location of the deformation gradient $\mathbf{F}$ in the definition, which is to the left, (4) is a metric whose inverse maps fiber lengths from the current configuration to the reference configuration, and (5) is second-order in current displacement gradients.

4.7 Polar Decomposition

Given the **rotation tensor** $\mathbf{R}$, the **material stretch tensor** $\mathbf{U}$, and **spatial stretch tensor** $\mathbf{v}$, the deformation gradient $\mathbf{F}$ has the multiplicative decomposition,

$$\mathbf{F} = \mathbf{R}\mathbf{U} = \mathbf{v}\mathbf{R} \iff F_{iJ} = R_{iK} U_{KJ} = v_{ik} R_{kJ}, \quad (4.16)$$

Here we have a slight abuse of notation, where intermediate configurations that have stretched but not yet rotated are denoted with capital letter indices. Thus the “ K ” subscript in $R_{iK}U_{KJ}$ is an intermediate stretched but non-rotated configuration.

The stretch tensors $\mathbf{U}$ and $\mathbf{v}$ are both symmetric and positive definite. The rotation tensor $\mathbf{R}$ is non-symmetric and orthogonal. Figure 4.1 shows the polar decomposition about a material point $\mathbf{X}$ and fibers $d\mathbf{X}$ in its vicinity mapped to the spatial point $\mathbf{x}$ with the same fibers mapped to $d\mathbf{x}$.

4.8 Principal Stretches and Axes

The stretch tensors $\mathbf{U}$ and $\mathbf{v}$ have the same eigenvalues, $(\lambda_1, \lambda_2, \lambda_3)$, called **principal stretches**. For non-trivial rotations, *i.e.*, $\mathbf{R} \neq \mathbf{1}$, $\mathbf{U}$ and $\mathbf{v}$ have unique eigenvectors, called **principal stretch directions**. The principal stretch directions of $\mathbf{U}$ are $\{\mathbf{N}_1, \mathbf{N}_2, \mathbf{N}_3\}$. The principal stretch directions of $\mathbf{v}$ are $\{\mathbf{n}_1, \mathbf{n}_2, \mathbf{n}_3\}$. The two sets of eigenvectors are related through rotation $\mathbf{R}$,

$$\mathbf{n}_1 = \mathbf{R}\mathbf{N}_1, \quad \mathbf{n}_2 = \mathbf{R}\mathbf{N}_2, \quad \mathbf{n}_3 = \mathbf{R}\mathbf{N}_3; \quad (4.17)$$

or generally,

$$\mathbf{n}_\alpha = \mathbf{R}\mathbf{N}_\alpha \quad \text{for } \alpha = 1, 2, 3. \quad (4.18)$$

4.9 Spectral Representation

The deformation gradient, its polar decomposition, and the Cauchy-Green deformations have spectral decompositions in terms of the principal stretches and stretch directions,

$$\mathbf{F} = \sum_{\alpha=1}^3 \lambda_{\alpha} \mathbf{n}_{\alpha} \otimes \mathbf{N}_{\alpha} \quad \mathbf{U} = \sum_{\alpha=1}^3 \lambda_{\alpha} \mathbf{N}_{\alpha} \otimes \mathbf{N}_{\alpha} \quad (4.19)$$

$$\mathbf{R} = \sum_{\alpha=1}^3 \mathbf{n}_{\alpha} \otimes \mathbf{N}_{\alpha} \quad \mathbf{v} = \sum_{\alpha=1}^3 \lambda_{\alpha} \mathbf{n}_{\alpha} \otimes \mathbf{n}_{\alpha} \quad (4.20)$$

$$\mathbf{C} = \sum_{\alpha=1}^3 \lambda_{\alpha}^2 \mathbf{N}_{\alpha} \otimes \mathbf{N}_{\alpha} \quad \mathbf{b} = \sum_{\alpha=1}^3 \lambda_{\alpha}^2 \mathbf{n}_{\alpha} \otimes \mathbf{n}_{\alpha} \quad (4.21)$$

4.10 Rates

Total Time Derivative

Let $\phi(\mathbf{x}, t)$ be an arbitrary spatial *scalar* field and let $\boldsymbol{\phi}(\mathbf{x}, t)$ be an arbitrary spatial *vector* field. For a body with spatial velocity $\mathbf{v}(\mathbf{x}, t)$, the total time derivative of the scalar field ϕ is given by

$$\dot{\phi}(\mathbf{x}, t) = \phi_{,t} + \nabla \phi \cdot \mathbf{v} \iff \dot{\phi} = \frac{\partial \phi}{\partial t} + \frac{\partial \phi}{\partial x_j} \frac{\partial x_j}{\partial t}. \quad (4.22)$$

The total time derivate of the vector field $\boldsymbol{\phi}$ is given by

$$\dot{\boldsymbol{\phi}}(\mathbf{x}, t) = \boldsymbol{\phi}_{,t} + [\nabla \boldsymbol{\phi}] \mathbf{v} \iff \dot{\phi}_i = \frac{\partial \phi_i}{\partial t} + \frac{\partial \phi_i}{\partial x_j} \frac{\partial x_j}{\partial t}. \quad (4.23)$$

Remark 4.10.4. Some texts prefer the notation $\mathbf{v} \cdot \nabla \boldsymbol{\phi}$ to $[\nabla \boldsymbol{\phi}] \mathbf{v}$, used in (4.23).

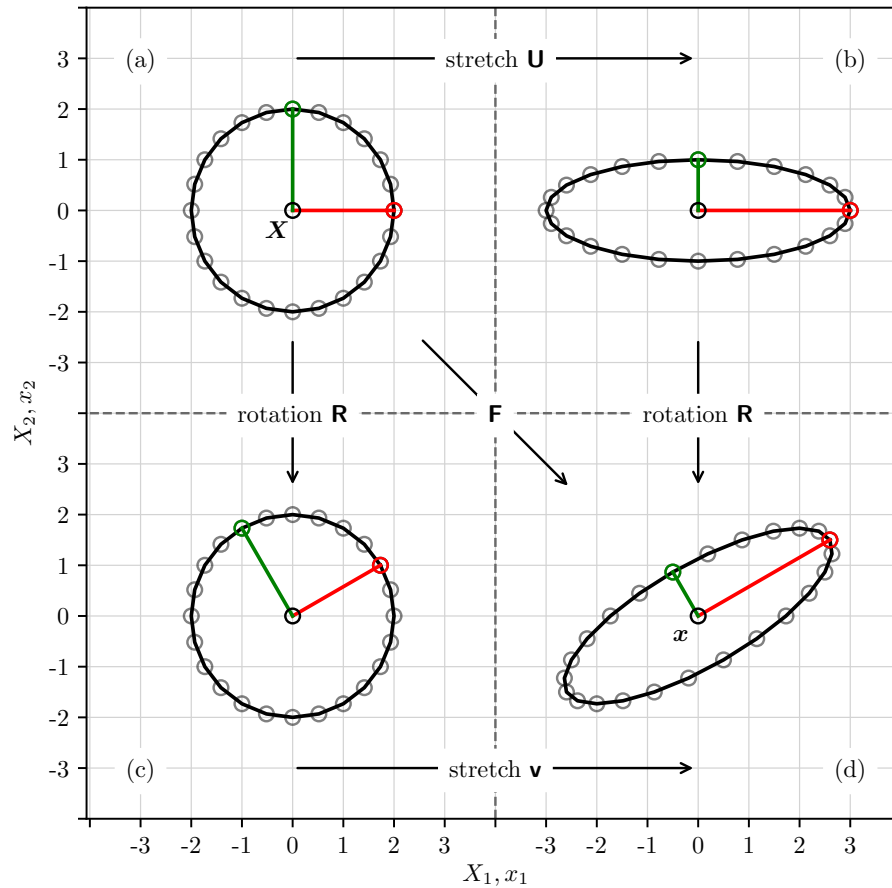


Figure 4.1: In the vicinity of $\mathbf{X}$, mapped to $\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X})$, the polar decomposition of deformation gradient $\mathbf{F}$ into stretch $\mathbf{U}$ then rotation $\mathbf{R}$; or, into rotation $\mathbf{R}$ then stretch $\mathbf{v}$: (a) Reference configuration, (b) stretched configuration, (c) rotated configuration, (d) current configuration. As shown, the eigenvalues of $\mathbf{U}$ (and $\mathbf{v}$) are $\lambda_1 = 1.5$, $\lambda_2 = 0.5$, $\lambda_3 = 1$ and the rotation $\mathbf{R}$ has a magnitude of 30° about the $\mathbf{e}_3$ axis. Source: `polar_decomposition.py`.

Chapter 5

Specific Motions

5.0.1 Inverse Motions

Next we explore specific cases of general motions. One type of specific motion is the **inverse motion**, which allows for a map from the current configuration back to the reference configuration.

The map from $\mathbf{X}$ to $\mathbf{x}$ is one-to-one and invertible. We write the inverse map, from $\mathbf{x}$ to $\mathbf{X}$ as

$$\mathbf{X} = \varphi^{-1}(\mathbf{x}, t) = \varphi_t^{-1}(\mathbf{x}), \quad (5.1)$$

which maps the current configuration back to the reference configuration.

5.0.2 Homogeneous Motions

Another type of specific motion is a **homogeneous motion**, which is linear with respect to the reference configuration $\mathbf{X}$. Consider a Taylor series expansion of the motion about a point $\mathbf{X}_0$,

$$\varphi(\mathbf{X}) = \varphi(\mathbf{X}_0) + \mathbf{F}(\mathbf{X}_0) \cdot \{\mathbf{X} - \mathbf{X}_0\} + \text{H.O.T.} \quad (5.2)$$

When the deformation gradient $\mathbf{F}$ is constant, thus the higher order terms (H.O.T.) are zero. Thus, a homogeneous motion requires that

$$\varphi(\mathbf{X}) = \varphi(\mathbf{X}_0) + \mathbf{F}(\mathbf{X}_0) \cdot \{\mathbf{X} - \mathbf{X}_0\} \quad \forall \quad \{\mathbf{X}\} \text{ and } \{\mathbf{X}_0\} \in \mathcal{B}. \quad (5.3)$$

Homogeneous motions will consist of several subtypes of isochoric motions, such as translations, rotations, and special case stretches. Figure 5.1 illustrates the inheritance relationship between motions and is explicated here:

Remark 5.0.5.

- Homogeneous motions are a subset of general motions. All homogeneous motions are motions, but not all motions are homogeneous motions.
- A homogeneous motion that is free from fixed points is a **translation**. When a body translates, no part of the body is fixed in space.
- A homogeneous motion that has a fixed point will be either a **rotation** or a **stretch**.
- Translations and rotations are rigid body motions, which are in contrast to deformable body motions, which occur as stretches.
- Stretches may be volume preserving (isochoric, $J = 1$), or non-isochoric ($J \neq 1$). All rigid body motions are isochoric.
- Isochoric deformations often occur as shear, but there is one special case deformation, absent of shear, that occurs when the volume is preserved through expansion along one principal axis, and contraction along the other two principal axes.
- The foregoing descriptions are explained in greater detail in the following sections.

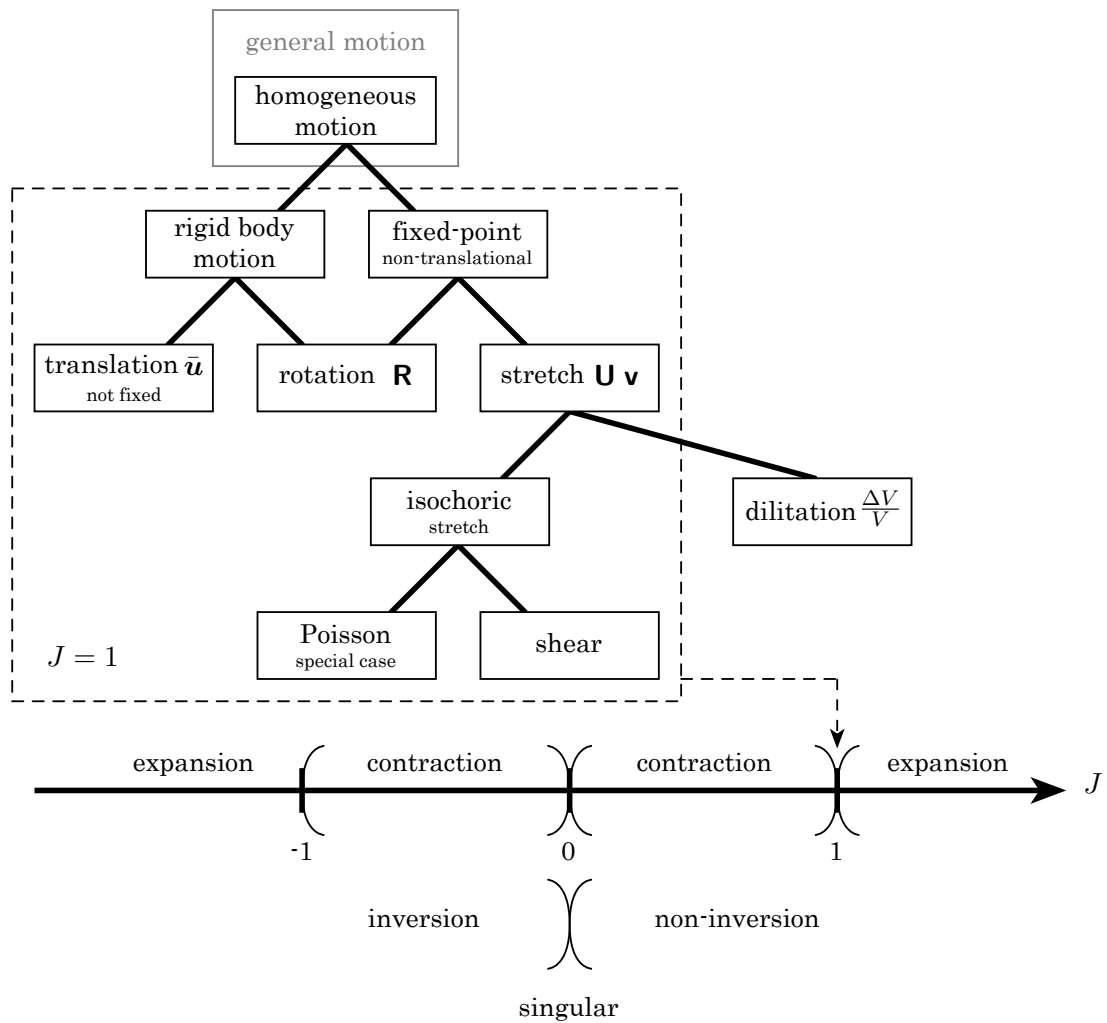


Figure 5.1: Taxonomy of general motions.

5.0.3 Translation

A translation is a homogeneous deformation of the form

$$\boldsymbol{\varphi}(\mathbf{X}) = \bar{\mathbf{u}} + \mathbf{X}. \quad (5.4)$$

This deformation occurs when the displacement $\mathbf{u}(\mathbf{X}, t)$ is a constant $\bar{\mathbf{u}}$, and thus not a function of reference position $\mathbf{X}$ or time t .

For the concrete example in Figure (5.2), let $\bar{\mathbf{u}} = \langle \bar{u}_1, \bar{u}_2 \rangle^\top = \langle 3, 2 \rangle^\top$. In this case, we see the placement $\boldsymbol{\varphi}$ moves right and up on the page, relative to the reference configuration $\mathbf{X}$, by an amount of 3 and 2, respectively. The reference configuration is shown in blue. The current configuration is shown in red.

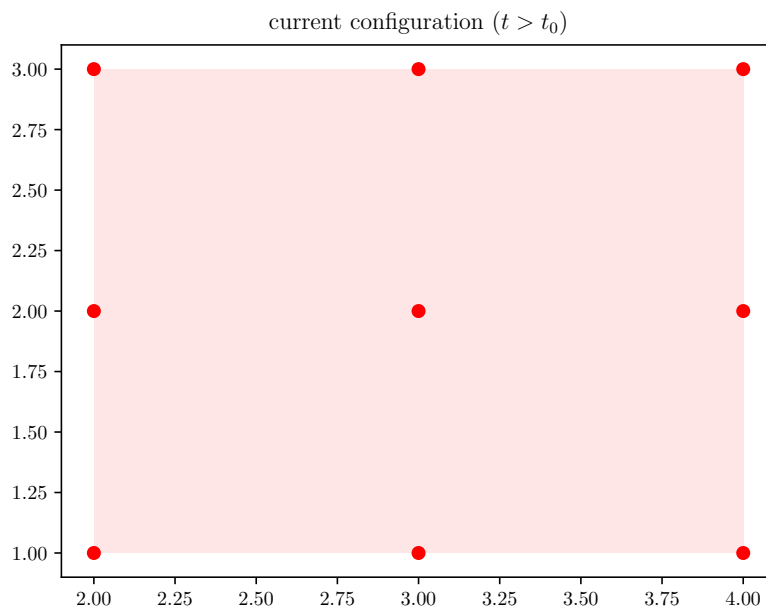


Figure 5.2: Illustration of a translational motion.

5.0.4 Fixed Point

A fixed point deformation is a homogeneous deformation of the form

$$\boldsymbol{\varphi}(\mathbf{X}) = \mathbf{X}_0 + \mathbf{F} \cdot \{\mathbf{X} - \mathbf{X}_0\}. \quad (5.5)$$

The deformation is fixed in the sense that $\boldsymbol{\varphi}(\mathbf{X}_0) = \mathbf{X}_0$.

5.0.5 Rotation

A rotation is a homogeneous deformation of the form

$$\boldsymbol{\varphi}(\mathbf{X}) = \mathbf{X}_0 + \mathbf{R} \cdot \{\mathbf{X} - \mathbf{X}_0\}, \quad (5.6)$$

where the rotation occurs about the point $\mathbf{X}_0$.

A rigid body motion, consisting of body translation and rotation, has the form

$$\boldsymbol{\varphi}(\mathbf{X}) = \bar{\mathbf{u}} + \mathbf{X}_0 + \mathbf{R} \cdot \{\mathbf{X} - \mathbf{X}_0\}, \quad (5.7)$$

where the translation is $\bar{\mathbf{u}}$ and the rotation occurs around point $\mathbf{X}_0$.

To motivate rotations of deformable bodies, we start with two simpler cases: general motion two *particles* instead of bodies, and general motion of *rigid* bodies instead of deformable bodies. These two specific cases will then be building blocks for the more general case of rotation of a deformable body, with and without stretch.

We first consider two particles, and describe absolute motion versus relative motion of particles. Then, we connect two particles to create a two-particle body, and show how a rigid body assumption leads to the absence of relative translation between the particles of a body. While the absence of relative translation leads to the absence of absolute **translation** between two particles, remarkably, two particles can still move relative to each other without violating the rigid body assumption and produce absolute translation through **rotation** of the rigid body.

Finally, we connect these two ideas—translation of particles and rotation of a rigid body—to describe the general case described as a particle moving on a rigid body. This formulation will then be further generalized to describe motion composed of *both* rotation and stretch of a deformable body.

Translation of Particles

Consider two particles P and Q located in the inertial reference frame $\mathcal{F}$ and located with position vectors $\mathbf{X}_P$ and $\mathbf{X}_Q$ from origin O .

For the time being, assume we know how particle Q moves in frame $\mathcal{F}$ and how particle P moves *relative* to particle Q . Given these two knowns, we wish to construct an absolute relationship for how particle P moves in the inertial reference frame $\mathcal{F}$.

To achieve this, we write the **loop vector**,

$$\mathbf{X}_P = \mathbf{X}_Q + \mathbf{X}_{P/Q} \quad (5.8)$$

in the reference configuration, which states that distance to point P from the origin O is equal to the distance to point Q from the origin O plus the distance to point P from

point Q .¹ Note, we could have written $\mathbf{X}_P$ explicitly as $\mathbf{X}_{P/O}$, but drop the notation of the origin O when it is used as the datum. This relationship extend to the current configuration through the absolute and relative displacement measures. The current placement of P , $\mathbf{x}_P$, is the superposition of the current placement of Q with the current placement of P *relative* to the current placement of Q . Explicitly,

$$\mathbf{x}_P(\mathbf{X}, t) = \mathbf{X}_Q + \mathbf{u}_Q(\mathbf{X}, t) + \mathbf{X}_{P/Q} + \mathbf{u}_{P/Q}(\mathbf{X}, t). \quad (5.9)$$

The velocity of point P is constructed by time-differentiating (5.9) to obtain

$$\dot{\mathbf{x}}_P = \frac{\partial \mathbf{u}_Q}{\partial \mathbf{X}_Q} \frac{\partial \mathbf{X}_Q}{\partial t} + \frac{\partial \mathbf{u}_Q}{\partial t} + \frac{\partial \mathbf{u}_{P/Q}}{\partial \mathbf{X}_{P/Q}} \frac{\partial \mathbf{X}_Q}{\partial t} + \frac{\partial \mathbf{u}_{P/Q}}{\partial t}. \quad (5.10)$$

Substituting (5.8) gives

$$\dot{\mathbf{x}}_P = \frac{\partial \mathbf{u}_Q}{\partial \mathbf{X}_Q} \frac{\partial \mathbf{X}_Q}{\partial t} + \frac{\partial \mathbf{u}_Q}{\partial t} + \frac{\partial \mathbf{u}_{P/Q}}{\partial \mathbf{X}_{P/Q}} \left(\frac{\partial \mathbf{X}_P}{\partial t} - \frac{\partial \mathbf{X}_Q}{\partial t} \right) + \frac{\partial \mathbf{u}_{P/Q}}{\partial t}. \quad (5.11)$$

Thus, the absolute velocity of point P simply the absolute velocity of point Q plus the relative velocity of point P taken with respect to point Q ,

$$\mathbf{v}_P = \mathbf{v}_Q + \mathbf{v}_{P/Q}, \quad (5.12)$$

Similarly, for the accelerations,

$$\mathbf{a}_P = \mathbf{a}_Q + \mathbf{a}_{P/Q}. \quad (5.13)$$

The canonical example of relative position, velocity, and acceleration is that of a person, P , walking down the aisle of a passenger train Q . The current position of the passenger in the inertial reference $\mathbf{x}_P$ is equal to the superposition of the current position of the train car in the inertial reference frame $\mathbf{x}_Q$ plus the position of the person *relative* to the train car, $\mathbf{x}_{P/Q}$. This relationship, and the corresponding velocity and acceleration relationship, are stated explicitly as

$$\mathbf{x}_P = \mathbf{x}_Q + \mathbf{x}_{P/Q}, \quad (5.14)$$

$$\mathbf{v}_P = \mathbf{v}_Q + \mathbf{v}_{P/Q}, \quad (5.15)$$

$$\mathbf{a}_P = \mathbf{a}_Q + \mathbf{a}_{P/Q}. \quad (5.16)$$

¹This term, $\mathbf{X}_{P/Q}$, is written in a way that is reminiscent of **conditional probabilities**, that is “the distance to point P *given* point Q .”

Rotation of a Rigid Body with Two Fixed Particles

Next, we construct a rigid body by rigidly connecting particle P and particle Q . Thus, we have a body $\mathcal{B}$ consisting of two particles. Unlike particles, which have only three translation degrees of freedom, rigid bodies have six degrees of freedom: three translations and three rotations.

In the previous section, translation of particle P required that either translation of particle Q , or translation of particle P *relative* to particle Q . This last term, translation of particle P relative to particle Q , can be interpreted as a positive or negative **stretch** between P and Q as motion from the reference configuration to the current configuration occurs.

In this current section, we have constructed a rigid body out of and between particles P and Q . Intuition then suggests positive or negative stretch between the two points can not occur, the distance between particles P and Q in body $\mathcal{B}$ is constant for all time.

Remarkably, motion of particle P can still occur not only as a consequence of translations of particle Q , but also from rotations of body $\mathcal{B}$ about an axis that contains Q . In fact, because of superposition of these two effects, even if particle Q never moves, particle P can move by virtue of rotations.

To come: discussions on $\boldsymbol{\omega}$, $\boldsymbol{\Omega}$, $\boldsymbol{\Omega}$, $\boldsymbol{\omega}$, $\widehat{\mathbf{W}} = \mathbf{W}_\times$, $\widehat{\mathbf{w}} = \mathbf{w}_\times$.

Result 5.0.1 A special case of a motion $\boldsymbol{\varphi}$ that is a rigid body motion has the form

$$\boldsymbol{\varphi}(\mathbf{X}, t) = \mathbf{R}(t)\mathbf{X} + \mathbf{b}(t) \iff \varphi_i = R_{iJ}X_J + b_i. \quad (5.17)$$

Result 5.0.2 The material velocity $\mathbf{V}$ of the rigid body motion has the form

$$\mathbf{V}(\mathbf{X}, t) = \dot{\mathbf{R}}(t)\mathbf{X} + \dot{\mathbf{b}}(t) \iff V_I = \delta_{Ii}\dot{R}_{iJ}X_J + \delta_{Ii}\dot{b}_i. \quad (5.18)$$

Result 5.0.3 The material acceleration $\mathbf{A}$ of the rigid body motion has the form

$$\mathbf{A}(\mathbf{X}, t) = \ddot{\mathbf{R}}(t)\mathbf{X} + \ddot{\mathbf{b}}(t) \iff A_I = \delta_{Ii}\ddot{R}_{iJ}X_J + \delta_{Ii}\ddot{b}_i. \quad (5.19)$$

It is more common to see the velocity and acceleration of the rigid body expressed in the current configuration, as opposed to the reference configuration, as in (5.18) and (5.19).

Result 5.0.4 The current velocity $\mathbf{v}$ of the rigid body motion has the form

$$\mathbf{v}(\mathbf{x}, t) = \boldsymbol{\omega}(t) \times [\mathbf{x} - \mathbf{b}(t)] + \dot{\mathbf{b}}(t) \iff v_i = \epsilon_{ijk} \omega_j [x_k - b_k] + \dot{b}_i \quad (5.20)$$

Remark 5.0.6. From time-to-time, and for the sake of brevity, we avoid writing explicit dependence on time, t . Thus, we write vector functions simply as function of $\mathbf{X}$ or $\mathbf{x}$, and where dependence on t is implied where appropriate.

Proof.

Rewrite rigid body motion (5.17) in the current configuration $\mathbf{x}$

$$\mathbf{x} = \mathbf{R} \varphi^{-1}(\mathbf{x}) + \mathbf{b} \quad (5.21)$$

or

$$\varphi^{-1}(\mathbf{x}) = \mathbf{R}^{-1} (\mathbf{x} - \mathbf{b}). \quad (5.22)$$

Since $\mathbf{R}^{-1} = \mathbf{R}^T$,

$$\varphi^{-1}(\mathbf{x}) = \mathbf{R}(t)^T (\mathbf{x} - \mathbf{b}). \quad (5.23)$$

Write the velocity in the current configuration, $\mathbf{v}(\mathbf{x}, t)$, using (5.18) and use explicit time dependence on t for clarity,

$$\mathbf{v}(\mathbf{x}, t) = \mathbf{V}(\varphi^{-1}(\mathbf{x}, t), t) = \dot{\mathbf{R}}(t) \varphi^{-1}(\mathbf{x}, t) + \dot{\mathbf{b}}(t). \quad (5.24)$$

Substitute (5.23),

$$\mathbf{v}(\mathbf{x}) = \underbrace{\dot{\mathbf{R}} \mathbf{R}^T}_{[\boldsymbol{\omega}]_{\times}} (\mathbf{x} - \mathbf{b}) + \dot{\mathbf{b}}, \quad (5.25)$$

where

$$[\boldsymbol{\omega}]_{\times} \mathbf{z} = \boldsymbol{\omega} \times \mathbf{z} = \dot{\mathbf{R}} \mathbf{R}^T \mathbf{z}, \quad (5.26)$$

for all vectors $\mathbf{z}$. With (5.26), (5.25) can be written as

$$\mathbf{v}(\mathbf{x}) = \boldsymbol{\omega} \times (\mathbf{x} - \mathbf{b}) + \dot{\mathbf{b}}, \quad (5.27)$$

which is the result. ■

Result 5.0.5 The current acceleration $\mathbf{a}$ of the rigid body motion has the form

$$\mathbf{a}(\mathbf{x}, t) = \boldsymbol{\alpha}(t) \times [\mathbf{x} - \mathbf{b}(t)] + \boldsymbol{\omega}(t) \times (\boldsymbol{\omega}(t) \times [\mathbf{x} - \mathbf{b}(t)]) + \ddot{\mathbf{b}}(t) \quad (5.28)$$

$$\iff a_i = \epsilon_{ijk} \alpha_j [x_k - b_k] + \epsilon_{ijk} \epsilon_{klm} \omega_j \omega_l [x_m - b_m] + \ddot{b}_i \quad (5.29)$$

Proof.

Start with (5.25) and differentiate with respect to time,

$$\mathbf{a}(\mathbf{x}, t) = \frac{d}{dt} \mathbf{v}(\mathbf{x}, t) = \frac{d}{dt} \left(\dot{\mathbf{R}} \mathbf{R}^T (\mathbf{x} - \mathbf{b}) + \dot{\mathbf{b}} \right). \quad (5.30)$$

Then expand,

$$\mathbf{a}(\mathbf{x}, t) = \ddot{\mathbf{R}} \mathbf{R}^T (\mathbf{x} - \mathbf{b}) + \dot{\mathbf{R}} \dot{\mathbf{R}}^T (\mathbf{x} - \mathbf{b}) + \dot{\mathbf{R}} \mathbf{R}^T (\dot{\mathbf{x}} - \dot{\mathbf{b}}) + \ddot{\mathbf{b}}. \quad (5.31)$$

The first two terms on the right-hand-side of (5.31) can be combined and simplified to

$$\left[\ddot{\mathbf{R}} \mathbf{R}^T + \dot{\mathbf{R}} \dot{\mathbf{R}}^T \right] (\cdot) = \frac{d}{dt} \left[\dot{\mathbf{R}} \mathbf{R}^T \right] (\cdot) = \frac{d}{dt} [\boldsymbol{\omega}]_{\times} (\cdot) = [\boldsymbol{\alpha}]_{\times} (\cdot), \quad (5.32)$$

where $(\cdot)$ stands in for the $(\mathbf{x} - \mathbf{b})$ vector. The third term on the right-hand-side of (5.31), with substitution of (5.25), can be rewritten as

$$\dot{\mathbf{R}} \mathbf{R}^T (\dot{\mathbf{x}} - \dot{\mathbf{b}}) = \dot{\mathbf{R}} \mathbf{R}^T (\dot{\mathbf{R}} \mathbf{R}^T (\mathbf{x} - \mathbf{b})) = \boldsymbol{\omega} \times (\boldsymbol{\omega} \times (\cdot)). \quad (5.33)$$

Combining these successive terms gives

$$\mathbf{a}(\mathbf{x}) = \boldsymbol{\alpha} \times (\mathbf{x} - \mathbf{b}) + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times (\mathbf{x} - \mathbf{b})) + \ddot{\mathbf{b}}, \quad (5.34)$$

which is the result. ■

Result 5.0.6 Rodrigues Rotation Formula. The rotation of the position vector $\mathbf{r}^{QP}$ into the position vector $\mathbf{r}^{Qp}$ by rotation $\theta \hat{\mathbf{n}}$ is given by

$$\mathbf{r}^{Qp} = (1 - \cos \theta) (\mathbf{r}^{QP} \cdot \hat{\mathbf{n}}) \hat{\mathbf{n}} + \cos \theta \mathbf{r}^{QP} + \sin \theta (\hat{\mathbf{n}} \times \mathbf{r}^{QP}) \quad (5.35)$$

Proof.

Let the shorthand notation $\mathbf{a} = \mathbf{r}^{QP}$ and $\mathbf{b} = \mathbf{r}^{Qp}$ be used such that the rotation $\theta \hat{\mathbf{n}}$ rotates vector $\mathbf{a}$ into vector $\mathbf{b}$. Furthermore, let the vectors $\mathbf{a}$ and $\mathbf{b}$ be decomposed into the sum of two subspace vectors, which are parallel and perpendicular to the $\hat{\mathbf{n}}$ vector, such that

$$\mathbf{a} = \mathbf{a}_{\parallel} + \mathbf{a}_{\perp} \iff \mathbf{a}_{\perp} = \mathbf{a} - \mathbf{a}_{\parallel} \text{ and } \mathbf{a}_{\parallel} = (\mathbf{a} \cdot \hat{\mathbf{n}}) \hat{\mathbf{n}}, \quad (5.36)$$

$$\mathbf{b} = \mathbf{b}_{\parallel} + \mathbf{b}_{\perp} \iff \mathbf{b}_{\perp} = \mathbf{b} - \mathbf{b}_{\parallel} \text{ and } \mathbf{b}_{\parallel} = (\mathbf{b} \cdot \hat{\mathbf{n}}) \hat{\mathbf{n}}. \quad (5.37)$$

From inspection of the rotation plane perpendicular to $\hat{\mathbf{n}}$, as shown in Fig. 5.3,

$$\mathbf{b}_{\perp} = \cos \theta \mathbf{a}_{\perp} + \sin \theta \hat{\mathbf{n}} \times \mathbf{a}_{\perp}. \quad (5.38)$$

Rewriting the last term gives,

$$\mathbf{b}_{\perp} = \cos \theta \mathbf{a}_{\perp} + \sin \theta \hat{\mathbf{n}} \times (\mathbf{a} - \mathbf{a}_{\parallel}) = \cos \theta \mathbf{a}_{\perp} + \sin \theta (\hat{\mathbf{n}} \times \mathbf{a} - \underbrace{\hat{\mathbf{n}} \times \mathbf{a}_{\parallel}}_{\mathbf{0}}). \quad (5.39)$$

where the $\hat{\mathbf{n}} \times \mathbf{a}_{\parallel} = \mathbf{0}$ since these two vectors are parallel. Then,

$$\mathbf{b} = \mathbf{b}_{\parallel} + \mathbf{b}_{\perp}, \quad (5.40)$$

$$= \mathbf{a}_{\parallel} + \mathbf{b}_{\perp} \text{ since } \mathbf{a}_{\parallel} = \mathbf{b}_{\parallel}, \quad (5.41)$$

$$= (\mathbf{a} \cdot \hat{\mathbf{n}}) \hat{\mathbf{n}} + \cos \theta \mathbf{a}_{\perp} + \sin \theta \hat{\mathbf{n}} \times \mathbf{a}, \quad (5.42)$$

$$= (\mathbf{a} \cdot \hat{\mathbf{n}}) \hat{\mathbf{n}} + \cos \theta (\mathbf{a} - \mathbf{a}_{\parallel}) + \sin \theta \hat{\mathbf{n}} \times \mathbf{a}, \quad (5.43)$$

$$= (\mathbf{a} \cdot \hat{\mathbf{n}}) \hat{\mathbf{n}} + \cos \theta \mathbf{a} - \cos \theta \mathbf{a}_{\parallel} + \sin \theta \hat{\mathbf{n}} \times \mathbf{a}, \quad (5.44)$$

$$= (\mathbf{a} \cdot \hat{\mathbf{n}}) \hat{\mathbf{n}} + \cos \theta \mathbf{a} - \cos \theta (\mathbf{a} \cdot \hat{\mathbf{n}}) \hat{\mathbf{n}} + \sin \theta \hat{\mathbf{n}} \times \mathbf{a}, \quad (5.45)$$

$$= (1 - \cos \theta)(\mathbf{a} \cdot \hat{\mathbf{n}}) \hat{\mathbf{n}} + \cos \theta \mathbf{a} + \sin \theta \hat{\mathbf{n}} \times \mathbf{a}, \quad (5.46)$$

which is the result. ■

Particle Moving on a Rigid Body

Rotation and Stretch of a Deformable Body

Result 5.0.7 The angular velocity $\boldsymbol{\omega}$ of a body $\tilde{B}$ that undergoes finite rotations and infinitesimal deformations may be approximated as a least squares projection of the body's configuration column space onto the body's velocity space as

$$\boldsymbol{\omega} = - \left(\mathbf{A}^\top \mathbf{A} \right)^{-1} \mathbf{A}^\top \cdot \{ \mathbf{v}_P - \mathbf{v}_Q \}, \text{ where} \quad (5.47)$$

$$\mathbf{A} = [\mathbf{x}_P - \mathbf{x}_Q]_\times. \quad (5.48)$$

Proof.

Let body $\tilde{B}$ undergo finite rotations and infinitesimal deformations in reference frame F . Using the velocity of two points Q and P , with a particle moving on rigid body B ,

$$\mathbf{v}_{P/F} = \mathbf{v}_{Q/F} + \mathbf{v}_{P/Q} + \boldsymbol{\omega}_{B/F} \times \mathbf{r}_{QP} \quad (5.49)$$

$$\mathbf{v}_P = \mathbf{v}_Q + \mathbf{v}_{P/Q} + \boldsymbol{\omega}_B \times \mathbf{r}_{QP} \quad \text{dropping } F \text{ where implied.} \quad (5.50)$$

Note the term $\mathbf{v}_{P/Q}$ is the *relative* velocity between points P and Q . Assuming infinitesimal deformations, it is reasonable to assume the time derivatives of these quantities are also small, thus $\mathbf{v}_{P/Q} \approx \mathbf{0}$. Thus for body $\tilde{B}$

$$\mathbf{v}_P - \mathbf{v}_Q \approx \boldsymbol{\omega}_B \times \mathbf{r}_{QP}, \quad (5.51)$$

$$= -\mathbf{r}_{QP} \times \boldsymbol{\omega} \quad (5.52)$$

$$= -(\boldsymbol{\varphi}(\mathbf{X}_P, t) - \boldsymbol{\varphi}(\mathbf{X}_Q, t)) \times \boldsymbol{\omega} \quad (5.53)$$

$$= -(\mathbf{X}_P + \mathbf{u}(\mathbf{X}_P, t) - (\mathbf{X}_Q + \mathbf{u}(\mathbf{X}_Q, t))) \times \boldsymbol{\omega} \quad (5.54)$$

if displacements $\mathbf{u}$ are used; or,

$$= -(\mathbf{x}_P - \mathbf{x}_Q) \times \boldsymbol{\omega} \quad \text{if current placements } \mathbf{x} \text{ are used,} \quad (5.55)$$

$$= -[\mathbf{x}_P - \mathbf{x}_Q]_\times \boldsymbol{\omega}, \quad (5.56)$$

where the skew-symmetric matrix, defined in Eq. (2.24), is used. Explicitly, this is

$$[\mathbf{x}_P - \mathbf{x}_Q]_\times = \begin{bmatrix} 0 & -(x_{P3} - x_{Q3}) & (x_{P2} - x_{Q2}) \\ (x_{P3} - x_{Q3}) & 0 & -(x_{P1} - x_{Q1}) \\ -(x_{P2} - x_{Q2}) & (x_{P1} - x_{Q1}) & 0 \end{bmatrix}, \quad (5.57)$$

which is solvable through a least-squares approach. ■

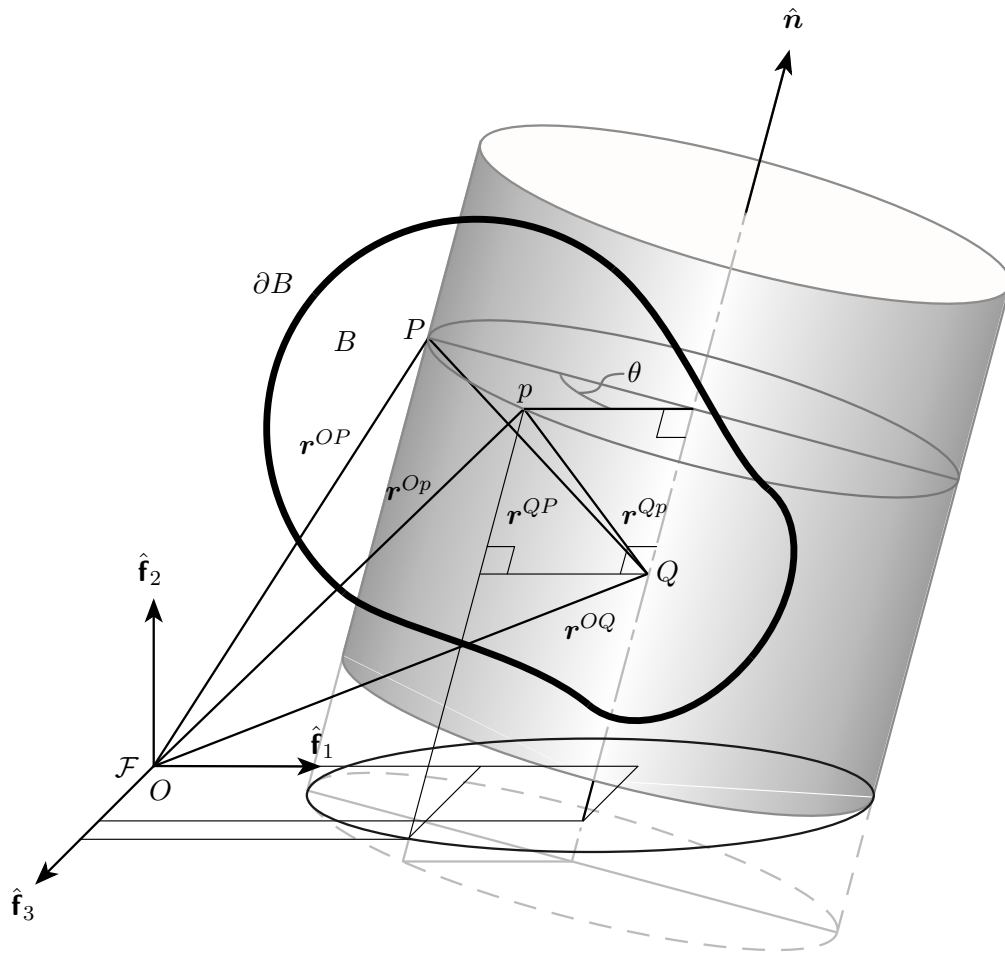


Figure 5.3: In reference frame $\mathcal{F}$, body B rotating about axis $\hat{\mathbf{n}}$ an amount θ , which moves point P in body B to point p .

Example 2.

To come: nonlinear rigid pendulum. The governing nonlinear ordinary differential equation of motion,

$$\ddot{\theta} + \frac{3g}{2L} \sin \theta = 0 \tag{5.58}$$

...

5.0.6 Isochoric (Poisson Special Case) Stretch

Next consider a slightly different form of $\mathbf{F}$, such that

$$\mathbf{F} = \begin{bmatrix} \lambda & 0 \\ 0 & \frac{1}{\lambda} \end{bmatrix}. \quad (5.59)$$

We keep the translation as $\mathbf{u} = \mathbf{0}$.

For concreteness, consider $\lambda_1 = 2$ (expansion) and $\lambda_2 = \frac{1}{2}$ (contraction), as shown in Fig. 5.4.

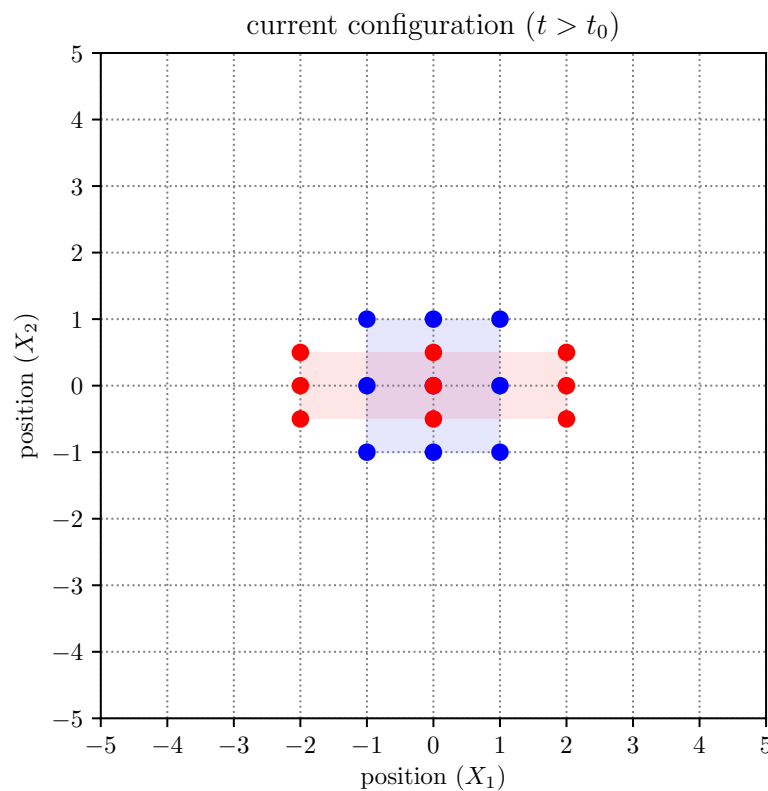


Figure 5.4: Illustration of an isochoric (deviatoric) motion.

The area in the reference configuration is $2 \times 2 = 4$, whereas the area in the current configuration is $4 \times 1 = 4$. For the sake of illustration, we have chosen stretches along the two axes such that the area does not change although the dimensions and thus shape have changed. This is an example of an **isochoric** motion. The roots *iso-* (*equal*) and *choro-* (*space*) describe a process from initial state 0 to final state 1 where the volume remains constant.

In three dimensions, this case becomes

$$\mathbf{F} = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{\lambda}} & 0 \\ 0 & 0 & \frac{1}{\sqrt{\lambda}} \end{bmatrix}. \quad (5.60)$$

Now, the motion as we have described it, doesn't yet show the *path* taken between the reference and current configuration. To get that detail, we would need to introduce the time parameter t , which we have suppressed for now. However, if we were to say the time interval is simply two points t_0 and t_1 , we can create the path between these two states 0 and 1, and connect with the isochoric process PV diagram in thermodynamics, shown in Fig. 5.5.

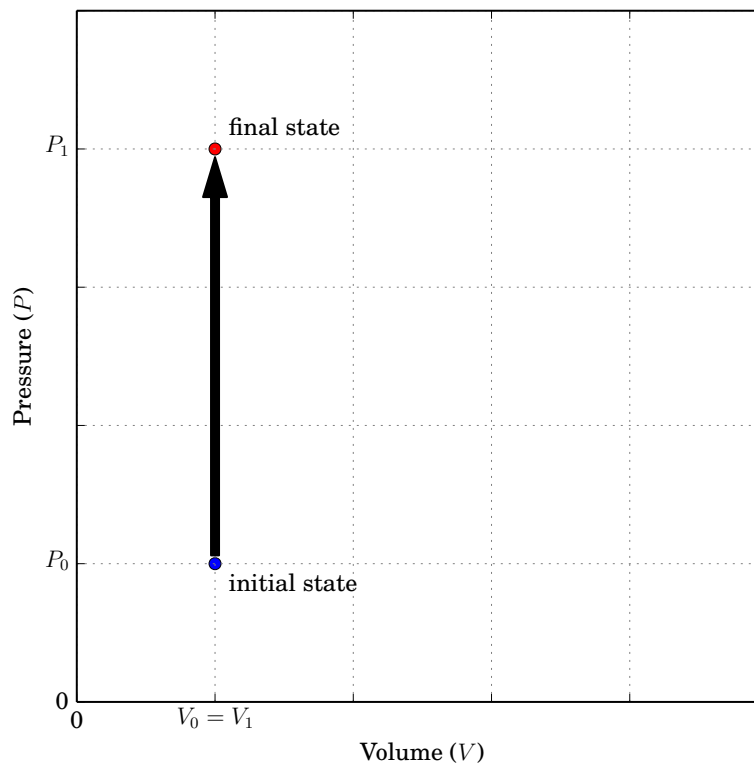


Figure 5.5: Isochoric process from initial state (reference configuration) 0 to final state (current configuration) 1.

We have yet to introduce and define pressure P , so this connection may be tenuous

at the moment. Nonetheless, we introduce this connection now to motivate the single idea that isochoric motions will play an important role in our later work.

5.0.7 Dilatational (Volumetric) Stretch

Consider now $\mathbf{F} = \lambda \mathbf{1}$ and $\mathbf{u} = \mathbf{0}$. This describes **volumetric** response, also known as a **dilatational** response. The parameter $\lambda \in \mathbb{R}^+$ causes the current configuration to change volume as

- $\lambda > 1$ expansion,
- $\lambda = 1$ identity,
- $0 < \lambda < 1$ contraction.

Notice that λ describes the “slope” of the mapping from reference to current configurations. In the case that $\lambda \rightarrow +\infty$, the volume grows infinite. In the case that $\lambda \rightarrow 0$, the volume shrinks to zero. For concreteness, consider $\lambda = 2$. The volumetric expansion is shown in Fig. 5.6. The reference configuration, shown in blue, has area of 4. The current configuration, shown in red, has area of 16.

Figure 5.6: Illustration of a dilatational (volumetric) motion.

5.1 Simple Shear

Figure 5.7 illustrates **simple shear**, with relative motion of planes normal to the vertical axis. For all configurations (reference and current):

- Horizontal fibers remain horizontal.
- The length of horizontal fibers remains constant.
- The vertical distance between the horizontal fibers remains constant.
- The body's volume is constant. The deformation is isochoric since $\det(\mathbf{F}) = 1$, see (5.63).

The relative motion is characterized by the non-dimensional ratio of length a per unit height, where $a \in \mathbb{R} \subset [0, \infty)$. The angle, $\gamma \in \mathbb{R} \subset [0, \pi/2)$, subtended by simple shear is $\gamma = \arctan(a)$. In the limit as $a \mapsto \infty$, the shear angle $\gamma \mapsto \pi/2$. For small values of a , the small-angle assumption is valid, with $\tan(\gamma) \approx \gamma \approx a$.

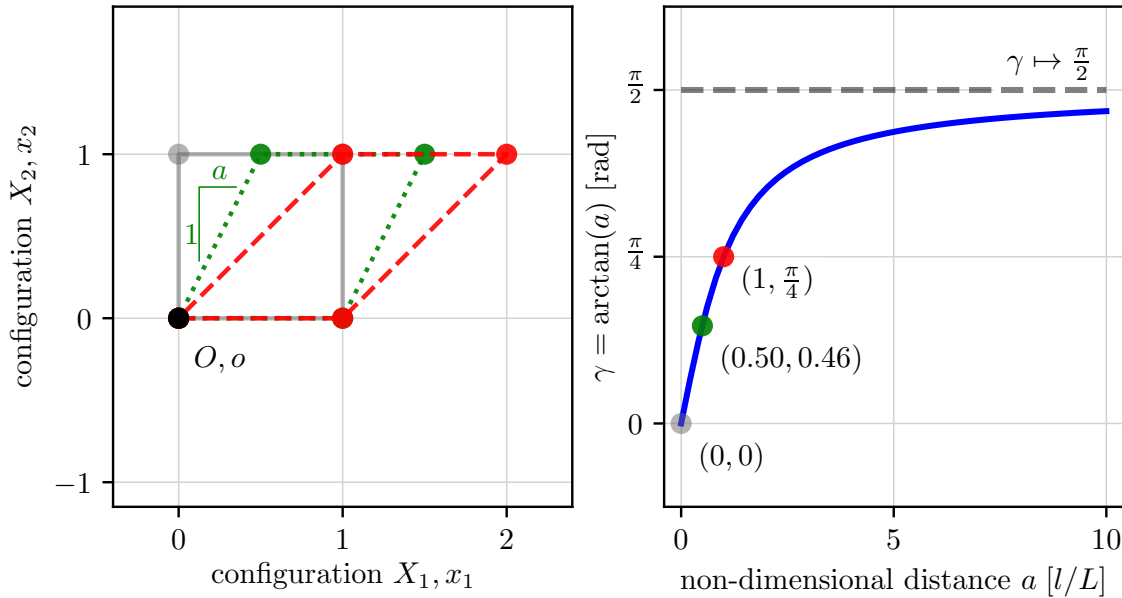


Figure 5.7: (Left) **Simple shear** of a unit cube in the reference configuration (gray) and two current configurations (dotted, green and dashed, red) and (right) with the shear angle, γ , created for all configurations that are parameterized by $a \in [0, 10]$. Source: `simple_shear.py`.

In Fig. 5.7 (left), the horizontal displacement per unit height, a , is illustrated by the dotted, green configuration. The (right) figure also shows the shear angle $\gamma = \arctan(a)$,

with the three (a, γ) pairs plotted in a color that matches the corresponding deformation in the left figure.

5.1.1 Rate-Independent

The simple shear deformation $\boldsymbol{\varphi}$ in rate-independent form is

$$\boldsymbol{\varphi}(\mathbf{X}) = (X_1 + aX_2) \mathbf{e}_1 + X_2 \mathbf{e}_2 + X_3 \mathbf{e}_3. \quad (5.61)$$

The deformation gradient $\mathbf{F}$ is

$$\mathbf{F} = \mathbf{e}_i \frac{\partial \varphi_i}{\partial X_J} \mathbf{E}_J = \mathbf{e}_1 \otimes \mathbf{E}_1 + a \mathbf{e}_1 \otimes \mathbf{E}_2 + \mathbf{e}_2 \otimes \mathbf{E}_2 + \mathbf{e}_3 \otimes \mathbf{E}_3 \quad (5.62)$$

$$= \{\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3\} \begin{bmatrix} 1 & a & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \left\{ \begin{array}{c} \mathbf{E}_1 \\ \mathbf{E}_2 \\ \mathbf{E}_3 \end{array} \right\} \iff \mathbf{F}_{iJ} = \begin{bmatrix} 1 & a & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (5.63)$$

Note that the volume remains constant for all deformations since $\det(\mathbf{F}) = 1$. The deformation thus belongs to the group of isochoric motions (see Fig. 5.1). The right Cauchy-Green strain, $\mathbf{C} = \mathbf{F}^T \mathbf{F}$, in simple shear, is

$$\begin{aligned} \mathbf{C} = & \mathbf{E}_1 \otimes \mathbf{E}_1 + a (\mathbf{E}_1 \otimes \mathbf{E}_2 + \mathbf{E}_2 \otimes \mathbf{E}_1) + \\ & (a^2 + 1) \mathbf{E}_2 \otimes \mathbf{E}_2 + \mathbf{E}_3 \otimes \mathbf{E}_3 \iff \mathbf{C}_{IJ} = \begin{bmatrix} 1 & a & 0 \\ a & a^2 + 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \end{aligned} \quad (5.64)$$

The principal directions of $\mathbf{C}$ are given with the three eigenvectors $\{\mathbf{N}_1, \mathbf{N}_2, \mathbf{N}_3\}$ and their respective eigenvalues $\lambda_1^2, \lambda_2^2, \lambda_3^2$,

$$\mathbf{C} = \sum_{\alpha=1}^3 \lambda_{\alpha}^2 \mathbf{N}_{\alpha} \otimes \mathbf{N}_{\alpha}. \quad (5.65)$$

For simple shear, the eigenvalues of $\mathbf{C}$ and their corresponding eigenvectors, satisfying $[\mathbf{C}]\{\mathbf{N}\} = \lambda^2 \{\mathbf{N}\}$, are

$$\begin{aligned} \lambda_1^2 &= \frac{1}{2} \left(a^2 + a \sqrt{a^2 + 4} + 2 \right), \quad \text{and} \quad \mathbf{N}_1 = \left\{ \frac{1}{2} \left(-a + \sqrt{a^2 + 4} \right), \ 1, \ 0 \right\}^T; \\ \lambda_2^2 &= 1, \quad \text{and} \quad \mathbf{N}_2 = \{0, \ 0, \ 1\}^T; \\ \lambda_3^2 &= \frac{1}{2} \left(a^2 - a \sqrt{a^2 + 4} + 2 \right), \quad \text{and} \quad \mathbf{N}_3 = \left\{ \frac{1}{2} \left(-a - \sqrt{a^2 + 4} \right), \ 1, \ 0 \right\}^T. \end{aligned} \quad (5.66)$$

The eigenvectors

The right stretch, $\mathbf{U} = \sqrt{\mathbf{C}}$, is

$$\mathbf{U}_{IJ} = \begin{bmatrix} \text{tk0} & \text{tk0} & \text{tk0} \\ \text{tk0} & \text{tk0} & \text{tk0} \\ \text{tk0} & \text{tk0} & \text{tk0} \end{bmatrix}. \quad (5.67)$$

The Green-Lagrange strain, $\mathbf{E} = \frac{1}{2}(\mathbf{C} - \mathbf{1})$, in simple shear is

$$\mathbf{E}_{IJ} = \frac{1}{2} \begin{bmatrix} 0 & a & 0 \\ a & a^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (5.68)$$

The left Cauchy-Green strain, $\mathbf{b} = \mathbf{F}\mathbf{F}^\top$, (and its inverse), in simple shear, are

$$\mathbf{b}_{ij} = \begin{bmatrix} a^2 + 1 & a & 0 \\ a & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \text{and} \quad \mathbf{b}_{ji}^{-1} = \begin{bmatrix} 1 & -a & 0 \\ -a & a^2 + 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (5.69)$$

The Euler-Almansi strain, $\mathbf{e} = \frac{1}{2}(\mathbf{1} - \mathbf{b}^{-1})$, for simple shear is

$$\mathbf{e}_{ij} = \frac{1}{2} \begin{bmatrix} 0 & a & 0 \\ a & -a^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (5.70)$$

5.1.2 Rate-Dependent

The simple shear deformation $\boldsymbol{\varphi}$ in rate-dependent form is

$$\boldsymbol{\varphi}(\mathbf{X}, t) = (X_1 + a(t)X_2) \mathbf{e}_1 + X_2 \mathbf{e}_2 + X_3 \mathbf{e}_3. \quad (5.71)$$

The deformation gradient $\mathbf{F}$ is unchanged from (5.63). Additionally, the rate and inverse, respectively, are found to be

$$\dot{\mathbf{F}}_{iJ} = \begin{bmatrix} 0 & \dot{a} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{F}_{Ji}^{-1} = \begin{bmatrix} 1 & -a & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (5.72)$$

From these two foregoing quantities, the spatial velocity gradient, $\boldsymbol{\nabla} \mathbf{v} = \dot{\mathbf{F}}\mathbf{F}^{-1}$, is

$$v_{i,j} = \begin{bmatrix} 0 & \dot{a} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (5.73)$$

From this, the rate of deformation, $\mathbf{d} = \frac{1}{2}[\boldsymbol{\nabla} \mathbf{v} + (\boldsymbol{\nabla} \mathbf{v})^\top]$, and spin, $\mathbf{w} = \frac{1}{2}[\boldsymbol{\nabla} \mathbf{v} - (\boldsymbol{\nabla} \mathbf{v})^\top]$, respectively, are

$$\mathbf{d}_{ij} = \frac{1}{2} \begin{bmatrix} 0 & \dot{a} & 0 \\ \dot{a} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{w}_{ij} = \frac{1}{2} \begin{bmatrix} 0 & \dot{a} & 0 \\ -\dot{a} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (5.74)$$

The eigenvalues of $\mathbf{d}$ and their corresponding eigenvectors, satisfying $[\mathbf{d}]\{\mathbf{n}\} = \lambda\{\mathbf{n}\}$, are

$$\begin{aligned}\lambda_1 &= \dot{a}/2 & \text{and} & \quad \mathbf{n}_1 = \begin{Bmatrix} 1 & 1 & 0 \end{Bmatrix}^\top = \mathbf{e}_1 + \mathbf{e}_2 \\ \lambda_2 &= 0, & \text{and} & \quad \mathbf{n}_2 = \begin{Bmatrix} 0 & 0 & 1 \end{Bmatrix}^\top = \mathbf{e}_3 \\ \lambda_3 &= -\dot{a}/2, & \text{and} & \quad \mathbf{n}_3 = \begin{Bmatrix} -1 & 1 & 0 \end{Bmatrix}^\top = -\mathbf{e}_1 + \mathbf{e}_2\end{aligned}\tag{5.75}$$

The pullback operator, $\varphi_*^{-1}[\cdot]$, applied to the rate of deformation tensor, $\mathbf{d}$, is the time derivative of the Green-Lagrange strain tensor,

The pullback operator, $\varphi_*^{-1}[\cdot]$, applied to the rate of deformation, $\mathbf{d}$, is the time derivative of the Green-Lagrange strain, $\dot{\mathbf{E}} = \mathbf{F}^\top \mathbf{d} \mathbf{F}$. This result can be checked by taking time derivatives directly of $\mathbf{C}$ and $\mathbf{E}$, respectively found to be

$$\dot{\mathbf{C}}_{IJ} = \begin{bmatrix} 0 & \dot{a} & 0 \\ \dot{a} & 2a\dot{a} & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad \dot{\mathbf{E}}_{IJ} = \frac{1}{2} \begin{bmatrix} 0 & \dot{a} & 0 \\ \dot{a} & 2a\dot{a} & 0 \\ 0 & 0 & 0 \end{bmatrix}.\tag{5.76}$$

The rotated rate of deformation, $\mathbf{D} = \mathbf{R}^\top \mathbf{d} \mathbf{R}$, is

$$\mathbf{D}_{IJ} = \begin{bmatrix} \text{tk0} & \text{tk0} & \text{tk0} \\ \text{tk0} & \text{tk0} & \text{tk0} \\ \text{tk0} & \text{tk0} & \text{tk0} \end{bmatrix},\tag{5.77}$$

and is related to the Green-Lagrange strain rate through $\dot{\mathbf{E}} = \mathbf{U} \mathbf{D} \mathbf{U}$ since

$$\dot{\mathbf{E}} = \mathbf{F}^\top \mathbf{d} \mathbf{F} = (\mathbf{R} \mathbf{U})^\top \mathbf{d} \mathbf{R} \mathbf{U} = \mathbf{U}^\top \mathbf{R}^\top \mathbf{d} \mathbf{R} \mathbf{U} = \mathbf{U}^\top \mathbf{D} \mathbf{U} = \mathbf{U} \mathbf{D} \mathbf{U}.\tag{5.78}$$

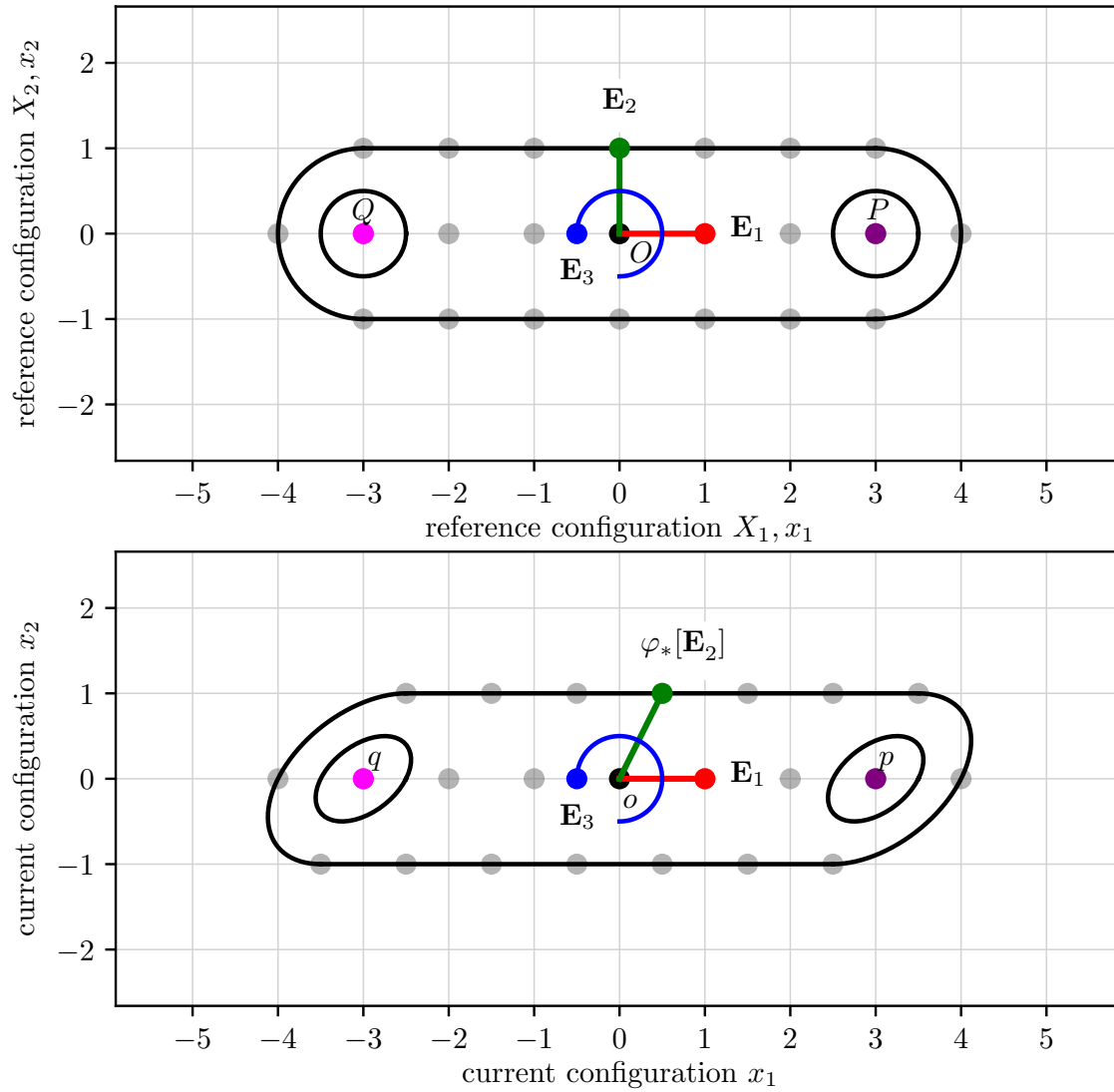


Figure 5.8: Simple shear of body from the reference configuration B to the current configuration $\varphi(B)$. Source: `simple_shear_on_body.py`.

Chapter 6

Quaternions

6.1 Multiplication

The multiplication for quaternion bases i, j, k is defined as

$$i j \triangleq k, \quad j k \triangleq i, \quad k i \triangleq j, \quad \text{and} \quad k j \triangleq -i, \quad j i \triangleq -k, \quad i k \triangleq -j. \quad (6.1)$$

- The positive combinations of the i, j, k triple can be constructed by repeating the i and j at then end of i, j, k sequence. Thus, the forward sequence is i, j, k, i, j .
- When two adjacent factors are read in forward order,¹ the result is the next sequential factor, multiplied by $+1$.
- The negative combinations of the i, j, k triple can be constructed by repeating the j and i at then end of k, j, i sequence. Thus, the reverse sequence is k, j, i, k, j .
- When two adjacent factors are read in backward order,² the result is the next sequential factor, multiplied by -1 .

Organizing the first factor into rows and the second factor into columns creates a superposition of a skew matrix and a negative identity matrix $-\mathbf{1}$, shown in Table 6.1.

The quaternion matrix $\mathbf{Q}$ is defined as

$$\mathbf{Q} \triangleq -\mathbf{1} - [\mathbf{q}]_{\times} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} - \begin{bmatrix} 0 & -k & j \\ k & 0 & -i \\ -j & i & 0 \end{bmatrix}. \quad (6.2)$$

¹Likewise, this is the *reverse* alphabetical order of a repeating triple, $i, j, k, i, j, k, \dots$

²Essentially, this is alphabetical order of a repeating triple, $k, j, i, k, j, i, \dots$

Table 6.1: Quaternion multiplication table.

	i	j	k
i	-1	k	$-j$
j	$-k$	-1	i
k	j	$-i$	-1

Chapter 7

Strain

7.1 Green-Lagrange Strain

The Green-Lagrange strain tensor,

$$\mathbf{E} \triangleq \frac{1}{2}(\mathbf{C} - \mathbf{1}) \iff E_{IJ} \triangleq \frac{1}{2}(C_{IJ} - \delta_{IJ}), \quad (7.1)$$

is closely related to the right Cauchy-Green strain tensor and is often used in defining constitutive law relationships because the measure, when linearized about the reference configuration, coincides with the small strain tensor of linear deformation elasticity, denoted $\boldsymbol{\epsilon}$ and defined in (7.10). This relationship can be seen as follows:

$$2\mathbf{E} \stackrel{(4.13)}{=} \mathbf{1} + \nabla_0 \mathbf{u} + (\nabla_0 \mathbf{u})^\top + (\nabla_0 \mathbf{u})^\top \nabla_0 \mathbf{u} - \mathbf{1}, \quad (7.2)$$

$$2\mathbf{E}_{\text{LIN}} = \nabla_0 \mathbf{u} + (\nabla_0 \mathbf{u})^\top, \quad (7.3)$$

where the higher-order (quadratic) term in (7.2) is set to zero to achieve (7.3).

7.2 Euler-Almansi Strain

The Euler-Almansi strain tensor,

$$\mathbf{e} \triangleq \frac{1}{2}(\mathbf{1} - \mathbf{b}^{-1}) \iff e_{ij} \triangleq \frac{1}{2}(\delta_{ij} - (b_{ij})^{-1}), \quad (7.4)$$

can likewise be used to approximate the small strain tensor ϵ by combining (4.15) and (4.5) as follows:

$$\begin{aligned}
 2\mathbf{e} &= \mathbf{1} - \left[\nabla_0 \boldsymbol{\varphi} (\nabla_0 \boldsymbol{\varphi})^\top \right]^{-1} \\
 &= \delta_{ij} - \left[\frac{\partial \varphi_i}{\partial X_J} \frac{\partial \varphi_j}{\partial X_J} \right]^{-1} = \delta_{ij} - \left[\frac{\partial X_J}{\partial \varphi_i} \frac{\partial X_J}{\partial \varphi_j} \right] \\
 &\stackrel{(4.2)}{=} \mathbf{1} - \left[(\mathbf{1} - \nabla \mathbf{u})(\mathbf{1} - \nabla \mathbf{u})^\top \right] \\
 &= \nabla \mathbf{u} + \nabla \mathbf{u}^\top - \nabla \mathbf{u} \nabla \mathbf{u}^\top
 \end{aligned} \tag{7.5}$$

$$2\mathbf{e}_{\text{LIN}} = \nabla \mathbf{u} + (\nabla \mathbf{u})^\top, \tag{7.6}$$

where higher-order (quadratic) term in (7.5) is set to zero to achieve (7.6).

7.3 Small Strain

When displacement gradients are small in the reference configuration

$$\nabla_0 \mathbf{u} \ll 1, \tag{7.7}$$

or in the current configuration,

$$\nabla \mathbf{u} \ll 1, \tag{7.8}$$

respectively, the nonlinear gradient terms are negligible and the finite strain theory simplifies to small strain theory, which occurs when finite strain measures are linearized to obtain $\mathbf{E}_{\text{LIN}}$ and $\mathbf{e}_{\text{LIN}}$ in the previous sections.

Note that we have restricted the gradients of displacement, and not the displacement $\mathbf{u}$ itself. Thus, displacements between the reference and current configurations can be large (finite), but the gradients of the displacement either in the reference or current configuration are small.

Section 7.6 will demonstrate that the small strain tensors are not suitable to describe motion that contains finite rotation. This makes sense because, in finite rotation, gradients of displacement are large, not small. To adequately describe motion that includes finite rotation, the fully nonlinear strain measure, such as the Seth-Hill strain family (see Section 7.5), must be used.

7.4 Infinitesimal Strain

If we further restrict the small strain theory such that the displacement is small compared to unity,

$$\mathbf{u} \ll 1, \tag{7.9}$$

the infinitesimal strain theory is obtained, which has no distinction between Lagrangian and Eulerian strain tensors.

In this case, the two small strain tensors, $\mathbf{E}_{\text{LIN}}$ and $\mathbf{e}_{\text{LIN}}$ converge to a single definition of strain, called the infinitesimal strain tensor $\boldsymbol{\epsilon}$, defined as

$$\boldsymbol{\epsilon} \triangleq \frac{1}{2}(\nabla \mathbf{u} + (\nabla \mathbf{u})^T). \quad (7.10)$$

Note that the $\nabla_0(\cdot)$ notation has been dropped since the distinction between the reference and current configurations is nonexistent. Also, note that the factor of $\frac{1}{2}$ appears because it then follows that the infinitesimal strain is simply the symmetric part of the displacement gradient,

$$\boldsymbol{\epsilon} = \text{SYM}(\nabla \mathbf{u}). \quad (7.11)$$

Finally, note that the finite Lagrangian and Eulerian strain tensors were defined with the factor of $\frac{1}{2}$ so that their expressions, once linearized and subject to a small displacement assumption, simplify to exactly the infinitesimal strain tensor $\boldsymbol{\epsilon}$.

7.5 Seth-Hill Strain Family

We return to finite strain definitions, and show that the Lagrangian and Eulerian strain tensors described above are two special cases of a family of finite strain tensors.

Seth and Hill showed that the Green-Lagrange and Euler-Almansi strain tensors are special cases of the so-called Seth-Hill family of strain measures, defined as

$$\mathbf{E}_{(m)} \triangleq \begin{cases} \frac{1}{m} (\mathbf{U}^m - \mathbf{1}) & \text{for } m \neq 0, \\ \ln \mathbf{U} & \text{for } m = 0. \end{cases} \quad (7.12)$$

The principal stretches $\lambda_I = \ell_k/L_0$, $0 < \lambda_I < \infty$, allow the strain measure $\mathbf{E}_{(m)}$ to be written as principal strains, as a function of principal stretch, $f(\lambda_I)$:

$$\begin{aligned} \mathbf{E}_{(m)} &= \sum_{I=1}^3 f(\lambda_I) \hat{\mathbf{E}}_I \otimes \hat{\mathbf{E}}_I \iff \\ \mathbf{E}_{(m)IJ} &= f(\lambda_1) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + f(\lambda_2) \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + f(\lambda_3) \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \end{aligned} \quad (7.13)$$

where the **stretch function**

$$f(\lambda_I) \triangleq \begin{cases} \frac{1}{m} (\lambda_I^m - 1) & \text{for } m \neq 0, \\ \ln(\lambda_I) & \text{for } m = 0. \end{cases} \quad (7.14)$$

For integer values of $m \in [-2, 2]$, five common strain measures result, listed in Table 7.1, in their three-dimensional and one-dimensional forms.

Table 7.1: Strains obtained from the Seth-Hill family.

m	Name	3D	1D
2	Green-Lagrange	$\mathbf{E}_{(2)} = \frac{1}{2} (\mathbf{U}^2 - \mathbf{1})$	$\mathbf{E}_{\text{LAG}} = \frac{1}{2} \left[\left(\frac{\ell}{L_0} \right)^2 - 1 \right]$
1	engineering (Biot, nominal)	$\mathbf{E}_{(1)} = \mathbf{U} - \mathbf{1}$	$\mathbf{E}_{\text{ENG}} = \frac{\ell - L_0}{L_0}$
0	log (Hencky, natural)	$\mathbf{E}_{(0)} = \ln \mathbf{U}$	$\mathbf{E}_{\text{LOG}} = \ln \left(\frac{\ell}{L_0} \right)$
-1	true	$\mathbf{E}_{(-1)} = \mathbf{1} - \mathbf{U}^{-1}$	$\mathbf{E}_{\text{TRUE}} = \frac{\ell - L_0}{\ell}$
-2	Euler-Almansi	$\mathbf{E}_{(-2)} = \frac{1}{2} (\mathbf{1} - \mathbf{U}^{-2})$	$\mathbf{E}_{\text{EUL}} = \frac{1}{2} \left[1 - \left(\frac{L_0}{\ell} \right)^2 \right]$

As will be seen in Section 10.1,

- The Green-Lagrange strain tensor will be the conjugate of the second Piola-Kirchhoff stress tensor,
- The deformation gradient will be the conjugate of the first Piola-Kirchhoff stress tensor,

- The engineering (Biot, nominal) strain tensor will be the conjugate of the Jaumann (Biot) stress tensor,
- The Euler-Almansi strain tensor will be the conjugate of the weighted convected stress tensor.

Figure 7.5 illustrates the one-dimensional strains as a function of stretch ratio $\lambda = \ell/L_0$. The figure illustrates several results:

- For small stretches, $\ell \approx L_0$,
 - The stretch ratio is near unity, $\lambda \approx 1$,
 - The strain values are small, $f(\lambda) \approx 0$,
 - The tangent of the strains with respect to the stretch ratio is near unity, $df/d\lambda \approx 1$.
- For elongations, $\lambda > 1$,
 - The strain monotonically increases since $df/d\lambda > 0$ when $\lambda > 0$.
- For extreme compressions, $\lambda \approx 0$,
 - The Green-Lagrange strain goes to a value of $-\frac{1}{2}$,
 - The engineering (Biot, nominal) strain tensor goes to a value of -1 ,
 - The log, true, and Eulerian strains tend to $-\infty$.
- The engineering (Biot, nominal) strain is a linear function of stretch λ ; all other measures are nonlinear functions of stretch λ .

7.5.1 Reasonable Requirements on the Stretch Function

This section is based on Neff (2013),¹ which presented “reasonable requirements” on $f(\lambda) : \mathbb{R}^+ \mapsto \mathbb{R}$.

f is smooth	true for all in Table 7.1
f is monotonically increasing	true for all
$f _{\lambda=1} = 0$	true for all
$f' _{\lambda=1} = 1$	true for all
As $\lambda \rightarrow \infty$, $f \rightarrow +\infty$	false for true and Euler-Almansi strains
As $\lambda \rightarrow 0^+$, $f \rightarrow -\infty$	false for Green-Lagrange and engineering strains
$-f(\lambda) = f\left(\frac{1}{\lambda}\right)$	true for Hencky and Bažant strains
$f(\lambda^\alpha) = \alpha f(\lambda)$	true for only the Hencky strain, $\alpha \in \mathbb{R}$

Note: the Bažant strain, defined as $f(\lambda) = \frac{1}{2} \left(\lambda - \frac{1}{\lambda} \right)$, is not considered herein.

¹Neff, P (2013). The Hencky strain measure the geodesic distance to $SO(n)$, at 6.

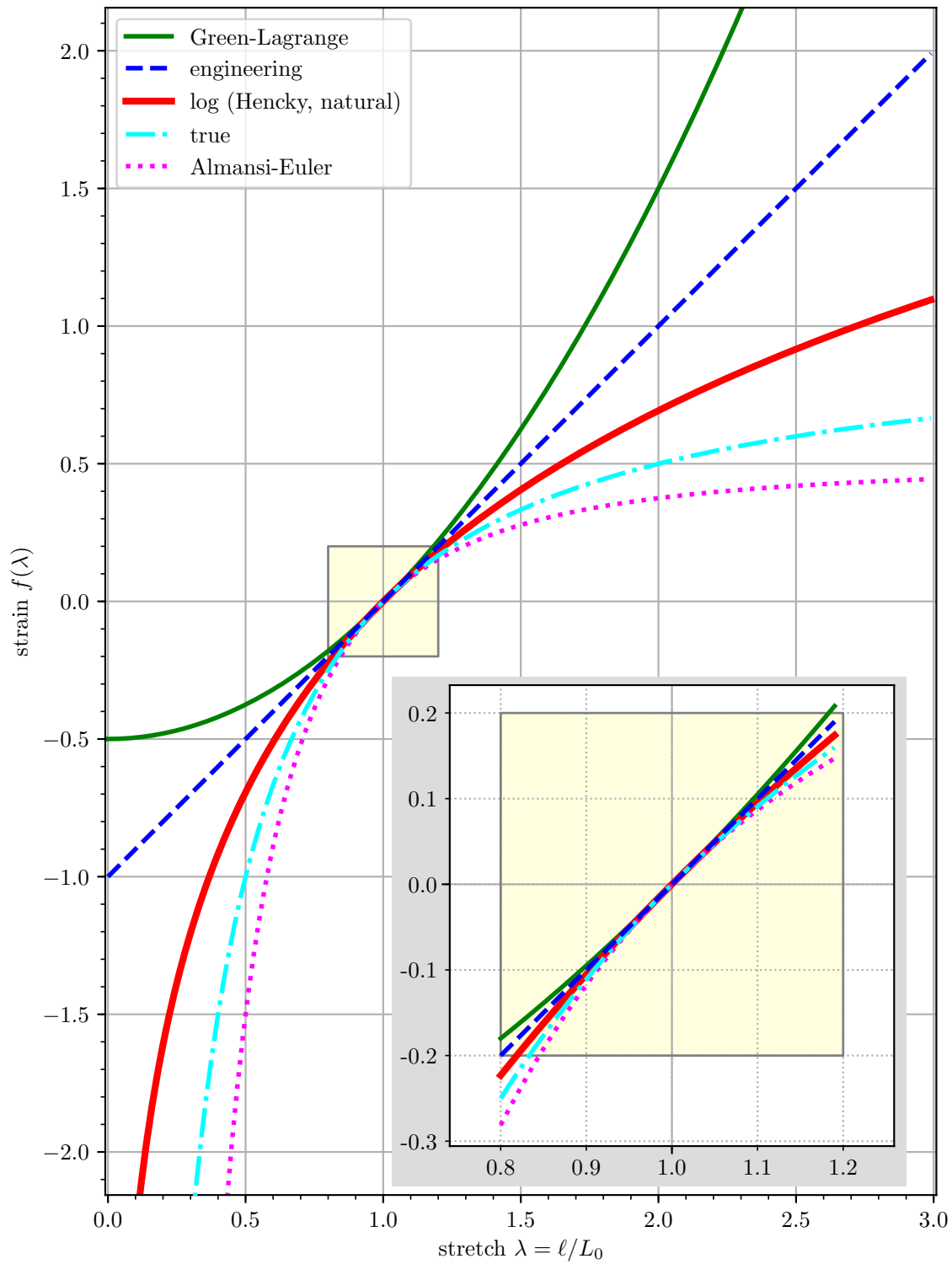


Figure 7.1: One dimensional strain as a function of stretch ratio.

7.6 Strain Tensors and Finite Rotations

Because it takes on nonzero values under finite rotation, the linearized strain tensor should not be used for geometrically nonlinear analysis. These nonzero values are completely artificial and strictly an artifact of using a linear strain definition with geometrically nonlinear motions. This result is shown as follows:

Let $\mathbf{R} = \mathbf{R}(t)$ be a two-dimensional, rigid body rotation parameterized by time t and scaled by constant $\omega = 2\pi$ radians per second. Then, the motion $\boldsymbol{\varphi}$ of a body can be written as

$$\boldsymbol{\varphi}(\mathbf{X}, t) = \mathbf{R}(t)\mathbf{X} \iff \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{bmatrix} \cos \omega t & \sin \omega t \\ -\sin \omega t & \cos \omega t \end{bmatrix} \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix}. \quad (7.15)$$

Then the deformation gradient $\mathbf{F} = \mathbf{F}(t)$ is a function of time alone,

$$\mathbf{F}(t) \iff F_{iJ} = \begin{bmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{bmatrix} = \begin{bmatrix} \cos \omega t & \sin \omega t \\ -\sin \omega t & \cos \omega t \end{bmatrix}. \quad (7.16)$$

The linearized strain tensor is found to be

$$2\mathbf{E}_{\text{LIN}} = \nabla_0 \mathbf{u} + (\nabla_0 \mathbf{u})^T, \quad (7.17)$$

$$= \mathbf{F} - \mathbf{1} + (\mathbf{F} - \mathbf{1})^T, \quad (7.18)$$

$$= F_{iJ} - \delta_{iJ} + F_{Ji} - \delta_{Ji}, \quad (7.19)$$

$$= \begin{bmatrix} \cos \omega t & \sin \omega t \\ -\sin \omega t & \cos \omega t \end{bmatrix} + \begin{bmatrix} \cos \omega t & -\sin \omega t \\ \sin \omega t & \cos \omega t \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad (7.20)$$

$$= 2(\cos \omega t - 1)\mathbf{1}. \quad (7.21)$$

Now, for small angles, $\omega t \approx 0$, which is for small deviations $x_1 \approx X_1$, $x_2 \approx X_2$, then $\mathbf{E}_{\text{LIN}} = \mathbf{0}$ for rigid body rotations. However, for arbitrary finite angles, $\omega t \neq 0$, and the linearized strain tensor $\mathbf{E}_{\text{LIN}}$ reports nonzero strain for rigid body rotations, which is nonsensical.

A correct strain tensor will report zero strain for rigid body rotations. One such strain tensor is the fully nonlinear Green-Lagrange strain tensor. This result is shown as follows:

$$2\mathbf{E} = \mathbf{F}^T \mathbf{F} - \mathbf{1} \quad (7.22)$$

$$= \begin{bmatrix} \cos \omega t & -\sin \omega t \\ \sin \omega t & \cos \omega t \end{bmatrix} \begin{bmatrix} \cos \omega t & \sin \omega t \\ -\sin \omega t & \cos \omega t \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{0}. \quad (7.23)$$

Chapter 8

Stress

8.1 Introduction

Stress can generally be thought of as a ratio of force to area. The greater the force for a given area, the higher the stress. The greater the area for a given force, the lower the stress.

Forces and areas are vector quantities, with magnitude and direction.

The force magnitude, a scalar value, must be zero or positive. Negative force values are disallowed because the force magnitude is the length of the force vector, which is independent of force direction.

The area magnitude, a scalar value, must be positive. A zero area value is disallowed because division by zero is undefined. Negative area values are disallowed because the area magnitude is the length of the area vector, which is independent of area direction.

The force direction is described by a unit vector in a particular reference frame. The unit vector endows the force vector signs, negative and positive, to indicate direction with the reference frame.

Similarly, the area direction is described by a unit vector in a particular reference frame. This reference frame is typically the same reference frame used to describe the force. However, if the two references are not the same, then a basis transformation may be used to put both vectors into the same reference frame.

8.2 Stress Measures

The **Kirchhoff stress** $\boldsymbol{\tau}$ (also known as the weighted Cauchy stress) is defined as

$$\boldsymbol{\tau} \triangleq J \boldsymbol{\sigma}. \quad (8.1)$$

The Kirchhoff stress is often used in metal plasticity formulations.

Table 8.1: Seth-Hill family of strains and their respective stress conjugates.

m	Name and Strain Tensor	Name and Stress Tensor
2	Green-Lagrange $\mathbf{E}_{(2)} = \frac{1}{2} (\mathbf{U}^2 - \mathbf{1})$	Piola-Kirchhoff Two $\mathbf{S} = J \mathbf{F}^{-1} \boldsymbol{\sigma} \mathbf{F}^{-\top}$
1	engineering (Biot, nominal) $\mathbf{E}_{(1)} = \mathbf{U} - \mathbf{1}$	Biot (Jaumann) $\mathbf{B} = \frac{1}{2} (\mathbf{S}\mathbf{U} + \mathbf{U}\mathbf{S})$
0	log (Hencky, natural) $\mathbf{E}_{(0)} = \ln \mathbf{U}$	to come
-1	true $\mathbf{E}_{(-1)} = \mathbf{1} - \mathbf{U}^{-1}$	Swainger $\mathbf{SW} = \frac{1}{2} J (\mathbf{U}\hat{\boldsymbol{\sigma}} + \hat{\boldsymbol{\sigma}}\mathbf{U})$
-2	Euler-Almansi $\mathbf{E}_{(-2)} = \frac{1}{2} (\mathbf{1} - \mathbf{U}^{-2})$	Almansi $\mathbf{A} = J \mathbf{F}^{\top} \boldsymbol{\sigma} \mathbf{F}$

The **Co-rotated Cauchy stress** (also known as the back-rotated Cauchy stress) is defined as

$$\hat{\boldsymbol{\sigma}} \triangleq \mathbf{R}^{\top} \boldsymbol{\sigma} \mathbf{R} \quad (8.2)$$

The **Piola-Kirchhoff One stress** $\mathbf{P}$ is defined as

$$\mathbf{P} \triangleq J \boldsymbol{\sigma} \mathbf{F}^{-\top} = \boldsymbol{\tau} \mathbf{F}^{-\top} \quad (8.3)$$

The **Piola-Kirchhoff Two stress** $\mathbf{S}$ is defined as

$$\mathbf{S} \triangleq J \mathbf{F}^{-1} \boldsymbol{\sigma} \mathbf{F}^{-\top} = \mathbf{F}^{-1} \boldsymbol{\tau} \mathbf{F}^{-\top}. \quad (8.4)$$

Thus

$$\mathbf{P} = \mathbf{F}\mathbf{S}. \quad (8.5)$$

8.3 Stress Rates

Stress rates arise when the stress depends on variables that have time dependency. Viscoelastic material behavior, for example, will exhibit strains, and thus stresses, with time dependence.

Chapter 9

Objectivity

9.1 Introduction

- For two reference frames, $\mathcal{F}$ and $\mathcal{F}^*$, the base vectors for $\mathcal{F}$, $\hat{\mathbf{f}}_1$, $\hat{\mathbf{f}}_2$, and $\hat{\mathbf{f}}_3$ transform through a rigid body rotation $\mathbf{R}$, such that

$$\hat{\mathbf{f}}_1^* = \mathbf{R}\hat{\mathbf{f}}_1, \quad \hat{\mathbf{f}}_2^* = \mathbf{R}\hat{\mathbf{f}}_2, \quad \text{and} \quad \hat{\mathbf{f}}_3^* = \mathbf{R}\hat{\mathbf{f}}_3. \quad (9.1)$$

- A vector quantity $\mathbf{a}$ is objective if

$$\mathbf{a}^* = \mathbf{R}\mathbf{a}. \quad (9.2)$$

- A tensor quantity $\mathbf{a}$ is objective if

$$\mathbf{a}^* = \mathbf{R}\mathbf{a}\mathbf{R}^\top. \quad (9.3)$$

Result 9.1.1 The Cauchy stress $\boldsymbol{\sigma}$ is objective.

Proof.

The traction vector $\mathbf{t}$ is the inner product of the Cauchy stress tensor $\boldsymbol{\sigma}$ and the outward surface normal vector $\hat{\mathbf{n}}$. The same traction vector in a different reference frame is $\mathbf{t}^* = \boldsymbol{\sigma}^*\hat{\mathbf{n}}^*$. Since vectors transform between reference frames by through a rotation tensor $\mathbf{R}$, $\mathbf{t}^* = \mathbf{R}\mathbf{t}$ and $\hat{\mathbf{n}}^* = \mathbf{R}\hat{\mathbf{n}}$. Substituting these two vector transformations in the second observer's frame of reference gives

$$\mathbf{R}\mathbf{t} = \boldsymbol{\sigma}^*\mathbf{R}\hat{\mathbf{n}} \quad (9.4)$$

but $\mathbf{R}\mathbf{t} = \mathbf{R}\boldsymbol{\sigma}\hat{\mathbf{n}}$, so $\mathbf{R}\boldsymbol{\sigma}\hat{\mathbf{n}} = \boldsymbol{\sigma}^*\mathbf{R}\hat{\mathbf{n}}$ for all $\hat{\mathbf{n}}$, thus

$$\boldsymbol{\sigma}^* = \mathbf{R}\boldsymbol{\sigma}\mathbf{R}^\top, \quad (9.5)$$

which is how second-order tensors transform when they are objective. Thus σ is objective. ■

9.2 Two-Point Tensors Transform as Vectors

- The deformation gradient is a two-point tensor, with one coordinate in the reference configuration and the other in the current configuration. The reference configuration is inherently independent since it is tied to the observer. The current configuration makes the deformation gradient transform as a vector, not a tensor. For more information, see Holzapfel at 189.
- The rotation is a two-point tensor, moving between the reference configuration and the current configuration. The rotation thus transforms as a vector, not a tensor. For more information, see Holzapfel at 190.
- The first Piola-Kirchhoff stress tensor is a two-point tensor. Like the deformation gradient and rotation, mentioned above, PK1 transforms as a vector, not a tensor. See Holzapfel at 191 for more detail.

9.3 Convected Rates of Change

9.4 Corotated Rates of Change

An objective vector $\mathbf{a}$

$$\mathbf{a}^* = \mathbf{R} \mathbf{a}, \quad (9.6)$$

and an objective tensor quantity $\mathbf{a}$

$$\mathbf{a}^* = \mathbf{R} \mathbf{a} \mathbf{R}^T, \quad (9.7)$$

generally do **not** maintain their objectivity through time differentiation, viz.,

$$\dot{\mathbf{a}}^* = \dot{\mathbf{R}} \mathbf{a} + \mathbf{R} \dot{\mathbf{a}}, \quad (9.8)$$

and

$$\dot{\mathbf{a}}^* = \dot{\mathbf{R}} \mathbf{a} \mathbf{R}^T + \mathbf{R} \dot{\mathbf{a}} \mathbf{R}^T + \mathbf{R} \mathbf{a} \dot{\mathbf{R}}^T. \quad (9.9)$$

But, material vector fields can be endowed with objectivity. For vector $\mathbf{a}$, the corotational rate $\overset{\nabla}{\dot{\mathbf{a}}}$ is defined as

$$\overset{\nabla}{\dot{\mathbf{a}}} \triangleq \dot{\mathbf{a}} - \mathbf{w} \mathbf{a}, \quad (9.10)$$

where $\mathbf{w}$ is the rotation rate.

For a second-order tensor $\mathbf{a}$, the corotational rate $\overset{\nabla}{\mathbf{a}}$ is defined as

$$\overset{\nabla}{\mathbf{a}} \triangleq \dot{\mathbf{a}} - \mathbf{w} \mathbf{a} + \mathbf{a} \mathbf{w}, \quad (9.11)$$

which is referred to as the **Jaumann-Zaremba rate**.

Chapter 10

Energy

10.1 Conjugates

Energy conjugates

Stress Power	
$\mathcal{P}_{\text{int}}$	$= \int_V \boldsymbol{\sigma} : \nabla \mathbf{v} \, dV = \int_V \boldsymbol{\sigma} : \mathbf{d} \, dV \text{ (since } \boldsymbol{\sigma} = \boldsymbol{\sigma}^T \text{)}$
	$= \int_{V_0} \boldsymbol{\tau} : \nabla \mathbf{v} \, dV_0 = \int_{V_0} \boldsymbol{\tau} : \mathbf{d} \, dV_0$
	$= \int_{V_0} \mathbf{B} : \dot{\mathbf{U}} \, dV_0$
	$= \int_{V_0} \mathbf{P} : \dot{\mathbf{F}} \, dV_0$
	$= \int_{V_0} \mathbf{S} : \dot{\mathbf{E}} \, dV_0 = \frac{1}{2} \int_{V_0} \mathbf{S} : \dot{\mathbf{C}} \, dV_0$

Table 10.1: Strain rates and their respective stress conjugates.

Chapter 11

Balance Laws

In this section, we posit the **Master Balance Law** in general form. Also called “conservation of anything,” this description assumes an arbitrary control volume $V \in \mathbb{R}^3$ with boundary ∂V . What follows is the so-called Eulerian formulation, illustrated in Fig. 11.1, which assumes the control volume, while arbitrary in its placement in space, is **fixed** in space.¹

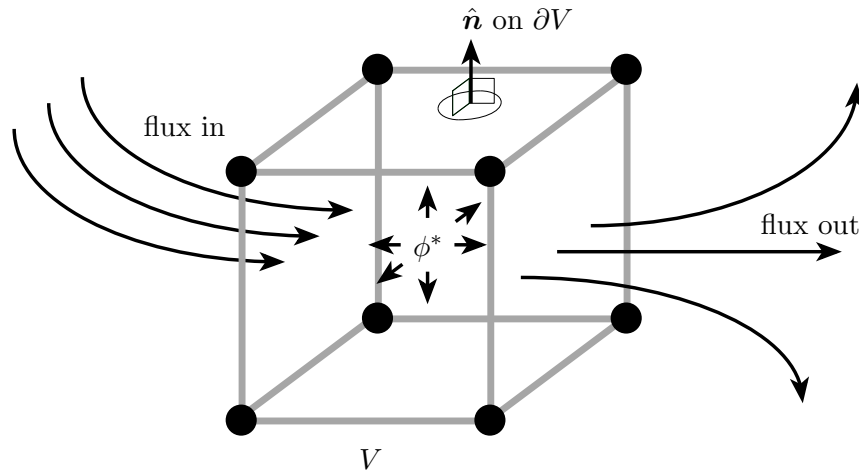


Figure 11.1: Concept of a fixed control volume of the Eulerian formulation.

Now, let

- $\phi(\mathbf{x}, t)$ be the description of some physical quantity per unit volume, which can in general be a scalar, vector, or tensor quantity,

¹The connection to the Lagrangian formulation, in which the control is not fixed in space, but instead, moves through space as the body to which it is attached moves through space, is examined in the sequel.

- $\phi^*(\mathbf{x}, t)$ be the generation (depletion) *rate* of the physical quantity per unit volume,
- $\hat{\phi}(\mathbf{x}, t)$ be the physical quantity per unit volume that flows across the boundary with rate (velocity) $\mathbf{v}(\mathbf{x}, t)$, and
- ∂V be the boundary of the control volume, with outward pointing surface normal unit vector $\hat{\mathbf{n}}(\mathbf{x}, t)$.

The time rate of change of the quantity $\phi(\mathbf{x}, t)$ within the differential control volume dV is then equal to

- **generation**—the rate at which the quantity is generated (or depleted) within the control volume by the point source (point sink), plus
- **flux**—the rate at which the quantity is either carried into (or out of) the volume across the volume's differential surface area dA .

These words, stated mathematically, are

$$\underbrace{\frac{d}{dt} \int_V \phi(\mathbf{x}, t) dV}_{\text{rate}} = \underbrace{\int_V \phi^*(\mathbf{x}, t) dV}_{\text{source}} - \underbrace{\int_{\partial V} \hat{\phi}(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t) \cdot \hat{\mathbf{n}} dA}_{\text{flux}}, \quad (11.1)$$

which is the general form of the **Master Balance Law**. We adopt the following sign convention to impart consistency:

- For positive time rate of change of the quantity $\phi(\mathbf{x}, t)$, the generation $\phi^*(\mathbf{x}, t)$ is trivially positive. For quantity $\hat{\phi}(\mathbf{x}, t)$ carried *into* the control volume, the inner product of $\mathbf{v}$ with $\hat{\mathbf{n}}$ is negative, thus a negative outside of the flux integral is used.
- Likewise, for a negative time rate of change of quantity $\phi(\mathbf{x}, t)$, the generation $\phi^*(\mathbf{x}, t)$ is negative. And, for the quantity $\hat{\phi}(\mathbf{x}, t)$ carried *out of* the control volume, the inner product of $\mathbf{v}$ with $\hat{\mathbf{n}}$ is positive, thus the negative outside of the flux integral provides sign consistency.

11.1 Balance Law Metaphor

Here we present the Master Balance Law by way of metaphor. Let the volume V be the geographic region of Oregon, with boundary ∂V that touches the state borders of Washington, Idaho, Nevada, and California in the United States.² Let ϕ be the number of people living in the state of Oregon. The change in population $\Delta\phi$ over some time interval Δt is equal to the net generation rate ϕ^* (number of births less the number of deaths in the time interval), plus the net population flux rate $\hat{\phi}\mathbf{v}$ (the number of people moving to Oregon less the number of people moving from Oregon within the time interval).

²In this oversimplified metaphor, we assume flux does not occur via the western seaboard nor via the air.

11.2 Conservation of Mass

Let us now apply the Master Balance Law to the physical quantity of mass, and develop a relationship for **conservation of mass**, a concept where mass is neither created nor destroyed. The physical quantity of mass per unit volume is density $\rho(\mathbf{x}, t)$. Thus, we substitute $\rho(\mathbf{x}, t)$ for $\phi(\mathbf{x}, t)$ in the foregoing Master Balance Law.

Assumption	Expression	
(1) No mass generation or depletion.	$\rho^*(\mathbf{x}, t) = 0$	(11.2)
(2) Mass can flow in and out of the boundaries.	$\rho(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t) \cdot \hat{\mathbf{n}}(\mathbf{x}, t) \neq 0$	

Result 11.2.1 Conservation of mass in the current configuration is stated as

$$\frac{\partial \rho(\mathbf{x}, t)}{\partial t} + \nabla \cdot (\rho(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t)) = 0 \iff \rho_{,t} + (\rho v_i)_{,i} = 0. \quad (11.3)$$

Proof.

For mass to be conserved, mass within a control volume is neither created nor destroyed, thus the source term is identically zero. Thus, increases (decreases) in the mass rate term are produced only by flux of mass in through (out from) the boundary. The idea, stated mathematically, is

$$\underbrace{\frac{d}{dt} \int_V \rho(\mathbf{x}, t) dV}_{\text{rate}} = \underbrace{\int_V \rho^*(\mathbf{x}, t) dV}_{\text{source} = 0} - \underbrace{\int_{\partial V} \rho(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t) \cdot \hat{\mathbf{n}}(\mathbf{x}, t) dA}_{\text{flux}}. \quad (11.4)$$

The Divergence Theorem is used to write the flux term surface integral as a volume integral, thus

$$\underbrace{\frac{d}{dt} \int_V \rho(\mathbf{x}, t) dV}_{\text{rate}} = 0 - \underbrace{\int_V \nabla \cdot (\rho(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t)) dV}_{\text{flux}}. \quad (11.5)$$

Because V is fixed in an inertial frame, the foregoing equation can be rewritten with the rate time derivative inside the integral, thus

$$\int_V \left(\frac{\partial \rho(\mathbf{x}, t)}{\partial t} + \nabla \cdot (\rho(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t)) \right) dV = 0. \quad (11.6)$$

Finally, arbitrariness of B_t requires that the integrand is identically zero, thus

$$\frac{\partial \rho(\mathbf{x}, t)}{\partial t} + \nabla \cdot (\rho(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t)) = 0 \iff \rho_{,t} + (\rho v_i)_{,i} = 0, \quad (11.7)$$

which is the result. ■

Remark 11.2.7. Conservation of mass is also called the **continuity equation**. Some texts will recast the continuity equation by defining a **material derivative** of density as follows: Since $\rho = \rho(\mathbf{x}, t)$, then

$$\dot{\rho} = \frac{\partial \rho}{\partial \mathbf{x}} \frac{\partial \mathbf{x}}{\partial t} + \frac{\partial \rho}{\partial t} \triangleq \frac{D\rho}{Dt}. \quad (11.8)$$

With arguments $(\mathbf{x}, t)$ suppressed for brevity, (11.3) can be expanded to give

$$\frac{\partial \rho}{\partial t} + \nabla \rho \cdot \mathbf{v} + \rho \nabla \cdot \mathbf{v} = \underbrace{\frac{\partial \rho}{\partial t} + \frac{\partial \rho}{\partial \mathbf{x}} \frac{\partial \mathbf{x}}{\partial t}}_{\triangleq \dot{\rho}} + \rho \nabla \cdot \mathbf{v} = 0, \quad (11.9)$$

or simply

$$\dot{\rho} + \rho \nabla \cdot \mathbf{v} = 0. \quad (11.10)$$

11.3 Conservation of Linear Momentum

11.4 Conservation of Angular Momentum

Assumption	Expression
(1) Point torques are zero.	$\mathbf{x} \times \mathbf{f}^{\text{ext}} = \mathbf{0}$

(11.11)

Result 11.4.1 The Cauchy stress tensor $\boldsymbol{\sigma}(\mathbf{x}, t)$ is symmetric, thus $\boldsymbol{\sigma} = \boldsymbol{\sigma}^T$.

11.5 Conservation of Energy

Assumption	Expression
(1) Energy is neither created nor destroyed.	

(11.12)

Chapter 12

Thermodynamics

12.1 Properties

Thermodynamics properties can be categorized as either **intensive** or **extensive**.

- Intensive properties are independent of, and therefore not proportional to, the size or sample of the material. Intensive properties are also referred to as **intrinsic** properties or, equivalently, as **bulk** properties.
- In contrast, extensive properties are dependent on, and therefore proportional to, the size or sample of the material. Extrinsic properties are also referred to as **extrinsic** properties.

Figure 12.1 demonstrates the independence of intrinsic properties and the dependence of extrinsic properties on the control volume, designated with the dashed lines. The original control volume, on the left side of the figure, is divided into two equal-sized control volumes, as shown on the right side of the figure. The intrinsic properties do not depend on the size of the control volume. The extrinsic properties, however, are now scaled by a factor of $\frac{1}{2}$.

It is often convenient to speak of extrinsic properties in a way that makes them *independent* of size or sample of the material. To achieve this, the extrinsic properties are divided by the mass of the sample. This ratio is referred to as the specific extrinsic property. Two common examples of specific properties are the **specific volume**,

$$\mathcal{V} \triangleq \frac{V}{m} = \frac{1}{\rho}, \quad (12.1)$$

and **specific energy**,

$$\mathcal{E} \triangleq \frac{E}{m}. \quad (12.2)$$

Intrinsic (intensive, bulk) pressure, density, temperature	P, ρ, T	P, ρ, T	P, ρ, T
Extrinsic (extensive) mass, volume, energy	m, V, E	$\frac{m}{2}, \frac{V}{2}, \frac{E}{2}$	$\frac{m}{2}, \frac{V}{2}, \frac{E}{2}$
Specific (extrinsic/mass) specific volume, specific energy	ν, ε	ν, ε	ν, ε

Figure 12.1: Thermodynamic properties categorized as intrinsic, extrinsic, and specific.

12.2 First Law of Thermodynamics

- A closed system can exchange heat and work with its surroundings, but cannot exchange mass.
- An open system can exchange heat, work, and mass with its surroundings.

The first law for a closed system states the change in internal energy ΔU is equal to the net energy Q added to the system as heat less the net work W done **by** the system.

$$\Delta U = Q - W. \quad (12.3)$$

This was the sign convention used by Joule and Clausius. A different sign convention, used by Planck, stated that the change in internal energy ΔU is equal to the net energy Q added to the system as heat plus the net work done **on** the system.

$$\Delta U = Q + W. \quad (12.4)$$

Thus, all net transfers **to** the system are **positive**. In contrast, all net transfers **from** the system are **negative**.

Then, the conservation of energy principle for open systems can be written as

$$\Delta E = \Delta(K + U) = Q + W, \quad (12.5)$$

where the total energy E is the sum of the kinetic energy K and the internal energy U . The change in total energy is equal to the net energy Q added to the system as heat plus the net work W done **on** the system.

Next, we introduce

- $d(\cdot)$ as the **exact differential**, which does not depend on the path taken through space of the thermodynamic parameters and depends only on the initial states, and
- $\delta(\cdot)$ as the **inexact differential**, which depends on the path taken through space of thermodynamic parameters.

The balance law then states

$$dE = \delta Q + \delta W. \quad (12.6)$$

First Law in Rate Form

We will be interested in the change in energy over a unit of time. Thus, it is useful to write the First Law of Thermodynamics in terms of time derivatives.

$$\frac{dE}{dt} = \delta \dot{Q} + \delta \dot{W} \quad (12.7)$$

Continuum Description

Now we write explicit continuum descriptions for each of the terms in the balance law

$$\frac{d}{dt}(K + U) = \delta \dot{Q} + \delta \dot{W}. \quad (12.8)$$

The kinetic energy is given by

$$K = \int_V \frac{1}{2} \rho(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t) \cdot \mathbf{v}(\mathbf{x}, t) dV \quad (12.9)$$

12.3 Second Law of Thermodynamics

The Second Law states that the infinitesimal increment of entropy dS of a system from an infinitesimal transfer of heat δQ , divided by the common temperature of the system and the surrounding that supplies the heat, satisfies

$$dS \geq \frac{\delta Q}{T}. \quad (12.10)$$

For a reversible process,

$$dS = \frac{\delta Q}{T}. \quad (12.11)$$

For an irreversible process,

$$dS > \frac{\delta Q}{T}. \quad (12.12)$$

12.4 First and Second Laws Combined

For a reversible process, the first and second laws can be combined as

$$\boxed{dE = TdS + \delta W, \quad \text{(1st and 2nd Law, Reversible Form).}} \quad (12.13)$$

12.5 Thermodynamic Potentials

12.5.1 Variables

P	pressure	(N/m ²)
V	volume	(m ³)
T	temperature	(K)
S	entropy	(J/K)

12.5.2 Potentials

Thermodynamic potentials consider a thermodynamically closed system, thus the energy $E = (K + U)$ is considered only as internal energy U .

U		internal energy	(J)
$\mathcal{F}$	$\triangleq U - TS$	Helmholtz free energy	(J)
$\mathcal{G}$	$\triangleq U + PV - TS$	Gibbs free energy	(J)
$\mathcal{H}$	$\triangleq U + PV$	enthalpy	(J)

The **Helmholtz** free energy $\mathcal{F}$ is the Legendre transform of the internal energy U , where the entropy S is replaced by the temperature T as the independent variable.

12.5.3 Differential Forms of Potentials

$$\begin{aligned} dU &= T dS - P dV \\ d\mathcal{F} &= -P dV - S dT \\ d\mathcal{G} &= -S dT + V dP \\ d\mathcal{H} &= V dP + T dS \end{aligned}$$

12.6 Maxwell Relations

The Maxwell relations are a statement of equality between *first* derivative of a function because of a quality possessed by the *second* derivative of that function. Specifically, the Maxwell relations equate *first* derivatives of the thermodynamic potential functions $U, \mathcal{F}, \mathcal{G}$, and $\mathcal{H}$, because the *second* derivatives of those same functions does not depend on the order of differentiation with respect to the thermodynamic variables P, V, T , and S .

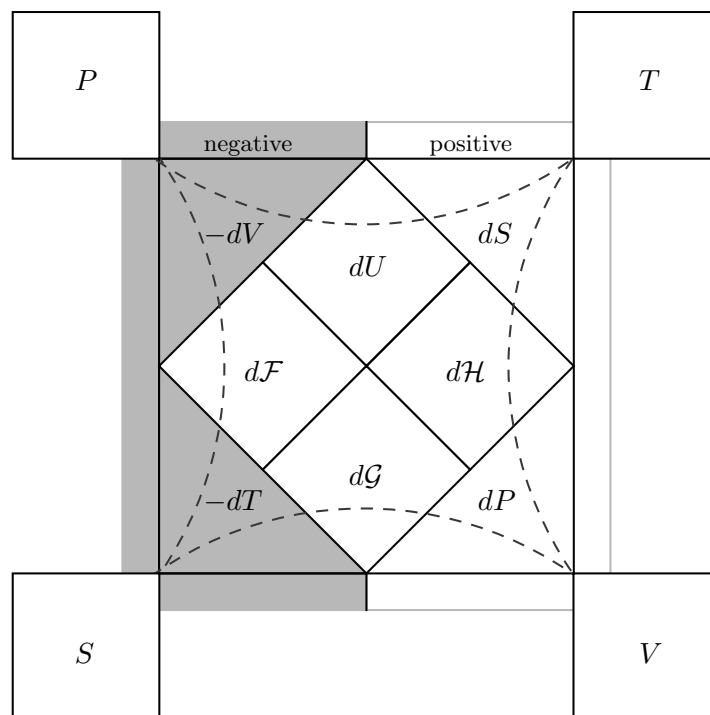


Figure 12.2: Mnemonic for the thermodynamic potentials in differential form.

The Maxwell relations make use of the **Schwarz theorem**, see (2.1). With this context in mind, consider now the *first* derivatives of the thermodynamic potentials with respect to their natural variables. Specifically, we can write the four thermodynamic potentials as functions of their natural variables:

$$U = U(S, V), \quad (12.14)$$

$$\mathcal{F} = \mathcal{F}(V, T), \quad (12.15)$$

$$\mathcal{G} = \mathcal{G}(T, P), \quad (12.16)$$

$$\mathcal{H} = \mathcal{H}(P, S). \quad (12.17)$$

Next, we take *first* derivatives of these potentials:

$$\left(\frac{\partial U}{\partial S}\right)_V = T, \quad \left(\frac{\partial U}{\partial V}\right)_S = -P, \quad (12.18)$$

$$\left(\frac{\partial \mathcal{F}}{\partial V}\right)_T = -P, \quad \left(\frac{\partial \mathcal{F}}{\partial T}\right)_V = -S, \quad (12.19)$$

$$\left(\frac{\partial \mathcal{G}}{\partial T}\right)_P = -S, \quad \left(\frac{\partial \mathcal{G}}{\partial P}\right)_T = V, \quad (12.20)$$

$$\left(\frac{\partial \mathcal{H}}{\partial P}\right)_S = V, \quad \left(\frac{\partial \mathcal{H}}{\partial S}\right)_P = T. \quad (12.21)$$

Finally, we take *second* derivatives of the potential functions with respect to the natural variables, which allows for the *first* derivatives of the natural variables with respect to the natural variables themselves to be equated, as follows:

$$\frac{\partial^2 U}{\partial V \partial S} \implies + \left(\frac{\partial T}{\partial V}\right)_S = - \left(\frac{\partial P}{\partial S}\right)_V \quad (12.22)$$

$$\frac{\partial^2 \mathcal{F}}{\partial T \partial V} \implies - \left(\frac{\partial P}{\partial T}\right)_V = - \left(\frac{\partial S}{\partial V}\right)_T \quad (12.23)$$

$$\frac{\partial^2 \mathcal{G}}{\partial P \partial T} \implies - \left(\frac{\partial S}{\partial P}\right)_T = + \left(\frac{\partial V}{\partial T}\right)_P \quad (12.24)$$

$$\frac{\partial^2 \mathcal{H}}{\partial S \partial P} \implies + \left(\frac{\partial V}{\partial S}\right)_P = + \left(\frac{\partial T}{\partial P}\right)_S \quad (12.25)$$

The foregoing equations are useful for eliminating entropy S , which is difficult to measure in a laboratory setting, in favor of the other natural variables pressure P , volume V , and temperature T , all of which are relatively easier to measure.

Chapter 13

Fluids

The central concepts for fluids are

- derivation of the governing equations of motion in the current configuration,
- a stress response function that is decomposed into volumetric and deviatoric parts,
- a volumetric stress response function that is either incompressible or compressible, and
- a deviatoric stress response that, through dependence on velocity gradients, is either free from shear (inviscid) or supports shear (viscid).

13.1 Incompressible, Inviscid (Ideal) Fluid

The incompressible, inviscid (ideal) has three assumptions:

Assumption	Expression	
(1) The reference mass density is uniform.	$\rho_0(\mathbf{X}) = \bar{\rho}_0 > 0$	
(2) The material is incompressible.	$\nabla \cdot \mathbf{v}(\mathbf{x}, t) = 0$	(13.1)
(3) The Cauchy stress is spherical only. The stress has no deviatoric response.	$\boldsymbol{\sigma}(\mathbf{x}, t) = -P(\mathbf{x}, t) \mathbf{1}$	

Result 13.1.1 The current mass density ρ is a positive constant, equal to the reference mass density. Thus $\rho(\mathbf{x}, t) = \rho_0(\mathbf{X}) = \bar{\rho}_0 > 0$.

Proof.

Use conservation of mass in Eulerian form (11.3),

$$\frac{\partial \rho(\mathbf{x}, t)}{\partial t} + \nabla \cdot [\rho(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t)] = 0. \quad (13.2)$$

Expand the divergence operator $\nabla \cdot [\cdot]$ on the product of density and velocity,

$$\underbrace{\frac{\partial \rho(\mathbf{x}, t)}{\partial t} + [\nabla \rho(\mathbf{x}, t)] \cdot \mathbf{v}(\mathbf{x}, t)}_{=\dot{\rho} \text{ (4.22)}} + \rho(\mathbf{x}, t) \underbrace{\nabla \cdot \mathbf{v}(\mathbf{x}, t)}_{=0 \text{ (13.1)}} = 0. \quad (13.3)$$

Thus,

$$\dot{\rho}(\mathbf{x}, t) = 0, \quad (13.4)$$

and expressed in the reference configuration variable $\mathbf{X}$,

$$\frac{\partial \rho(\varphi(\mathbf{X}, t), t)}{\partial t} = 0. \quad (13.5)$$

Since the time variation is zero at the initial time $t = 0$,

$$\rho(\varphi(\mathbf{X}, 0), 0) = \rho(\mathbf{X}, 0) = \rho_0(\mathbf{X}) = \bar{\rho}_0, \quad (13.6)$$

$$\therefore \rho(\mathbf{x}, t) = \bar{\rho}_0 > 0. \quad (13.7)$$

Thus the current mass density $\rho(\mathbf{x}, t)$ is uniform and constant for all time. ■

Result 13.1.2 The **Euler equations** for an **ideal fluid** can then be written as

$$\bar{\rho}_0 \left[\frac{\partial \mathbf{v}}{\partial t} + [\nabla \mathbf{v}] \cdot \mathbf{v} \right] = -\nabla P + \bar{\rho}_0 \mathbf{b} \iff \bar{\rho}_0 [v_{i,t} + v_{i,j} v_j] = -P_{,i} + \bar{\rho}_0 b_i \quad (13.8)$$

with

$$\nabla \cdot \mathbf{v} = 0 \iff v_{i,i} = 0 \quad (13.9)$$

where functional dependence on the current configuration, $f = f(\mathbf{x}, t)$, is assumed, but not explicitly written for the sake of brevity.

Proof.

The appearance of $\rho(\mathbf{x}, t)$ in the current configuration balance of linear momentum is substituted with $\bar{\rho}_0$,

$$\bar{\rho}_0 \dot{\mathbf{v}}(\mathbf{x}, t) = \nabla \cdot \boldsymbol{\sigma}(\mathbf{x}, t) + \bar{\rho}_0 \mathbf{b}(\mathbf{x}, t). \quad (13.10)$$

The $\nabla \cdot \boldsymbol{\sigma}$ term reduces to $\nabla \cdot (-P\mathbf{1}) = -\nabla P$. The spatial body force per unit mass is $\mathbf{b}(\mathbf{x}, t)$. The material time derivative is written explicitly as

$$\dot{\mathbf{v}}(\mathbf{x}, t) = \frac{\partial \mathbf{v}(\mathbf{x}, t)}{\partial t} + [\nabla \mathbf{v}(\mathbf{x}, t)] \cdot \mathbf{v}(\mathbf{x}, t). \quad (13.11)$$

The Euler equations for an ideal fluid can then be written as

$$\bar{\rho}_0 \left[\frac{\partial \mathbf{v}(\mathbf{x}, t)}{\partial t} + [\nabla \mathbf{v}(\mathbf{x}, t)] \cdot \mathbf{v}(\mathbf{x}, t) \right] = -\nabla P(\mathbf{x}, t) + \bar{\rho}_0 \mathbf{b}(\mathbf{x}, t), \quad (13.12)$$

with the incompressibility constraint,

$$\nabla \cdot \mathbf{v}(\mathbf{x}, t) = 0. \quad (13.13)$$

used to obtain the foregoing result. ■

Remark 13.1.8.

- The ideal fluid does not support shear stresses.
- There is no explicit relationship between pressure P and density $\bar{\rho}_0$.
- There is no explicit relationship between pressure P and velocity $\mathbf{v}$.
- Air flows with $v < 100$ m/s (224 mph, 328 fps) or $M < 0.3$ are well-approximated as incompressible.¹

13.2 Compressible, Inviscid Fluid

The compressible, inviscid fluid eliminates the constant density and incompressibility assumptions of an ideal fluid. To accommodate the compressibility behavior, the pressure becomes a function of density, $P = P(\rho(\mathbf{x}, t), t)$, with the added constraint that changes in pressure with respect to changes in density must be positive. Figure 13.1 illustrates this relationship (compare to Figure 18.2).

Assumption	Expression
(1) The Cauchy stress is spherical only, and is a function of density. The stress has no deviatoric response.	$\boldsymbol{\sigma}(\mathbf{x}, t) = -P(\rho(\mathbf{x}, t), t) \mathbf{1}$ (13.14)
(2) The change in pressure with respect to change in density is positive.	$\frac{\partial P}{\partial \rho} > 0$

Remark 13.2.9. Refer to Figure 13.1. We discuss the $P(\rho)$ relationship near a reference density ρ_0 , at reference pressure P_0 . Near ρ_0 , the slope is positive, increases as ρ increases, and decreases as ρ decreases. The slope remains positive for all values of ρ , which is a positive, real value.

The character of $P(\rho)$ at the extreme values of ρ , zero and infinity, are shown in general terms only. The density of the fluid can tend toward zero, but as the pressure

¹Pletcher RH, Tannehill JC, Anderson D. Computational fluid mechanics and heat transfer. Hemisphere Publishing Corporation, 1984, at 183.

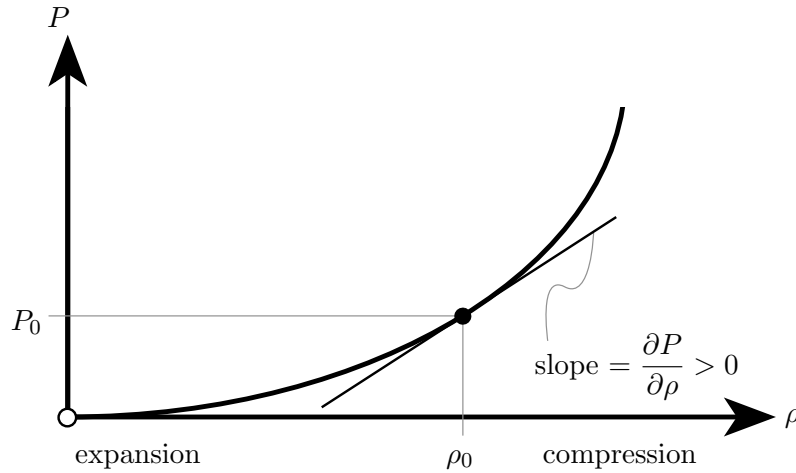


Figure 13.1: Pressure as a function of density for a compressible, inviscid fluid.

tends toward zero (a vacuum), the fluid will disassociate and the continuity assumption underlying continuum mechanics ceases to be valid. Thus, the $P(\rho)$ curve is plotted with a hole where the P and ρ axes intersect.

Conversely, as the density of the fluid tends toward infinity, so too does the pressure. While no vertical density asymptote is shown in Figure 13.1, it is implied because density increasing without bounds seems nonphysical. The sharp upward bend of $P(\rho)$ suggests this asymptotic behavior.

The general behavior that the slope of the $P(\rho)$ curve always be positive follows from the notion that an increase in pressure on a fluid should cause increase in density, and vice versa. Thus, the slope of the curve should never be zero or negative.

Finally, unlike the first derivative $\partial P/\partial \rho > 0$, the second derivative $\partial^2 P/\partial \rho^2$ has no requirements of positivity, even though $P(\rho)$ is shown in Figure 13.1 as a convex function.

Result 13.2.1 The **elastic fluid equations** can then be written as

$$\rho \left[\frac{\partial \mathbf{v}}{\partial t} + [\nabla \mathbf{v}] \mathbf{v} \right] = -\frac{\partial P}{\partial \rho} \nabla \rho + \rho \mathbf{b} \iff \rho [v_{i,t} + v_{i,j} v_j] = -\frac{\partial P}{\partial \rho} \rho_{,i} + \rho b_i \quad (13.15)$$

with

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \iff \rho_{,t} + (\rho v_i)_{,i} = 0 \quad (13.16)$$

where functional dependence on the current configuration, $f = f(\mathbf{x}, t)$, is assumed, but not explicitly written for the sake of brevity.

Remark 13.2.10.

- The elastic fluid is compressible, and thus allows volume and density changes.
- The elastic fluid does not support shear stresses.
- There is an explicit relationship between pressure P and density ρ , and the first derivative of this relationship must be greater than zero.
- There is no explicit relationship between pressure P and velocity $\mathbf{v}$.
- The elastic fluid is the starting point for the discussion of shock waves in condensed matter (see Chapter 18).

13.2.1 The Wave Equation

Now we add additional assumptions to the assumptions of a compressible, inviscid fluid, listed in (13.14).

Assumption	Expression	
(1) Body forces are negligible compared to internal and inertial forces.	$\mathbf{b}(\mathbf{x}, t) = \mathbf{0}$	
(2) Change in pressure with respect to change in density is a positive (squared) constant.	$\frac{\partial P}{\partial \rho} = c^2$	
(3) Uniform density as in (13.7)	$\rho(\mathbf{x}, t) = \bar{\rho}_0$	(13.17)
(4) Quiescent initial velocity	$\mathbf{v}(\mathbf{x}, 0) = \mathbf{0}$	
(5) The velocity field is a perturbation ϵ of the initial velocity, where $0 \leq \epsilon \ll 1$	$\mathbf{v}(\mathbf{x}, t) = \epsilon \mathbf{v}(\mathbf{x}, 0)$	
(6) The velocity field has continuous second partial derivatives, see (2.1)	$\frac{\partial}{\partial t} (\nabla \cdot \mathbf{v}) = \nabla \cdot \left(\frac{\partial \mathbf{v}}{\partial t} \right)$	

With these additional assumptions, the **elastic fluid equations** become the **wave equations**.

Result 13.2.2 The **wave equation** can be written as

$$\boxed{\frac{\partial^2 \rho(\mathbf{x}, t)}{\partial t^2} = c^2 \Delta \rho(\mathbf{x}, t) \iff \rho_{,tt} = c^2 \rho_{,ii}} \quad (13.18)$$

where

$$c^2 = \left. \frac{\partial P}{\partial \rho} \right|_{\rho=\bar{\rho}_0} \quad (13.19)$$

and c is called the **sound speed** at $\bar{\rho}_0$.

Proof.

Use conservation of momentum in the current configuration (13.15) with the assumptions in (13.17) of negligible body force, constant $\partial P / \partial \rho$, and uniform density,

$$\bar{\rho}_0 \left[\frac{\partial \mathbf{v}(\mathbf{x}, t)}{\partial t} + [\nabla \mathbf{v}(\mathbf{x}, t)] \mathbf{v}(\mathbf{x}, t) \right] = -c^2 \nabla \rho(\mathbf{x}, t). \quad (13.20)$$

Eliminate the $[\nabla \mathbf{v}] \mathbf{v}$ term because $\mathbf{v}(\mathbf{x}, t) = \epsilon \mathbf{v}(\mathbf{x}, 0) \approx \mathbf{0}$, and take the divergence,

$$\bar{\rho}_0 \left[\nabla \cdot \frac{\partial \mathbf{v}(\mathbf{x}, t)}{\partial t} \right] = -c^2 \nabla \cdot \nabla \rho(\mathbf{x}, t). \quad (13.21)$$

Use conservation of mass from (13.16)

$$\frac{\partial \rho(\mathbf{x}, t)}{\partial t} + \nabla \cdot (\bar{\rho}_0 \mathbf{v}(\mathbf{x}, t)) = 0, \quad (13.22)$$

and take the time derivative,

$$\frac{\partial^2 \rho(\mathbf{x}, t)}{\partial t^2} + \bar{\rho}_0 \frac{\partial}{\partial t} (\nabla \cdot \mathbf{v}(\mathbf{x}, t)) = 0. \quad (13.23)$$

Use the continuous second partial derivative assumption of (13.17) to rewrite the foregoing equation as

$$\frac{\partial^2 \rho(\mathbf{x}, t)}{\partial t^2} = -\bar{\rho}_0 \nabla \cdot \left(\frac{\partial \mathbf{v}(\mathbf{x}, t)}{\partial t} \right). \quad (13.24)$$

Finally, substitute the foregoing equation into (13.21) gives

$$\frac{\partial^2 \rho(\mathbf{x}, t)}{\partial t^2} = c^2 \nabla \cdot \nabla \rho(\mathbf{x}, t) = c^2 \Delta \rho(\mathbf{x}, t), \quad (13.25)$$

which is the wave equation. ■

13.3 Incompressible, Viscid Fluid (Newtonian Fluid)

13.4 Compressible, Viscid Fluid

Chapter 14

Elasticity

14.1 Nonlinear Elastodynamics

The bulk modulus K , a measure of local compressibility of a substance, is defined as the ratio of infinitesimal change in pressure dP caused by an infinitesimal change in volume dV , at a given volume V

$$K \triangleq -V \frac{dP}{dV}. \quad (14.1)$$

We require $K > 0$, and since $V > 0$, then $dP/dV < 0$. This results in behaviors observed physically, namely that, all else being equal,

- increase in pressure causes decrease in volume, and vice versa, $\uparrow P \iff \downarrow V$, and
- increase in volume causes decrease in pressure, and vice versa, $\uparrow V \iff \downarrow P$.

Using the definition of density $\rho \triangleq m/V$,

$$\frac{d\rho}{dV} = \frac{-m}{V^2}, \quad (14.2)$$

and

$$\frac{dP}{dV} = \frac{dP}{d\rho} \frac{d\rho}{dV}, \quad (14.3)$$

the bulk modulus K may also be written as

$$K = \rho \frac{dP}{d\rho}. \quad (14.4)$$

Since $K > 0$ and $\rho > 0$, then $dP/d\rho > 0$. This result physically means that, all else being equal,

- increases in pressure cause increases in density (and vice versa), and
- increases in density cause increases in pressure (and vice versa).

14.2 Linear Elastodynamics

Linear elastodynamics begins with nonlinear elastodynamics, and adds progressively simplifying (and thus, restrictive) assumptions. The first assumption will be regarding linearity of displacement gradients.

- **Assumption 1:** Second-order and higher displacement gradients are negligible,

which restricts the stress-response function to be a special case of the fully general stress response function

$$\mathbf{P} = \mathbf{P}(\mathbf{F}). \quad (14.5)$$

Rewriting the foregoing stress-response function in terms of the displacement gradient,

$$\mathbf{P}(\mathbf{F}) = \mathbf{P}(\mathbf{1} + \nabla_0 \mathbf{u}), \quad (14.6)$$

gives an expression that can be cast in terms of a Taylor series expansion,

$$f(x_0 + \Delta x) = f(x_0) + f'(x_0)\Delta x + \frac{f''(x_0)}{2!}(\Delta x)^2 + \text{H.O.T.}, \quad (14.7)$$

$$= \sum_{j=0}^{\infty} \frac{f^{(j)}(x_0)}{j!} (\Delta x)^j. \quad (14.8)$$

Then

$$\mathbf{P}(\mathbf{1} + \nabla_0 \mathbf{u}) = \mathbf{P}(\mathbf{1}) + \frac{\partial \mathbf{P}(\mathbf{1})}{\partial \mathbf{F}} \nabla_0 \mathbf{u} + \mathcal{O}[(\nabla_0 \mathbf{u})^2]. \quad (14.9)$$

Next, we make another assumption,

- **Assumption 2:** The initial state is stress free, thus $\mathbf{P}(\mathbf{1}) = \mathbf{0}$.

The fourth-order elasticity tensor $\mathbb{C}$ is defined as,

$$\mathbb{C} \triangleq \frac{\partial \mathbf{P}(\mathbf{1})}{\partial \mathbf{F}}. \quad (14.10)$$

Symmetry of the elasticity tensor provides

$$\mathbb{C} : \nabla_0 \mathbf{u} = \mathbb{C} : \text{SYM}(\nabla_0 \mathbf{u}). \quad (14.11)$$

Finally,

$$\text{SYM}(\nabla_0 \mathbf{u}) = \frac{1}{2}(\nabla_0 \mathbf{u} + \nabla_0 \mathbf{u}^T) = \boldsymbol{\epsilon}. \quad (14.12)$$

These results give

$$\mathbf{P}(\mathbf{1} + \nabla_0 \mathbf{u}) = \mathbf{0} + \mathbb{C} : \boldsymbol{\epsilon} + \text{H.O.T.}, \quad (14.13)$$

or simply

$$\mathbf{P}(\mathbf{1} + \nabla_0 \mathbf{u}) \stackrel{(\text{LIN})}{=} \mathbf{P}_{\text{LIN}}(\boldsymbol{\epsilon}) = \mathbb{C} : \boldsymbol{\epsilon}. \quad (14.14)$$

Next, we can make another limiting assumption,

- **Assumption 3:** The material is **isotropic**,

which allows the stress-response function to be written in terms of two Lamé parameters λ and μ as

$$\mathbf{P}_{\text{LIN}}(\boldsymbol{\epsilon}) = \lambda \operatorname{tr}(\boldsymbol{\epsilon}) \mathbf{1} + 2 \mu \boldsymbol{\epsilon}. \quad (14.15)$$

Next, we make another assumption,

- **Assumption 4:** The material is **homogeneous**,

thus ρ_0 , λ , and μ are constants with respect to the position $\mathbf{X}$ in the body.

In the conservation of momentum equation, the divergence of the stress tensor appears. Thus, we find

$$\nabla_0 \cdot \mathbf{P}_{\text{LIN}}(\boldsymbol{\epsilon}) = \nabla_0 \cdot (\lambda \operatorname{tr}(\boldsymbol{\epsilon}) \mathbf{1} + 2 \mu \boldsymbol{\epsilon}) \quad (14.16)$$

$$= \lambda \nabla_0 (\nabla_0 \cdot \mathbf{u}) + \mu [\nabla_0 \cdot (\nabla_0 \mathbf{u}) + \nabla_0 (\nabla_0 \cdot \mathbf{u})], \quad (14.17)$$

$$= (\lambda + \mu) \nabla_0 (\nabla_0 \cdot \mathbf{u}) + \mu \nabla_0 \cdot (\nabla_0 \mathbf{u}). \quad (14.18)$$

Now, since

$$\nabla_0 \cdot (\nabla_0 \mathbf{u}) = \nabla_0 (\nabla_0 \cdot \mathbf{u}) - (\nabla_0 \times \nabla_0 \times \mathbf{u}), \quad (14.19)$$

we can write

$$\nabla_0 \cdot \mathbf{P}_{\text{LIN}}(\boldsymbol{\epsilon}) = \underbrace{(\lambda + 2\mu) \nabla_0 (\nabla_0 \cdot \mathbf{u})}_{\text{dilatational}} - \underbrace{\mu (\nabla_0 \times \nabla_0 \times \mathbf{u})}_{\text{rotational}} \quad (14.20)$$

Finally, we make another assumption

- **Assumption 5:** The **body force** is **negligible**, thus $\mathbf{b}_0 = \mathbf{0}$.

This final assumption allows the **progressive wave equation for solids** to be written as

$$\rho_0 \ddot{\mathbf{u}} = \underbrace{(\lambda + 2\mu) \nabla_0 (\nabla_0 \cdot \mathbf{u})}_{\text{dilatational}} - \underbrace{\mu (\nabla_0 \times \nabla_0 \times \mathbf{u})}_{\text{rotational}} \quad (14.21)$$

There are two wave speeds: the longitudinal sound speed c_L and the transverse sound speed c_T .

$$c_L = \sqrt{\frac{\lambda + 2\mu}{\rho_0}} \quad (\text{longitudinal or P-wave}), \quad (14.22)$$

$$c_T = \sqrt{\frac{\mu}{\rho_0}} \quad (\text{transverse or S-wave}). \quad (14.23)$$

Remark 14.2.11. The longitudinal sound speed is faster than the transverse sound speed.

The bulk modulus K is related to the Lamé constants λ and μ through

$$K = \lambda + \frac{2\mu}{3}. \quad (14.24)$$

Alternatively, the bulk modulus K in terms of E and ν is given as

$$K = \frac{E}{2(1-2\nu)}. \quad (14.25)$$

Substituting for K in the longitudinal sound speed gives

$$c_L = \sqrt{\frac{K + \frac{4}{3}\mu}{\rho_0}} \quad (\text{longitudinal or P-wave}). \quad (14.26)$$

Remark 14.2.12. Often, the Lamé constant μ will be denoted G , which is the shear modulus. Preference for μ or G is a matter of taste.

Chapter 15

Materials

15.1 Linear Elasticity

15.2 Finite Elasticity

15.2.1 Kirchhoff-St. Venant

15.2.2 Rivlin

Let I_1, I_2, I_3 be the three invariants of the two Cauchy-Green deformation tensors. The invariants are defined in terms of principal stretch ratios $\lambda_1, \lambda_2, \lambda_3$,

$$I_1 = \lambda_1^2 + \lambda_2^2 + \lambda_3^2, \quad (15.1)$$

$$I_2 = \lambda_1^2 \lambda_2^2 + \lambda_2^2 \lambda_3^2 + \lambda_3^2 \lambda_1^2, \quad (15.2)$$

$$I_3 = \lambda_1^2 \lambda_2^2 \lambda_3^2 = J^2. \quad (15.3)$$

Rivlin¹ specified a strain energy density function W as power series of the invariants,

$$W = \sum_{i+j+k=1}^{\infty} C_{ijk} (I_1 - 3)^i (I_2 - 3)^j (I_3 - 1)^k. \quad (15.4)$$

For **incompressible** materials, $I_3 = 1$, and the Rivlin equation reduces to

$$W = \sum_{i+j=1}^{\infty} C_{ij} (I_1 - 3)^i (I_2 - 3)^j. \quad (15.5)$$

Remark 15.2.13. The presence of the $-3, -3$, and -1 terms in parenthesis allow for zero strain energy density when there is zero stretch.

Remark 15.2.14. When a material's bulk modulus K is at least two orders of magnitude larger than the shear modulus G , the incompressibility modeling assumption may be appropriate.

Incompressible Rivlin

Using on the first two terms of the incompressible Rivlin gives the **Mooney-Rivlin** strain energy density function:

$$W = C_{10}(I_1 - 3) + C_{01}(I_2 - 3). \quad (15.6)$$

Incompressible Neo-Hookean

Using only the first term of the Mooney-Rivlin equation gives the **Incompressible Neo-Hookean** strain energy density function:

$$W = C_{10}(I_1 - 3). \quad (15.7)$$

¹Rivlin RS. Solution of Some Problems in the Exact Theory of Visco-Elasticity, J Rational Mechanics and Analysis, 5:1 (1956), 179-188.

The invariants in terms of the right Cauchy-Green tensor $\mathbf{C}$ and left Cauchy-Green tensor $\mathbf{b}$ are

$$I_1 = \text{tr}(\mathbf{C}) = C_{KK} = \text{tr}(\mathbf{b}) = b_{kk}, \quad (15.8)$$

$$\begin{aligned} I_2 &= \frac{1}{2} (I_1^2 - \mathbf{C} : \mathbf{C}) = \frac{1}{2} (I_1^2 - C_{IK} C_{KI}) \\ &= \frac{1}{2} (I_1^2 - \mathbf{b} : \mathbf{b}) = \frac{1}{2} (I_1^2 - b_{ik} b_{ki}), \end{aligned} \quad (15.9)$$

$$I_3 = \det(\mathbf{C}) = \det C_{IJ} = \det(\mathbf{b}) = \det b_{ij} = J^2. \quad (15.10)$$

For small strains and incompressible Mooney-Rivlin:

$$E = 6(C_{10} + C_{01}), \quad (15.11)$$

$$G = 2(C_{10} + C_{01}). \quad (15.12)$$

For small strains and incompressible Neo-Hookean:

$$C_{01} = 0, \quad (15.13)$$

$$C_{10} = G/2, \implies E = 3G. \quad (15.14)$$

Compressible Rivlin

Next, allow the material to be compressible, characterized through bulk modulus K . Thus, $I_3 \neq 1$. The **Mooney-Rivlin equation** for compressible materials is given by

$$W = W_{\text{VOL}}(I_3) + C_{10}(I_1 - 3) + C_{01}(I_2 - 3), \quad (15.15)$$

$$= W_{\text{VOL}}(I_3) + C_{10}(I_1 - 3) + C_{01}(I_2 - 3), \quad (15.16)$$

where W_{VOL} is the hydrostatic (volumetric) term.

To fully uncouple the volumetric from the deviatoric responses, let the **volume preserving (isochoric) distortion invariants** be calculated as

$$\bar{I}_1 = \frac{I_1}{I_3^{(1/3)}} = \frac{I_1}{J^{(2/3)}}, \quad (15.17)$$

$$\bar{I}_2 = \frac{I_2}{I_3^{(2/3)}} = \frac{I_2}{J^{(4/3)}}, \quad (15.18)$$

$$J = \sqrt{I_3}. \quad (15.19)$$

Then, for **compressible Rivlin**:

$$W = \underbrace{\frac{K}{2}(J - 1)^2}_{\text{VOL}} + \underbrace{C_{10}(\bar{I}_1 - 3) + C_{01}(\bar{I}_2 - 3)}_{\text{DEV}}. \quad (15.20)$$

Finally, for **compressible Neo-Hookean**:

$$W = \underbrace{\frac{K}{2}(J-1)^2}_{\text{VOL}} + \underbrace{C_{10}(\bar{I}_1 - 3)}_{\text{DEV}}. \quad (15.21)$$

In the finite compression limit, as the volume goes to zero ($J \rightarrow 0$), the factor $(J-1)^2$ in the volumetric term tends toward unity. This behavior—finite strain energy density in the limit of compression—is not desirable. Instead, as the volume goes toward zero, the strain energy density so go toward $+\infty$. Simo and Hughes² provided an alternative form of the compressible Neo-Hookean strain energy density that exhibits this desired response,

$$W = \underbrace{\frac{K}{2} \left[\frac{J^2 - 1}{2} - \ln J \right]}_{\text{VOL}} + \underbrace{\frac{G}{2}(\bar{I}_1 - 3)}_{\text{DEV}}. \quad (15.22)$$

Figure 15.1 shows the volumetric term for the infinitesimal and finite cases, with the relative contribution of the two terms that add to compose the finite case.

Remark 15.2.15. The Simo and Hughes notation, while slightly different from above, is equivalent. For the deviatoric term, since

$$\bar{\mathbf{F}} = \frac{\mathbf{F}}{J^{(1/3)}}, \quad (15.23)$$

and

$$\bar{\mathbf{C}} = \bar{\mathbf{F}}^T \bar{\mathbf{F}} = \frac{\mathbf{F}^T}{J^{(1/3)}} \frac{\mathbf{F}}{J^{(1/3)}} = \frac{\mathbf{F}^T \mathbf{F}}{J^{(2/3)}} = \frac{\mathbf{C}}{J^{(2/3)}}, \quad (15.24)$$

then,

$$\text{tr}(\bar{\mathbf{C}}) = \text{tr} \left(\frac{\mathbf{C}}{J^{(2/3)}} \right) = J^{(-2/3)} \text{tr}(\mathbf{C}) = \frac{I_1}{J^{(2/3)}} = \bar{I}_1, \quad (15.25)$$

which harmonizes the notation used here with the deviatoric term from Simo and Hughes.

²Simo JC, Hughes TJR. Computational inelasticity (Vol. 7). Springer Science & Business Media, 2006, page 361, Eq. (10.4.19).

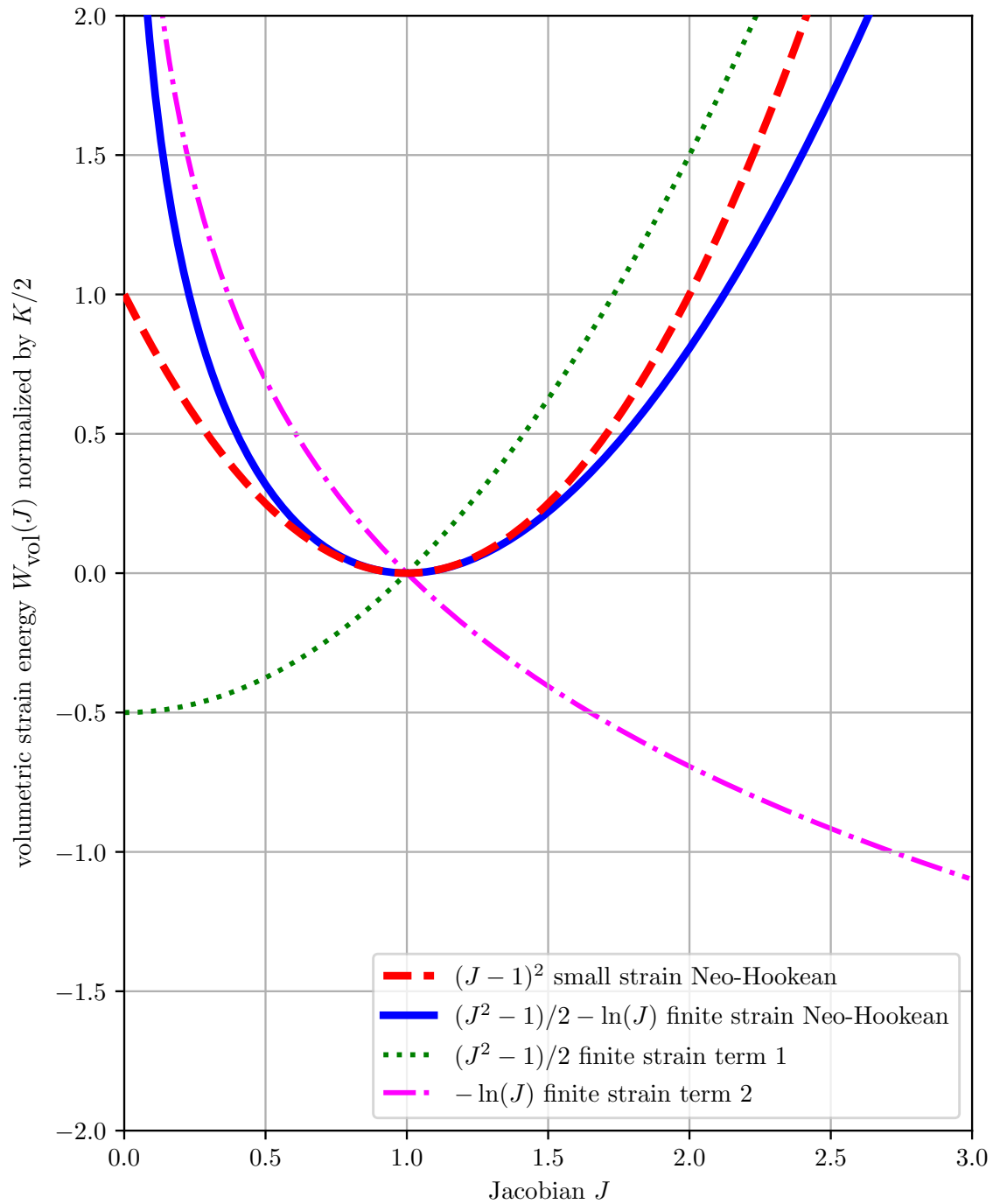


Figure 15.1: Neo-Hookean volumetric strain energy density $W_{\text{VOL}}(J)$ normalized by $K/2$ for small strain and finite strain cases.

Remark 15.2.16. Consider the relative effects of the $(J^2 - 1)/2$ term versus the $-\ln(J)$ term in (15.22):

- For finite compression, the $-\ln(J)$ dominates the $(J^2 - 1)/2$ term, causing the strain energy function to go toward $+\infty$, as desired.
- For finite extension, the $(J^2 - 1)/2$ term dominates the $-\ln(J)$ term, causing the strain energy function to go toward $+\infty$, as desired.
- The addition of the two terms evaluated at $J = 1$ gives $W = 0$, as desired. For zero volumetric deformation, there should be zero strain energy density.
- The slope of the strain energy density function, $dW/dJ = J - 1/J$, evaluated at $J = 1$ is zero. For zero volumetric deformation, there should be zero initial stress.
- The second derivative of the strain energy density function, $d^2W/dJ^2 = (1 + 1/J^2)$ is positive for all $J \in (0, +\infty)$. Thus, the volumetric strain energy density function W_{VOL} is convex.

Chapter 16

Plasticity

We first focus on plasticity in one dimension, as opposed to fully three-dimensional plasticity, to emphasize concepts while being as mathematically simple as possible. After the concepts are clear, the extension of the mathematics from one- to three-dimensions can be made naturally.

16.1 Perfect Plasticity

The extension of elasticity to plasticity is made by way of *perfect* plasticity. Conceptually, perfect plasticity in the constitutive stress-strain relationship is obtained by “capping” the elastic response function at some yield stress (to be defined later as σ_y). Beyond the yield stress, the stress response function is perfectly horizontal. Increases in strain produce no increase in stress beyond the yield stress.

16.1.1 Deformation Measures

Let the body Ω have strain measure $\varepsilon : \Omega \rightarrow \mathbb{R}$ in one dimension be defined as a ratio of change in length, that is the difference between the new (deformed, current configuration) length ℓ of body Ω and the original (undeformed, reference configuration) length L , divided by the original length,

$$\varepsilon \triangleq \frac{\ell - L}{L}. \quad (16.1)$$

The total strain ε in the body be defined as an additive decomposition of elastic strain ε^e and plastic strain ε^p ,

$$\varepsilon \triangleq \varepsilon^e + \varepsilon^p. \quad (16.2)$$

16.1.2 Constitutive Law

Then, the traditional stress-strain relationship, $\sigma = E\varepsilon^e$, can be rewritten as

$$\sigma = E(\varepsilon - \varepsilon^p). \quad (16.3)$$

A change in the configuration of the plastic element (see Fig. 16.1) can only occur if the notion of **slip rate** is allowed.

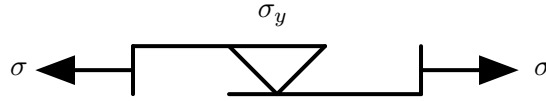


Figure 16.1: Phenomenological model of a friction element. A unit cross-sectional area is assumed. Motion in the friction element occurs only if the friction force, σ_y , is exceeded. The motion, in rate form, is referred to as the slip rate. Note that we employ a slight abuse of notation by using σ to describe a force F , since the cross-sectional area A is unity, $F = \sigma A \implies \sigma \cdot 1 = \sigma$.

That is, *assume* that $\varepsilon^p(t)$ is a function of time on interval $[0, T] \subset \mathbb{R}$. Then from (16.2) and (16.3), time evolution expressions $\varepsilon(t)$ and $\sigma(t)$ exist, respectively. Finally, the presence of evolution equations allows the constitutive law in (16.3) to be written as

$$\dot{\sigma} = E(\dot{\varepsilon} - \dot{\varepsilon}^p). \quad (16.4)$$

This form of the constitutive equation will become useful for quantifying plastic flow, as seen below in (16.13).

16.1.3 Yield Function

The stress σ in the frictional device can never greater in absolute value than the yield stress (also known as flow stress) $\sigma_y \in \mathbb{R}^+$. The **yield function** Φ is then defined as

$$\Phi(\sigma) \triangleq |\sigma| - \sigma_y \leq 0. \quad (16.5)$$

For stress levels within the elastic domain, only elastic straining can occur, thus

$$\text{If } \Phi(\sigma) < 0 \Rightarrow \dot{\varepsilon}^p = 0. \quad (16.6)$$

For stress levels on the boundary of the elastic domain (at the yield stress), either plastic strain via plastic loading or non-plastic strain via elastic unloading occurs.

$$\begin{aligned} \text{If } \Phi(\sigma) = 0 \\ \Rightarrow \begin{cases} \dot{\varepsilon}^P \neq 0 & \text{for plastic loading,} \\ \dot{\varepsilon}^P = 0 & \text{for elastic unloading.} \end{cases} \end{aligned} \quad (16.7)$$

16.1.4 Plastic Flow Rule, Loading/Unloading Conditions

We next define how, on the yield surface, the plastic strain evolves through time, which is called the **plastic flow rule**. A reasonable first approximation for the plastic strain evolution is to posit

$$\dot{\varepsilon}^P \triangleq \dot{\gamma} \text{sign}(\sigma), \quad (16.8)$$

where $\dot{\gamma} \geq 0$ is the **plastic multiplier** or absolute value of the plastic slip rate, and where $\text{sign} : \mathbb{R} \rightarrow \{-1, +1\}$ is the *signum* function defined as

$$\text{sign}(a) = \begin{cases} +1 & \text{if } a \geq 0, \\ -1 & \text{if } a < 0. \end{cases} \quad (16.9)$$

Both σ and $\dot{\gamma}$ are restricted by **unilateral constraints**,

$$\begin{aligned} \Phi(\sigma) &\leq 0 \\ \dot{\gamma} &\geq 0 \end{aligned} \quad (16.10)$$

Now if $\Phi(\sigma) < 0$, then $\dot{\gamma} = 0$. Similarly, if $\dot{\gamma} > 0$, then $\Phi(\sigma) = 0$. These two observations require

$$\dot{\gamma} \Phi(\sigma) = 0, \quad (16.11)$$

which is referred to as the **complementarity condition**, and when taken with the unilateral constraints, describes the so-called **Kuhn-Tucker conditions**. The unilateral constraints and the complementarity condition are illustrated in Fig. 16.2.

Together, the unilateral constraints and complementarity condition create the normal plasticity law, shown in Fig. 16.3.

Next, consider any time interval $[t, t + \Delta t]$ during which *plastic flow occurs*, the yield function remains constant, $\Phi(\sigma) = 0$. Similarly, if during this time interval, we have elastic response only, and $\dot{\Phi}(\sigma) < 0$, then necessarily $\dot{\gamma} = 0$. These relationships create the **persistency (or consistency) condition**

$$\left. \begin{aligned} \text{If } \dot{\gamma} > 0 &\Rightarrow \dot{\Phi} = 0, \\ \text{If } \dot{\Phi} < 0 &\Rightarrow \dot{\gamma} = 0, \end{aligned} \right\} \Rightarrow \dot{\gamma} \dot{\Phi} = 0. \quad (16.12)$$

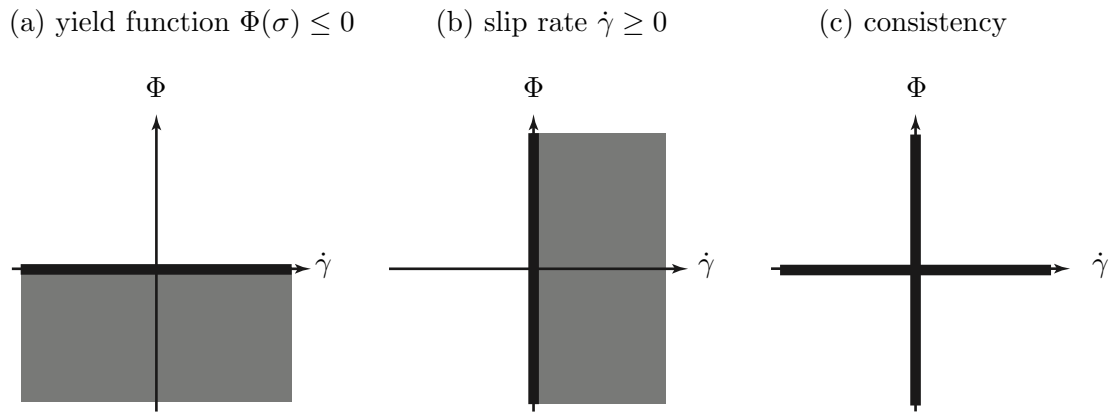


Figure 16.2: Unilateral conditions (a) $\Phi(\sigma) \leq 0$ and (b) slip rate $\dot{\gamma} \geq 0$. The complementarity condition, $\dot{\gamma} \Phi(\sigma) = 0$, is shown in (c).

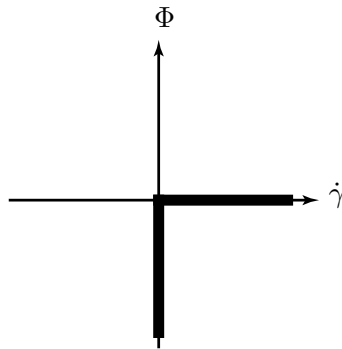


Figure 16.3: The normal plastic law described as the intersection of the unilateral constraints and the complementarity condition.

16.1.5 Determination of Plastic Multiplier

As of yet, the value of the plastic multiplier $\dot{\gamma}$ has remained undetermined. For non-trivial $\dot{\gamma}$, we know $\dot{\Phi} = 0$ by the persistency condition. An explicit form for $\dot{\Phi}$ may be found using the chain rule with (16.4) and (16.8),

$$\begin{aligned}
 \frac{\partial \Phi}{\partial t} &= \frac{\partial \Phi}{\partial \sigma} \frac{\partial \sigma}{\partial t} \\
 &= \frac{\partial}{\partial \sigma} (|\sigma| - \sigma_y) E (\dot{\epsilon} - \dot{\epsilon}^P) \\
 &= \text{sign}(\sigma) E (\dot{\epsilon} - \dot{\gamma} \text{sign}(\sigma)),
 \end{aligned} \tag{16.13}$$

or, compactly

$$\dot{\Phi} = \text{sign}(\sigma)E\dot{\varepsilon} - E\dot{\gamma} \stackrel{\text{set}}{=} 0, \quad (16.14)$$

gives

$$\dot{\gamma} = \dot{\varepsilon} \text{sign}(\sigma) \quad (16.15)$$

which, when compared with (16.8), shows that for perfect plasticity during plastic flow,

$$\varepsilon = \varepsilon^P. \quad (16.16)$$

The results from (16.15) and (16.16) are remarkable for the following reasons: During plastic flow,

- None of the strain ε is *elastic* strain ε^e ; 100% of the strain ε is *plastic* strain ε^P . This feature will not be true, in general, for more sophisticated yield functions (for example, that have plastic hardening).
- The plastic flow parameter $\dot{\gamma}$ is equal to the strain ε times the sign of the stress. For more sophisticated yield functions, the plastic flow parameter will not be identically equal to the signed strain, but will be related thereto via factors of the elastic and hardening parameters, as shown subsequent developments and in particular in (16.23).

16.2 Hardening Plasticity

Hardening plasticity is considered an extension of perfect plasticity. With perfect plasticity, the slope of stress-strain relationship at yielding is identically zero. With hardening plasticity, the slope becomes positive.¹

16.2.1 Remarks on Loss of Stability

Perfect plasticity allows for the notion of unbounded strains when stress is at the yield stress. The slope of the stress-strain curve for strains beyond the elastic strain region is identically zero. This zero slope causes numerical problems for the consistent tangent, which is inverted and multiplied by the external load to obtain displacements. Three strategies to circumvent this zero tangent are mentioned here: arc-length, displacement control, and dynamic formulation.

The most notable of the continuation methods is called the **arc-length method**, which is a non-trivial area of study within computational mechanics. Arc-length methods

¹For completeness, we note that softening plasticity occurs when the slope becomes negative.

will be used eventually, but their explication is delayed for now, in favor development of hardening plasticity.

Despite being more slightly more complex than perfect plasticity, hardening plasticity does not likewise suffer from loss of ellipticity and therefore is actually easier to solve since algorithmic enhancements, such as arc-length, are (usually²) unnecessary.

Next, algorithmic formulations that utilize **displacement control** instead of force control can avoid loss of ellipticity if the displacement boundary conditions reduce the number of equations in the tangent to provide rank sufficiency. However, displacement control is less-favored than force control since we often know imposed forces and seek to discover the displacement response. The setting of known inhomogeneous displacements and unknown forces is less frequently encountered in practice. On the topic of load control versus displacement control, de Borst *et al.*,³ offer similar insights and additional context.

Finally, elastoplastic **dynamics**, and so-called dynamic relaxation as a subset thereof, are additional strategies to circumvent the above-mentioned problem as these approaches condition the consistent tangent with mass matrices. The benefit of mass matrices comes at the cost of increased expense to compute and store the same, and to incorporate a time stepping algorithm, such as the Newmark methods.

16.2.2 Yield Function

The departure of hardening plasticity from perfect plasticity begins with the yield function. The **yield function** from perfect plasticity in (16.5) is now modified to include a **linear hardening modulus** $H \in \mathbb{R}$ and the **internal hardening variable** $\alpha : [0, T] \rightarrow \mathbb{R}$,

$$\Phi(\sigma, \varepsilon^p) \triangleq |\sigma| - (\sigma_y + H\alpha) \leq 0. \quad (16.17)$$

for strain-hardening plasticity. The variable $\alpha : [0, T] \rightarrow \mathbb{R}$ is called the **internal hardening variable** and is a nonnegative function of the amount of plastic flow (slip). The simplest evolution for α can be written as

$$\dot{\alpha} = |\dot{\varepsilon}^p|. \quad (16.18)$$

The **plastic flow rule**,

$$\dot{\varepsilon}^p = \dot{\gamma} \text{sign}(\sigma), \quad (16.19)$$

²If the hardening is “soft” enough that the hardening approximates perfect plasticity, hardening plasticity can result in lack of conditioning. Usually, however, hardening is sufficiently stiff and therefore loss of ellipticity is not a concern.

³De Borst, R., Crisfield, M. A., Remmers, J. J., & Verhoosel, C. V. (2012). Nonlinear finite element analysis of solids and structures. John Wiley & Sons, at page 50–51.

remains unchanged. The **Kuhn-Tucker conditions** are rewritten to include dependence on α ,

$$\dot{\gamma} \geq 0 \quad \Phi(\sigma, \alpha) \leq 0 \quad \dot{\gamma} \Phi(\sigma, \alpha) = 0, \quad (16.20)$$

and the **persistence condition** is written as

$$\dot{\gamma} \dot{\Phi}(\sigma, \alpha) = 0. \quad (16.21)$$

16.2.3 Determination of Plastic Multiplier

Like perfect plasticity, strain-hardening plasticity uses the time derivative of the yield function to generate an expression for the plastic multiplier $\dot{\gamma}$, which ultimately leads to a relationship between stress rates and strain rates, and the elasto-plastic tangent.

With the yield function in (16.17),

$$\begin{aligned} \frac{\partial \Phi}{\partial t} &= \frac{\partial \Phi}{\partial \sigma} \frac{\partial \sigma}{\partial t} + \frac{\partial \Phi}{\partial \alpha} \frac{\partial \alpha}{\partial t}, \\ &= \text{sign}(\sigma) \dot{\sigma} - H \dot{\alpha}, \\ &= \text{sign}(\sigma) E (\dot{\epsilon} - \dot{\epsilon}^P) - H |\dot{\epsilon}^P|, \\ &= \text{sign}(\sigma) E \dot{\epsilon} - E \dot{\gamma} - H \dot{\gamma} \stackrel{\text{set}}{=} 0, \end{aligned} \quad (16.22)$$

gives

$$\dot{\gamma} = \dot{\epsilon} \text{sign}(\sigma) \frac{E}{E + H}. \quad (16.23)$$

The ratio $\frac{E}{E+H}$ in (16.23) may be conceived as the apportionment of total strain (rate) to plastic slip:

- If $H = 0$, then all strain is plastic slip, and the perfect plasticity results obtained in previous sections are obtained as a special case.
- If $H = E$, then half of all strain goes into plastic slip.
- If $H \gg E$, then very little $\dot{\epsilon}$ goes into plastic slip.

16.2.4 Elasto-Plastic Tangent

The elasto-plastic tangent is found by observing $\dot{\sigma} = \frac{d\sigma}{d\epsilon} \dot{\epsilon}$. Starting with the elastic stress-strain relationship $\sigma = E(\epsilon - \epsilon^P)$, in the rate form and substituting for $\dot{\epsilon}$ from (16.23) and for $\dot{\epsilon}^P$ from the plastic flow rule (16.19) gives

$$\begin{aligned} \dot{\sigma} &= E (\dot{\epsilon} - \dot{\epsilon}^P), \\ &= \dot{\gamma} \text{sign}(\sigma) (E + H) - \dot{\gamma} \text{sign}(\sigma) E, \\ &= \dot{\gamma} \text{sign}(\sigma) H. \end{aligned} \quad (16.24)$$

Substituting for $\dot{\gamma}$ from (16.23) gives

$$\dot{\sigma} = \frac{EH}{E+H} \dot{\varepsilon}, \quad (16.25)$$

where E_{ep} , the elasto-plastic tangent, is given by

$$E_{\text{ep}} = \frac{EH}{E+H}. \quad (16.26)$$

The concept of the elasto-plastic tangent is readily understood with the phenomenological model shown in Fig. 16.4.

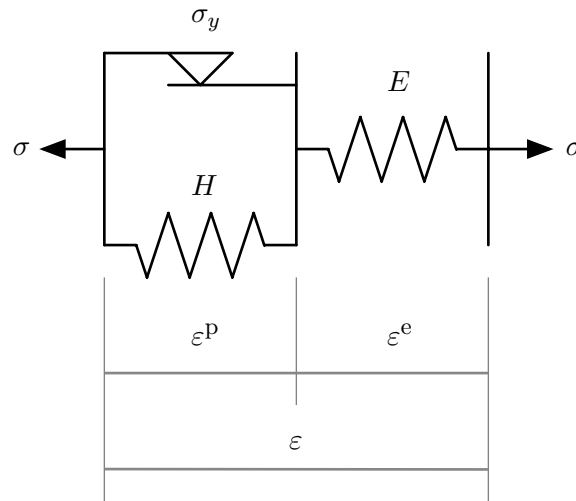


Figure 16.4: Phenomenological model of strain-hardening plasticity with applied stress σ , yield stress σ_y , hardening modulus H and elastic modulus E . The total strain, ε is the sum of plastic strain ε^p and elastic strain ε^e .

The stress σ is applied to the model. Initially, the applied stress σ creates only elastic strain ε^e since $\sigma \leq \sigma_y$. When $\sigma > \sigma_y$, strain composed of *both* elastic strain ε^e and plastic strain ε^p is created. The linear moduli E and H act as springs in series. The relationship for two springs with spring constants, k_1 and k_2 , in series immediately shows the relationship between the equivalent spring constant k_{eq} and the elasto-plastic tangent E_{ep}

$$\frac{1}{k_{\text{eq}}} = \frac{1}{k_1} + \frac{1}{k_2} \implies k_{\text{eq}} = \frac{k_1 k_2}{k_1 + k_2} \quad (16.27)$$

$$\begin{array}{ccc} \Updownarrow & & \Updownarrow \\ E_{\text{ep}} & = & \frac{EH}{E+H} \end{array} \quad (16.28)$$

The hardening modulus H is shown in Fig. 16.5(b) such that, during plastic flow, the incremental stress $\partial\sigma$ relative to the incremental plastic strain $\partial\varepsilon^P$ is given by

$$\frac{\partial\sigma}{\partial\varepsilon^P} = H. \quad (16.29)$$

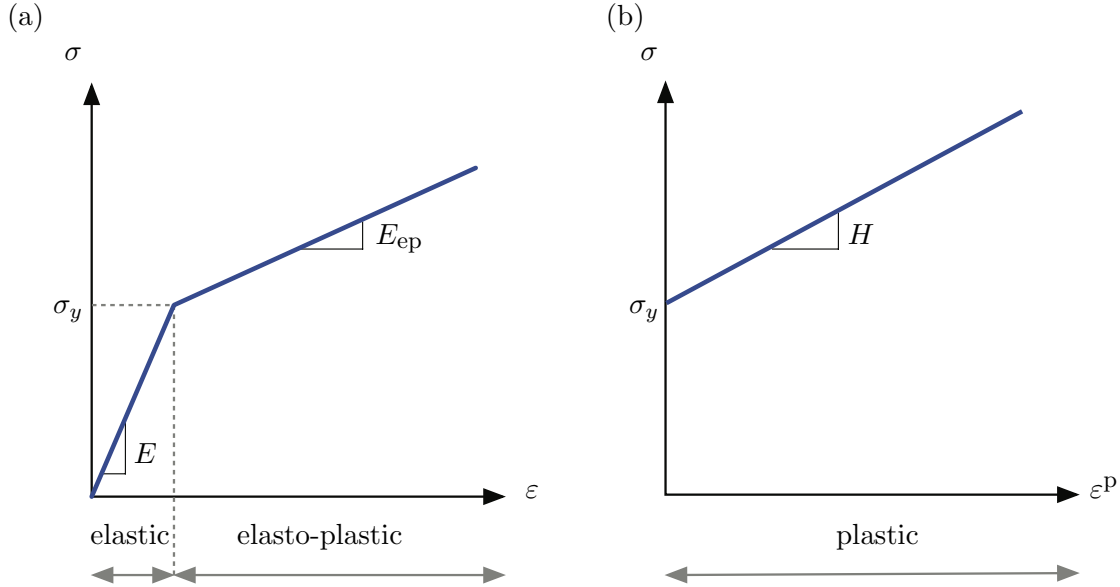


Figure 16.5: (a) Strain-hardening stress-strain relationship with elastic and elasto-plastic strain regions delineated, elastic modulus E , elasto-plastic modulus E_{ep} , and yield stress σ_y , and (b) stress-(plastic) strain relationship with hardening modulus H .

16.3 Elasto-Plastic 2D Truss Element

Consider the two-dimensional truss member shown in Fig. 16.6. The coordinates in the reference configuration are noted with $\mathbf{X}^1$ and $\mathbf{X}^2$ for nodes 1 and 2, respectively. These nodes are moved into the current configuration at $\mathbf{x}^1$ and $\mathbf{x}^2$ through nodal displacements $\mathbf{u}^1$ and $\mathbf{u}^2$.

The strain energy of the truss is the strain energy density integrated over the truss's volume,

$$\mathcal{W} = \int_{\Omega} \boldsymbol{\sigma} : \boldsymbol{\varepsilon} d\Omega. \quad (16.30)$$

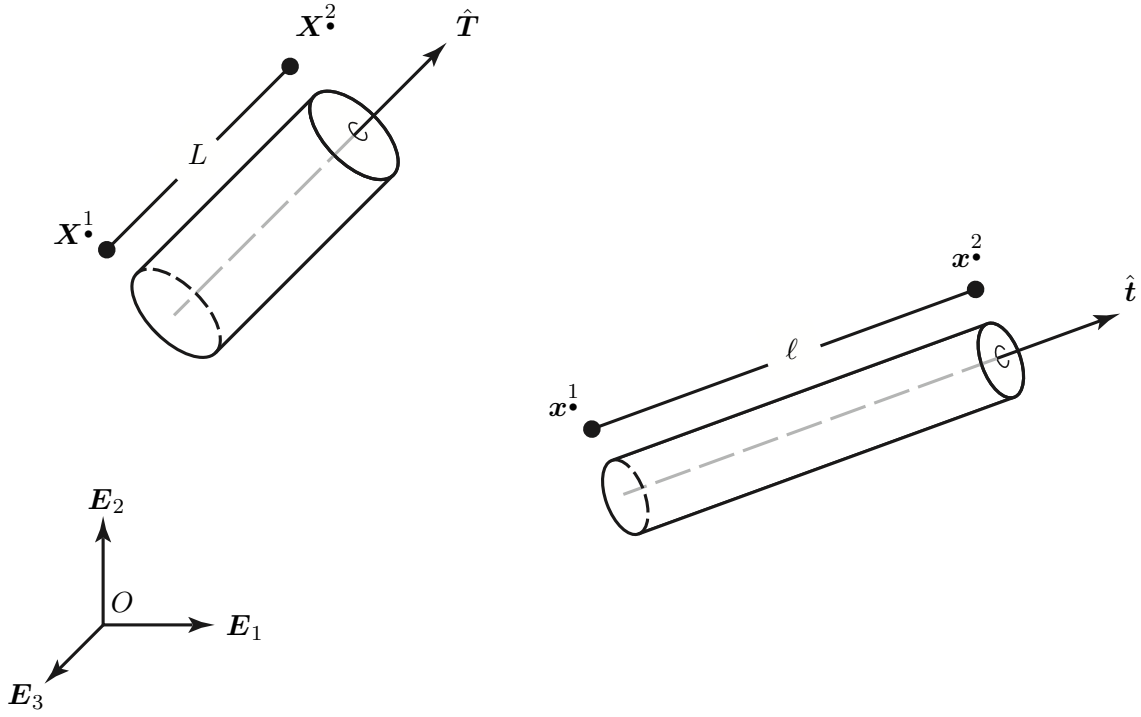


Figure 16.6: Nonlinear truss in reference and current configurations.

The virtual work done by the truss is

$$\delta \mathcal{W} = \int_{\Omega} \delta \varepsilon : \boldsymbol{\sigma} \, d\Omega. \quad (16.31)$$

The virtual work computed in the reference configuration is then

$$\delta \mathcal{W} = \int_L \delta \varepsilon : \boldsymbol{\sigma} A \, dL, \quad (16.32)$$

$$\stackrel{(1D)}{=} \delta \left(\frac{\ell - L}{L} \right) \sigma A L = \frac{\delta \ell}{L} \sigma A L = \delta \ell \sigma A, \quad (16.33)$$

$$\stackrel{(2.70)}{=} (-\delta \mathbf{u}^1 + \delta \mathbf{u}^2) \cdot \hat{\mathbf{t}} \, \sigma A, \quad (16.34)$$

$$\stackrel{(\text{lin})}{=} (-\delta \mathbf{u}^1 + \delta \mathbf{u}^2) \cdot \hat{\mathbf{t}} \, E \left(\frac{\ell - L}{L} \right) A, \quad (16.35)$$

$$= (-\delta \mathbf{u}^1 + \delta \mathbf{u}^2) \cdot \hat{\mathbf{t}} \, k(\ell - L), \quad (16.36)$$

where the result from Hovey⁴ page 45, Eq. (2.70), $\delta \ell = (-\delta \mathbf{u}^1 + \delta \mathbf{u}^2) \cdot \hat{\mathbf{t}}$, has been used, and agrees with Eq. (2.76) of the same manuscript. The virtual work expressed in terms of

⁴Hovey CB (2000). Stress and Motion in Diarthrodial Joints Using Coupled Nonlinear Elastodynamics, Rigid Body Dynamics, and Large-Slip Contact-Impact, Doctoral Thesis, Stanford University, Stanford, CA.

nodal displacements and the *nonlinear* internal force vector in (16.34) is then

$$\delta \mathcal{W} = \boldsymbol{\delta} \cdot \mathbf{f}^{\text{int}} = \left\{ \begin{array}{c} \delta \mathbf{u}^1 \\ \delta \mathbf{u}^2 \end{array} \right\}^T \sigma A \left\{ \begin{array}{c} -\hat{\mathbf{t}} \\ \hat{\mathbf{t}} \end{array} \right\}. \quad (16.37)$$

With $\delta \ell$ at hand, and noting that

$$\delta \hat{\mathbf{t}} = \delta \left(\frac{\mathbf{x}_2 - \mathbf{x}_1}{l} \right) = \delta \left((\mathbf{x}_2 - \mathbf{x}_1) l^{-1} \right), \quad (16.38)$$

$$= \frac{1}{l} [\mathbf{1} - \hat{\mathbf{t}} \otimes \hat{\mathbf{t}}] (-\delta \mathbf{u}_1 + \delta \mathbf{u}_2), \quad (16.39)$$

the second variation of the nonlinear form in (16.34), or equivalently in (16.37), is found through

$$\Delta(\delta \mathcal{W}) = \Delta \left[(-\delta \mathbf{u}^1 + \delta \mathbf{u}^2) \cdot \sigma A \hat{\mathbf{t}} \right], \quad (16.40)$$

$$\begin{aligned} &= (-\delta \mathbf{u}^1 + \delta \mathbf{u}^2) \cdot \Delta \sigma A \hat{\mathbf{t}} \\ &\quad + (-\delta \mathbf{u}^1 + \delta \mathbf{u}^2) \cdot \sigma A \Delta \hat{\mathbf{t}}. \end{aligned} \quad (16.41)$$

Now, adopting a compact notation where $\delta \mathbf{u} = (-\delta \mathbf{u}^1 + \delta \mathbf{u}^2)$ and $\Delta \mathbf{u} = (-\Delta \mathbf{u}^1 + \Delta \mathbf{u}^2)$,

$$\begin{aligned} \Delta(\delta \mathcal{W}) &= \delta \mathbf{u} \cdot \left\{ \frac{\partial \sigma}{\partial \varepsilon} \Delta \varepsilon A \hat{\mathbf{t}} + \sigma A \frac{1}{l} [\mathbf{1} - \hat{\mathbf{t}} \otimes \hat{\mathbf{t}}] \Delta \mathbf{u} \right\}, \end{aligned} \quad (16.42)$$

$$= \delta \mathbf{u} \cdot \left\{ \frac{\partial \sigma}{\partial \varepsilon} \left(\frac{\Delta \ell}{L} \right) A \hat{\mathbf{t}} + \frac{\sigma A}{l} [\mathbf{1} - \hat{\mathbf{t}} \otimes \hat{\mathbf{t}}] \Delta \mathbf{u} \right\}. \quad (16.43)$$

Expanding $\delta \mathbf{u}$ and $\Delta \mathbf{u}$ gives

$$\begin{aligned} \Delta(\delta \mathcal{W}) &= (-\delta \mathbf{u}^1 + \delta \mathbf{u}^2) \\ &\quad \cdot \left[\frac{\partial \sigma}{\partial \varepsilon} \frac{A}{L} \hat{\mathbf{t}} \otimes \hat{\mathbf{t}} + \frac{\sigma A}{l} [\mathbf{1} - \hat{\mathbf{t}} \otimes \hat{\mathbf{t}}] \right] \\ &\quad \left\{ (-\Delta \mathbf{u}^1 + \Delta \mathbf{u}^2) \right\}. \end{aligned} \quad (16.44)$$

This leads to the element stiffness equations

$$\Delta \delta \mathcal{W} =$$

$$\delta \cdot \mathbf{K} \Delta = \left\{ \begin{array}{c} \delta \mathbf{u}^1 \\ \delta \mathbf{u}^2 \end{array} \right\}^T \left[\begin{array}{cc} \boldsymbol{\kappa} & -\boldsymbol{\kappa} \\ -\boldsymbol{\kappa} & \boldsymbol{\kappa} \end{array} \right] \left\{ \begin{array}{c} \Delta \mathbf{u}^1 \\ \Delta \mathbf{u}^2 \end{array} \right\}, \quad (16.45)$$

where

$$\boldsymbol{\kappa} = \left[\frac{\partial \sigma}{\partial \varepsilon} \frac{A}{L} \hat{\mathbf{t}} \otimes \hat{\mathbf{t}} + \frac{\sigma A}{l} [\mathbf{1} - \hat{\mathbf{t}} \otimes \hat{\mathbf{t}}] \right]. \quad (16.46)$$

Note that the explicit form of $\partial\sigma/\partial\varepsilon$ will depend on the plastic multiplier $\dot{\gamma}$. If $\dot{\gamma} = 0$, then $\partial\sigma/\partial\varepsilon = E$, as shown in the elastic region of Fig. 16.5 (a). If $\dot{\gamma} > 0$, then $\partial\sigma/\partial\varepsilon = E_{\text{ep}}$, as shown in the elasto-plastic region of Fig. 16.5 (a).

16.4 From 1D to 3D

In one-dimensional plasticity, we spoke of the Cauchy stress scalar $\sigma \in \mathbb{R}$ with positive values describing tension (resulting in elongations) and compression (resulting in contractions). Now, we generalize this concept to speak of the Cauchy stress tensor $\boldsymbol{\sigma}$ as a collection of all the normal and shear Cauchy stress components present in three dimensions.

We decompose $\boldsymbol{\sigma}$ into volumetric and deviatoric components, denoted

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}_{\text{VOL}} + \boldsymbol{\sigma}_{\text{DEV}}. \quad (16.47)$$

Next, we *define* each of these two components. The **volumetric Cauchy stress tensor** is defined as

$$\boldsymbol{\sigma}_{\text{VOL}} \triangleq \frac{1}{3} \text{tr}(\boldsymbol{\sigma}) \mathbf{1} = \frac{1}{3} \sigma_{kk} \delta_{ij}. \quad (16.48)$$

It is convenient to define the mean stress, $\bar{\sigma}$, as

$$\bar{\sigma} \triangleq \frac{1}{3} (\sigma_{11} + \sigma_{22} + \sigma_{33}). \quad (16.49)$$

Then

$$\boldsymbol{\sigma}_{\text{VOL}} \triangleq \bar{\sigma} \mathbf{1} = \bar{\sigma} \delta_{ij}. \quad (16.50)$$

The **deviatoric Cauchy stress tensor**, $\mathbf{s}$, is defined as

$$\mathbf{s} \triangleq \boldsymbol{\sigma} - \boldsymbol{\sigma}_{\text{VOL}} = \boldsymbol{\sigma} - \bar{\sigma} \mathbf{1}. \quad (16.51)$$

It is often convenient to define **pressure** $P \in \mathbb{R}$ as

$$P \triangleq -\frac{1}{3} \text{tr}(\boldsymbol{\sigma}) \quad \begin{cases} P > 0 \text{ compression,} \\ P < 0 \text{ tension.} \end{cases} \quad (16.52)$$

Note the negative sign in the definition. The negative sign allows us to harmonize our concepts of (compressive, e.g., underwater) pressure as a positive, scalar with tension/compression (positive/negative) convention of the Cauchy stress tensor.

16.4.1 Uniaxial Tension Test

With these definitions at hand, we are now poised to demonstrate how the results of a uniaxial tension test at the plastic limit are used to generalize a theory of small-strain plasticity to three dimensions.

At yielding, the Cauchy stress tensor $\boldsymbol{\sigma}$ has the simple form

$$\boldsymbol{\sigma}|_y = \begin{bmatrix} \sigma_y & 0 & 0 \\ & 0 & 0 \\ & & 0 \end{bmatrix}. \quad (16.53)$$

Remark 16.4.17. Note that for symmetric matrices, we will avoid filling in the lower part of the matrix both for brevity, and to reinforce the idea of symmetry.

The volumetric part of the Cauchy stress tensor at yield is then

$$\boldsymbol{\sigma}_{\text{VOL}}|_y = \frac{\sigma_y}{3} \begin{bmatrix} 1 & 0 & 0 \\ & 1 & 0 \\ & & 1 \end{bmatrix}. \quad (16.54)$$

The deviatoric part of the Cauchy stress tensor at yield is then

$$\mathbf{s}|_y = \frac{3\sigma_y}{3} \begin{bmatrix} 1 & 0 & 0 \\ & 1 & 0 \\ & & 1 \end{bmatrix} - \frac{\sigma_y}{3} \begin{bmatrix} 1 & 0 & 0 \\ & 1 & 0 \\ & & 1 \end{bmatrix} = \frac{\sigma_y}{3} \begin{bmatrix} 2 & 0 & 0 \\ & -1 & 0 \\ & & -1 \end{bmatrix}. \quad (16.55)$$

So, we now have the form of the deviatoric Cauchy stress tensor $\mathbf{s}$ at yield. How can this result for a uniaxial tension test be generalized for three dimensions? The thought experiment is to envision $\mathbf{s}$ for *any and all* matrix forms, not just the form obtained in (16.55). Any and all matrix form for $\mathbf{s}$ will have one commonality: their matrix invariants.

The Cauchy Stress Tensor Invariants

The invariants of the Cauchy stress tensor, $\boldsymbol{\sigma}$, satisfy the characteristic equation (see (2.27) for details),

$$0 = -\lambda^3 + I_1\lambda^2 - I_2\lambda + I_3, \quad (16.56)$$

where the three invariants, I_1 , I_2 , and I_3 are

$$I_1 = \text{tr}(\boldsymbol{\sigma}), \quad (16.57)$$

$$I_2 = \frac{1}{2} [\text{tr}(\boldsymbol{\sigma})\text{tr}(\boldsymbol{\sigma}) - \boldsymbol{\sigma} : \boldsymbol{\sigma}] = \frac{1}{2} [I_1^2 - \boldsymbol{\sigma} : \boldsymbol{\sigma}], \quad (16.58)$$

$$I_3 = \det(\boldsymbol{\sigma}). \quad (16.59)$$

The Deviatoric Cauchy Stress Tensor Invariants

The invariants of the deviatoric Cauchy stress tensor, $\mathbf{s}$, satisfy the characteristic equation

$$0 = -\lambda^3 + J_1\lambda^2 - J_2\lambda + J_3, \quad (16.60)$$

where the three invariants, J_1 , J_2 , and J_3 are

$$J_1 = \text{tr}(\mathbf{s}) = 0, \quad (16.61)$$

$$J_2 = \frac{1}{2} [\text{tr}(\mathbf{s})\text{tr}(\mathbf{s}) - \mathbf{s} : \mathbf{s}] = \frac{1}{2} [J_1^2 - \mathbf{s} : \mathbf{s}] = -\frac{1}{2} \mathbf{s} : \mathbf{s}, \quad (16.62)$$

$$J_3 = \det(\mathbf{s}). \quad (16.63)$$

will have a first invariant, J_1 equal to zero since the trace of a deviatoric second order tensor is zero.

Then the first invariant of $\mathbf{s}$,

$$J_1(\mathbf{s}) \triangleq \text{tr}(\mathbf{s}) = s_{kk} = 0, \quad (16.64)$$

is uninteresting because it has zero value for all stress states of $\boldsymbol{\sigma}$. The second invariant of $\mathbf{s}$,

$$J_2(\mathbf{s}) \triangleq \frac{1}{2} (\text{tr}(\mathbf{s} \cdot \mathbf{s})) = \frac{1}{2} s_{ij} s_{ji} \quad (16.65)$$

is interesting because we arrive at a quadratic form of $\mathbf{s}$. Returning to the thought experiment, we can now envision of form of $\mathbf{s}$ by way of $J_2(\mathbf{s})$ as valid for all values Cauchy stress tensors $\boldsymbol{\sigma}$ and with plasticity occurring when $J_2(\mathbf{s})$ reaches $J_2(\mathbf{s}|_y)$. Said equivalently, we can imagine a yield function, as defined for scalars in (16.5), recast now for the entire stress tensor $\boldsymbol{\sigma}$ as

$$\Phi(\boldsymbol{\sigma}) \triangleq J_2(\mathbf{s}(\boldsymbol{\sigma})) - J_2(\mathbf{s}|_y) \leq 0. \quad (16.66)$$

Now

$$J_2(\mathbf{s}|_y) = \frac{1}{2} \frac{\sigma_y^2}{9} (4 + 1 + 1) = \frac{\sigma_y^2}{3}. \quad (16.67)$$

Thus

$$\Phi(\boldsymbol{\sigma}) \triangleq J_2(\mathbf{s}(\boldsymbol{\sigma})) - \frac{\sigma_y^2}{3} \leq 0. \quad (16.68)$$

Now, on the yield surface, $\Phi = 0$, which means

$$J_2(\mathbf{s}(\boldsymbol{\sigma}|_y)) = \frac{\sigma_y^2}{3} \implies \sqrt{3J_2(\mathbf{s}(\boldsymbol{\sigma}|_y))} = \sigma_y. \quad (16.69)$$

This gives rise to the form of the yield function of so-called J_2 **plasticity**,

$$\boxed{\Phi(\boldsymbol{\sigma}) \triangleq \sqrt{3J_2(\mathbf{s}(\boldsymbol{\sigma}))} - \sigma_y \leq 0.} \quad (16.70)$$

The quantity $\sqrt{3J_2}$ is often referred to as the **von Mises stress** or **equivalent tensile stress** and denoted as σ_{VM} . Thus, the yield function may be equivalently written as

$$\boxed{\Phi(\boldsymbol{\sigma}) \triangleq \sigma_{\text{VM}}(\boldsymbol{\sigma}) - \sigma_y \leq 0.} \quad (16.71)$$

Result 16.4.1 The squared von Mises stress can be written explicitly in terms of the Cauchy stress components as follows:

$$\sigma_{\text{VM}}^2 = 3J_2 = \frac{(\sigma_{11} - \sigma_{22})^2 + (\sigma_{22} - \sigma_{33})^2 + (\sigma_{33} - \sigma_{11})^2}{2} + 3(\sigma_{12}^2 + \sigma_{23}^2 + \sigma_{31}^2). \quad (16.72)$$

Proof.

Recall $J_2(\mathbf{s}) = \frac{1}{2} \mathbf{s} : \mathbf{s} = \frac{1}{2} \text{tr}(\mathbf{s} \cdot \mathbf{s})$. Also, $\mathbf{s} \triangleq \boldsymbol{\sigma} - \bar{\sigma} \mathbf{1}$, where $\bar{\sigma} \triangleq \frac{1}{3}(\sigma_{11} + \sigma_{22} + \sigma_{33})$. Then,

$$\sigma_{\text{VM}}^2 = 3J_2 \quad (16.73)$$

$$= \frac{3}{2} \text{tr} \left(\begin{bmatrix} (\sigma_{11} - \bar{\sigma}) & \sigma_{12} & \sigma_{13} \\ & (\sigma_{22} - \bar{\sigma}) & \sigma_{23} \\ \text{sym.} & & (\sigma_{33} - \bar{\sigma}) \end{bmatrix} \begin{bmatrix} (\sigma_{11} - \bar{\sigma}) & \sigma_{12} & \sigma_{13} \\ & (\sigma_{22} - \bar{\sigma}) & \sigma_{23} \\ \text{sym.} & & (\sigma_{33} - \bar{\sigma}) \end{bmatrix} \right), \quad (16.74)$$

$$= \frac{3}{2} [(\sigma_{11} - \bar{\sigma})^2 + (\sigma_{22} - \bar{\sigma})^2 + (\sigma_{33} - \bar{\sigma})^2 + 2\sigma_{12}^2 + 2\sigma_{23}^2 + 2\sigma_{31}^2], \quad (16.75)$$

$$= \frac{3}{2} [\sigma_{11}^2 + \sigma_{22}^2 + \sigma_{33}^2 - 2\bar{\sigma}(\sigma_{11} + \sigma_{22} + \sigma_{33}) + 3\bar{\sigma}^2] + 3(\sigma_{12}^2 + \sigma_{23}^2 + \sigma_{31}^2), \quad (16.76)$$

$$= \frac{3}{2} \left[\sigma_{11}^2 + \sigma_{22}^2 + \sigma_{33}^2 - \frac{2}{3}(\sigma_{11} + \sigma_{22} + \sigma_{33})^2 + \frac{3}{9}(\sigma_{11} + \sigma_{22} + \sigma_{33})^2 \right] + 3(\sigma_{12}^2 + \sigma_{23}^2 + \sigma_{31}^2), \quad (16.77)$$

$$= \frac{1}{2} [3(\sigma_{11}^2 + \sigma_{22}^2 + \sigma_{33}^2) - (\sigma_{11} + \sigma_{22} + \sigma_{33})^2] + 3(\sigma_{12}^2 + \sigma_{23}^2 + \sigma_{31}^2), \quad (16.78)$$

$$= \frac{1}{2} [2(\sigma_{11}^2 + \sigma_{22}^2 + \sigma_{33}^2 - \sigma_{11}\sigma_{22} - \sigma_{22}\sigma_{33} - \sigma_{33}\sigma_{11})] + 3(\sigma_{12}^2 + \sigma_{23}^2 + \sigma_{31}^2), \quad (16.79)$$

$$= \frac{(\sigma_{11} - \sigma_{22})^2 + (\sigma_{22} - \sigma_{33})^2 + (\sigma_{33} - \sigma_{11})^2}{2} + 3(\sigma_{12}^2 + \sigma_{23}^2 + \sigma_{31}^2), \quad (16.80)$$

which is the result. ■

16.5 Plasticity Implementation

16.5.1 Incremental Initial Value Problem for Elastoplasticity with Isotropic Hardening

To motivate the time stepping components of the implementation algorithm, consider smooth function $f(x) : \mathbb{R} \rightarrow \mathbb{R}$. With initial condition $x(0) = x_0$ and the notation $x_n \approx x(t_n)$, f can be approximated as

$$\dot{x}(t) = f(x(t)), \quad (16.81)$$

$$\approx \frac{x_{n+1} - x_n}{\Delta t} = f(x_{n+\theta}), \quad \text{where } \theta \in [0, 1], \quad \text{and} \quad (16.82)$$

$$x_{n+\theta} = (1 - \theta)x_n + \theta x_{n+1}. \quad (16.83)$$

The value θ controls the stability and accuracy of the time stepping algorithm.

$$\begin{aligned} \theta = 0 & \quad \text{forward (explicit) Euler,} \\ \theta = \frac{1}{2} & \quad \text{midpoint rule,} \\ \theta = 1 & \quad \text{backward (implicit) Euler.} \end{aligned} \quad (16.84)$$

With these notions at hand and use of the implicit Euler, we use (16.82) for evolution in the plastic flow rule.

$$\dot{\varepsilon}^p \approx \frac{\varepsilon_{n+1}^p - \varepsilon_n^p}{\Delta t} = [\dot{\gamma} \text{sign}(\sigma)]_{n+\theta}, \quad (16.85)$$

$$\begin{aligned} & \quad (\text{now with } \theta = 1) \\ & = \dot{\gamma}_{n+1} \text{sign}(\sigma_{n+1}). \end{aligned} \quad (16.86)$$

Using (16.82) again for $\dot{\gamma}$ gives

$$\dot{\gamma} \approx \frac{\gamma_{n+1} - \gamma_n}{\Delta t} = \dot{\gamma}_{n+\theta}, \quad (16.87)$$

$$\begin{aligned} & \quad (\text{now with } \theta = 1) \\ & = \dot{\gamma}_{n+1}. \end{aligned} \quad (16.88)$$

Thus

$$\dot{\varepsilon}^p \approx \frac{\varepsilon_{n+1}^p - \varepsilon_n^p}{\Delta t} = \frac{\gamma_{n+1} - \gamma_n}{\Delta t} \text{sign}(\sigma_{n+1}). \quad (16.89)$$

Adopting the notation $\gamma_{n+1} - \gamma_n = \Delta\gamma$,

$$\boxed{\varepsilon_{n+1}^p = \varepsilon_n^p + \Delta\gamma \text{sign}(\sigma_{n+1})}. \quad (16.90)$$

Analogous arguments with the hardening variable evolution $\dot{\alpha} = \dot{\gamma}$ lead to

$$\boxed{\alpha_{n+1} = \alpha_n + \Delta\gamma}. \quad (16.91)$$

16.5.2 Trial Elastic State and Plastic Return-Mapping

Consider now a trial state that is purely elastic, with all plastic flow “frozen” in time. The incremental displacement $\Delta \mathbf{u}$ creates an incremental strain $\Delta \varepsilon \equiv \varepsilon_{n+1} - \varepsilon_n$. Then the trial state functions are denoted as

$$\sigma_{n+1}^{\text{tr}} = E (\varepsilon_{n+1} - \varepsilon_n^{\text{p}}) \equiv \sigma_n + E \Delta \varepsilon \quad (16.92)$$

$$\Phi_{n+1}^{\text{tr}} = |\sigma_{n+1}^{\text{tr}}| - (\sigma_y + H \alpha_n) \quad (16.93)$$

$$\varepsilon_{n+1}^{\text{p tr}} = \varepsilon_n^{\text{p}} \quad (16.94)$$

$$\alpha_{n+1}^{\text{tr}} = \alpha_n \quad (16.95)$$

With the known initial conditions $\{\varepsilon_n^{\text{p}}, \alpha_n\}$, the strain is incremented $\Delta \varepsilon$ and the resulting trial yield function is tested for the occurrence of a strictly elastic step or an elasto-plastic step. Specifically,

$$\begin{aligned} &\text{if } \Phi_{n+1}^{\text{tr}} \leq 0, \text{ then } \Delta \gamma = 0 \text{ and} \\ &\left. \begin{aligned} \sigma_{n+1} &= \sigma_{n+1}^{\text{tr}} \\ \Phi_{n+1} &= \Phi_{n+1}^{\text{tr}} \\ \varepsilon_{n+1}^{\text{p}} &= \varepsilon_{n+1}^{\text{p tr}} \\ \alpha_{n+1} &= \alpha_{n+1}^{\text{tr}} \end{aligned} \right\} \text{ (elastic step).} \end{aligned} \quad (16.96)$$

Otherwise, the return-mapping algorithm is used for the elasto-plastic step,

$$\begin{aligned} &\text{if } \Phi_{n+1}^{\text{tr}} > 0, \text{ then } \Delta \gamma > 0 \text{ and} \\ &\left. \begin{aligned} \sigma_{n+1} &= \sigma_{n+1}^{\text{tr}} - E \Delta \gamma \text{ sign}(\sigma_{n+1}^{\text{tr}}) \\ \Phi_{n+1} &= 0 \\ \varepsilon_{n+1}^{\text{p}} &= \varepsilon_n^{\text{p}} + \Delta \gamma \text{ sign}(\sigma_{n+1}^{\text{tr}}) \\ \alpha_{n+1} &= \alpha_n + \Delta \gamma \end{aligned} \right\} \text{ (ep step).} \end{aligned} \quad (16.97)$$

The results in (16.97) are derived as follows. First, find the following two useful relationships between σ_{n+1}^{tr} and σ_{n+1} .

$$\sigma_{n+1} = E (\varepsilon_{n+1} - \varepsilon_{n+1}^{\text{p}}) \quad (16.98)$$

$$= E (\varepsilon_{n+1} - \varepsilon_n^{\text{p}} + \varepsilon_n^{\text{p}} - \varepsilon_{n+1}^{\text{p}}), \quad (16.99)$$

$$\stackrel{(16.90, 16.92)}{=} \sigma_{n+1}^{\text{tr}} - E \Delta \gamma \text{ sign}(\sigma_{n+1}^{\text{tr}}). \quad (16.100)$$

Thus

$$\text{sign}(\sigma_{n+1}) (|\sigma_{n+1}| + E \Delta \gamma) = \text{sign}(\sigma_{n+1}^{\text{tr}}) |\sigma_{n+1}^{\text{tr}}|. \quad (16.101)$$

Since $E > 0$ and $\Delta \gamma > 0$, we require

$$\boxed{\text{sign}(\sigma_{n+1}) = \text{sign}(\sigma_{n+1}^{\text{tr}})}, \quad (16.102)$$

and

$$\boxed{|\sigma_{n+1}| + E\Delta\gamma = |\sigma_{n+1}^{\text{tr}}|}. \quad (16.103)$$

Next, determine the incremental plastic multiplier $\Delta\gamma$ using the discrete versions of the yield function

$$\Phi_{n+1} \stackrel{(16.17)}{=} |\sigma_{n+1}| - (\sigma_y + H\alpha_{n+1}) \stackrel{\text{set}}{=} 0, \quad (16.104)$$

$$\begin{aligned} &\stackrel{(16.103)}{=} |\sigma_{n+1}^{\text{tr}}| - E\Delta\gamma - (\sigma_y + H\alpha_n) \\ &\quad - H(\alpha_{n+1} - \alpha_n), \end{aligned} \quad (16.105)$$

$$\stackrel{(16.93,16.91)}{=} \Phi_{n+1}^{\text{tr}} - E\Delta\gamma - H\Delta\gamma, \quad (16.106)$$

gives

$$\boxed{\Delta\gamma = \frac{\Phi_{n+1}^{\text{tr}}}{E + H}}. \quad (16.107)$$

Then all variables Eq. (16.97) are known. Equation (16.97)₁, rewritten as,

$$\sigma_{n+1} = \left[1 - \frac{E\Delta\gamma}{|\sigma_{n+1}^{\text{tr}}|} \right] \sigma_{n+1}^{\text{tr}}, \quad (16.108)$$

shows that the final stress state σ_{n+1} may be interpreted as the **projection** of the trial stress state σ_{n+1}^{tr} on the yield function. Figure 16.7 (b) illustrates this concept.

16.5.3 Linearization

The Taylor series expansion of real function $f(x)$ about a point $x = a$ is applied to the residual function $\mathbf{r}(\mathbf{u})$ as follows,

$$\begin{aligned} f(x) &= f(a) \\ &\quad + f'(a)(x - a) \\ &\quad + \frac{f''(a)}{2!}(x - a)^2 \\ &\quad + \dots \\ &\quad + \frac{f^{(n)}(a)}{n!}(x - a)^n \end{aligned} \quad (16.109)$$

Thus, with combining all higher order terms, abbreviated as (H.O.T.),

$$\mathbf{r}(\mathbf{u} + \Delta\mathbf{u}) = \mathbf{r}(\mathbf{u}) + \frac{\partial \mathbf{r}(\mathbf{u})}{\partial \mathbf{u}} \Delta\mathbf{u} + \text{H.O.T.}, \quad (16.110)$$

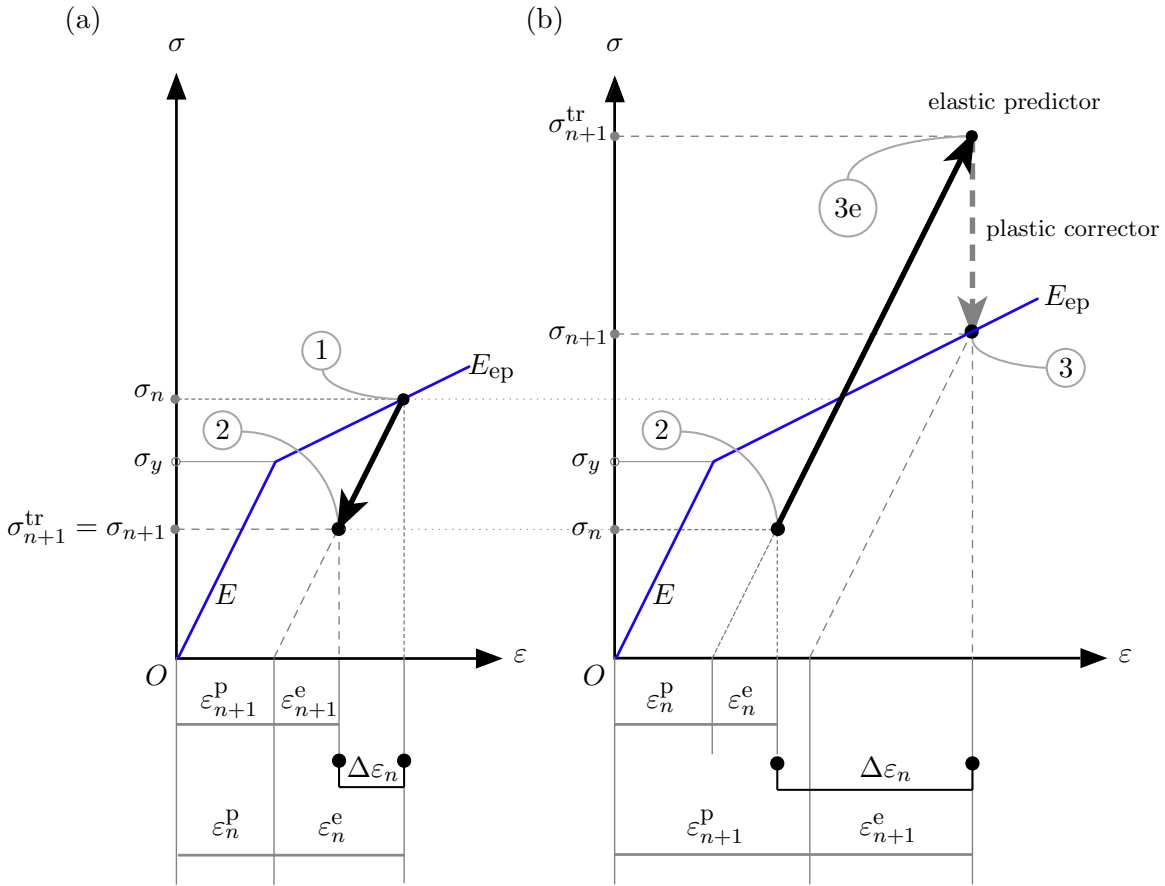


Figure 16.7: (a) Unloading step from state 1 to state 2 and (b) loading step from state 2, to state 3e (elastic prediction), and to state 3.

and

$$\mathbf{r}^{(i+1)} = \mathbf{r}^{(i)} + \frac{\partial \mathbf{r}^{(i)}}{\partial \mathbf{u}^{(i)}} \Delta \mathbf{u}^{(i)} + \text{H.O.T.} \quad (16.111)$$

This iterative approach is the groundwork for the Newton-Raphson iterative procedure. At the initial iteration, $i = 0$, the initial guess is set equal to the known quantity at the previously converged time step, $\mathbf{u}_{n+1}^{(0)} = \mathbf{u}_n$. Then, linearization about the current iterative step is taken, and updates are determined, as explicated in Box 1. Figure 16.8 shows a graphical representation of this procedure.

$\mathbf{u}_{n+1}^{(0)} = \mathbf{u}_n$ initial guess as converged solution at t_n

$$\left[\frac{-\partial \mathbf{r}(\mathbf{u}_{n+1}^{(0)})}{\partial \mathbf{u}_{n+1}^{(0)}} \right]^{-1} \mathbf{r}(\mathbf{u}_{n+1}^{(0)}) = \Delta \mathbf{u}_{n+1}^{(0)}$$

$$\Rightarrow \Delta \mathbf{u}_{n+1}^{(0)} + \mathbf{u}_{n+1}^{(0)} = \mathbf{u}_{n+1}^{(1)} \quad \Rightarrow \frac{|\mathbf{r}(\mathbf{u}_{n+1}^{(1)})|}{|\mathbf{f}_{n+1}^{\text{ext}}|} \stackrel{?}{\leq} \text{TOL}$$

$$\left[\frac{-\partial \mathbf{r}(\mathbf{u}_{n+1}^{(1)})}{\partial \mathbf{u}_{n+1}^{(1)}} \right]^{-1} \mathbf{r}(\mathbf{u}_{n+1}^{(1)}) = \Delta \mathbf{u}_{n+1}^{(1)}$$

$$\Rightarrow \Delta \mathbf{u}_{n+1}^{(1)} + \mathbf{u}_{n+1}^{(1)} = \mathbf{u}_{n+1}^{(2)} \quad \Rightarrow \frac{|\mathbf{r}(\mathbf{u}_{n+1}^{(2)})|}{|\mathbf{f}_{n+1}^{\text{ext}}|} \stackrel{?}{\leq} \text{TOL}$$

$\vdots$ $\vdots$ $\vdots$

$$\dots \quad \dots \quad \Rightarrow \frac{|\mathbf{r}(\mathbf{u}_{n+1}^{(*)})|}{|\mathbf{f}_{n+1}^{\text{ext}}|} \leq \text{TOL}$$

$\mathbf{u}_{n+1} = \mathbf{u}_{n+1}^{(*)}$ update with converged solution

Box 1: Newton-Raphson iterative procedure, explicated for the first and second iterations.

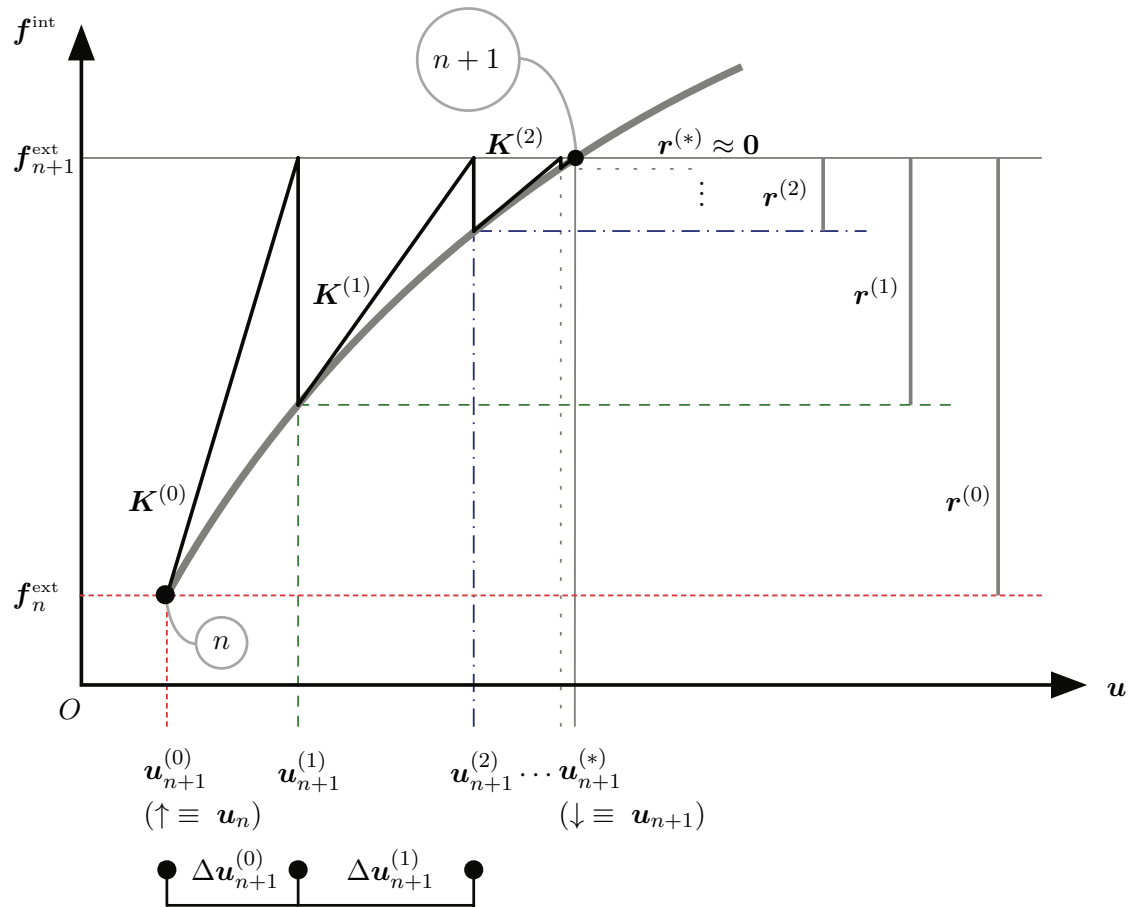


Figure 16.8: Newton-Raphson iterative procedure, shown graphically: The first iteration $K^{(0)}$ approximates the curve as tangent line at n . Successive tangents are constructed $K^{(i)}$ until the root at $n+1$ is found.

Time step loop ($n = 0$, WHILE $n \leq \text{NSTEPS}$)1. $i = 0$, CONVERGED = FALSE2. $\mathbf{u}_{n+1}^{(i)} = \mathbf{u}_n$ (prediction)3. **Iteration loop** (WHILE $i \leq \text{IMAX}$ AND CONVERGED IS FALSE)

- if $\frac{\left| \mathbf{r} \left(\mathbf{u}_{n+1}^{(i)} \right) \right|}{\left| \mathbf{f}_{n+1}^{\text{ext}} \right|} \leq \text{TOL} \quad \left| \mathbf{f}_{n+1}^{\text{ext}} \right| \neq 0$

- then

(a) CONVERGED = TRUE

(b) $\mathbf{u}_{n+1}^{(*)} = \mathbf{u}_{n+1}^{(i)}$ (c) $\varepsilon_{n+1}^{\text{P} (*)} = \varepsilon_{n+1}^{\text{P} (i)}$ (d) $\alpha_{n+1}^{(*)} = \alpha_{n+1}^{(i)}$

- else

(a) $\left[\frac{-\partial \mathbf{r} \left(\mathbf{u}_{n+1}^{(i)} \right)}{\partial \mathbf{u}_{n+1}^{(i)}} \right]^{-1} \mathbf{r} \left(\mathbf{u}_{n+1}^{(i)} \right) = \Delta \mathbf{u}_{n+1}^{(i)}$ (form tangent and solve)(b) $\mathbf{u}_{n+1}^{(i+1)} = \mathbf{u}_{n+1}^{(i)} + \Delta \mathbf{u}_{n+1}^{(i)}$ (iteration update)(c) $i \leftarrow i + 1$ 4. $\mathbf{u}_{n+1} = \mathbf{u}_{n+1}^{(*)}$ (time step update)5. $\varepsilon_{n+1}^{\text{P}} = \varepsilon_{n+1}^{\text{P} (*)}$ 6. $\alpha_{n+1} = \alpha_{n+1}^{(*)}$ 7. $n \leftarrow n + 1$

Box 2: Predictor/corrector time stepping algorithm applied to elasto-plastic equations of motion.

Chapter 17

Contact

17.1 One Dimensional Contact

17.1.1 Concepts of Gap and Penetration

Consider two bodies $\mathcal{A}$ and $\mathcal{B}$ in moving in inertial reference from $\mathcal{F}$ with origin O and basis vector $\hat{\mathbf{f}}_1$, as shown in Fig. 17.1. Let point P be located on the right-hand-side of body $\mathcal{A}$, denoting the first contact surface $\partial\mathcal{A}$. Let point Q be located on the left-hand-side of body $\mathcal{B}$, denoting the second contact surface $\partial\mathcal{B}$. Let the contact boundaries be denoted as $\Gamma_c = \partial\mathcal{A} \cup \partial\mathcal{B}$. Let the time evolution from $t = 0$ (initial) to $t = T$ (final) be denoted as $\mathbb{T} = [0, T] \subset \mathbb{R}^+$. At the initial time $t = 0$, these two points are located by X^{OP} and X^{OQ} , respectively.

At $t = 0$, the initial gap $g_0 : [\Gamma_c \times 0] \mapsto \mathbb{R}^+$ between the two points P and Q is given by

$$g(t = 0) = g_0 = X^{OQ} - X^{OP} \triangleq X^{PQ}, \quad g_0 > 0. \quad (17.1)$$

In general, the gap $g : [\Gamma_c \times \mathbb{T}] \mapsto \mathbb{R}$ between the points is given by

$$g(\boldsymbol{\varphi}(\mathbf{X}), t) = \varphi_t^{OQ} - \varphi_t^{OP} = (X^{OQ} + u_t^Q) - (X^{OP} + u_t^P) = g_0 + u_t^Q - u_t^P, \quad (17.2)$$

where u_t is a displacement, φ_t is a current configuration, $\mathbf{u} = \langle u_t^P, u_t^Q \rangle^T$, and $\boldsymbol{\varphi} = \langle \varphi_t^{OP}, \varphi_t^{OQ} \rangle^T$. For arbitrary displacements u_t^P and u_t^Q , the gap can be either positive, negative, or zero. For example, at $t = t_2$, the gap is negative. It is useful, as will seen in subsequent discussions on the penalty formulation (*see ??*), to define penetration p as the negative of the gap. The penetration $p : [\Gamma_c \times \mathbb{T}] \mapsto \mathbb{R}$ is defined as

$$p(\mathbf{u}, t) \triangleq -g(\mathbf{u}, t) = -\varphi_t^{OQ} + \varphi_t^{OP} = -(X^{OQ} + u_t^Q) + (X^{OP} + u_t^P) = -g_0 - u_t^Q + u_t^P. \quad (17.3)$$

Note that

$$p(t = 0) = p_0 = -g_0, \quad p_0 < 0. \quad (17.4)$$

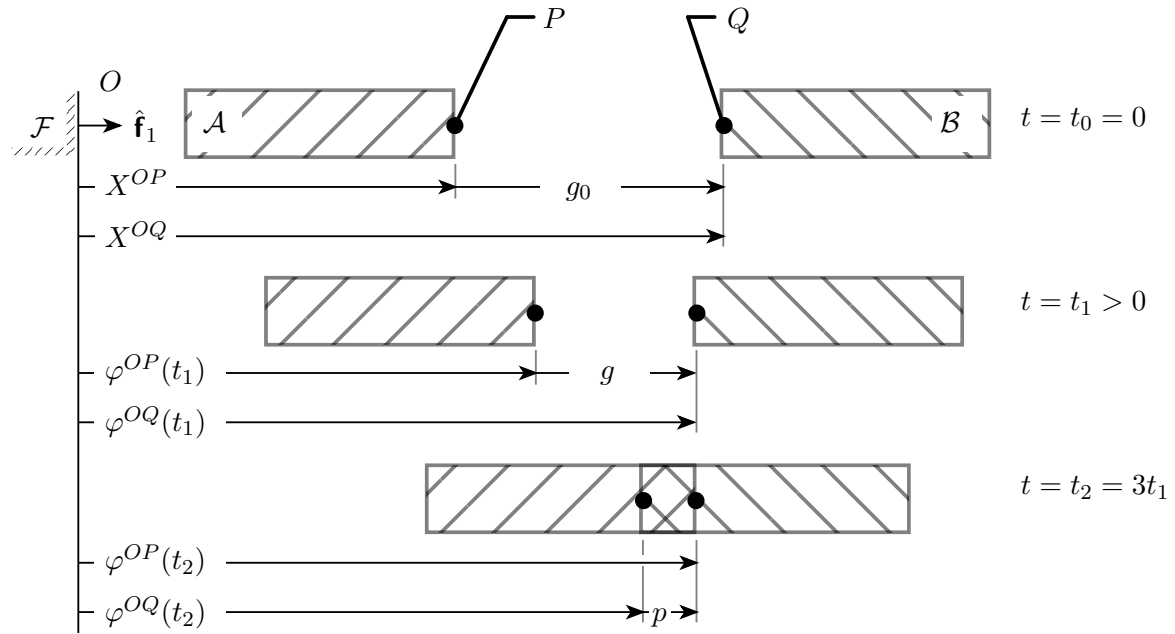


Figure 17.1: Caption to come.

17.1.2 Non-Penetration

The penetration of body $\mathcal{A}$ and body $\mathcal{B}$, shown (for example) at $t = t_2$ in Fig. 17.1, is prevented by constraining the gap g (or, equivalently, the penetration p) as follows:

$$g \geq 0, \quad (17.5)$$

$$p \leq 0. \quad (17.6)$$

These two inequality conditions are **unilateral constraints** on the motion φ . To illustrate the unilateral constraint concept in a concrete sense, consider $\mathcal{B}$ in Fig. 17.1 to be stationary for all time. Thus $u_t^Q = 0$, and we can abbreviate the displacement of body $\mathcal{A}$ as $u_t^P = u$. The gap and penetration functions are then

$$g(u) = g_0 - u, \quad (17.7)$$

$$p(u) = -g(u) = u - g_0 = u + p_0. \quad (17.8)$$

Assume, for concreteness $g_0 = 1$ m. The gap and penetration functions are then shown in Fig. 17.2.

17.1.3 Example with Spring and Weight

We start by considering a spring mass system, with spring constant k , mass m , gravitational field $-g \hat{\mathbf{f}}_2$, with the spring initially unstretched at $t = t_0 = 0$, as shown in Fig. 17.3. While

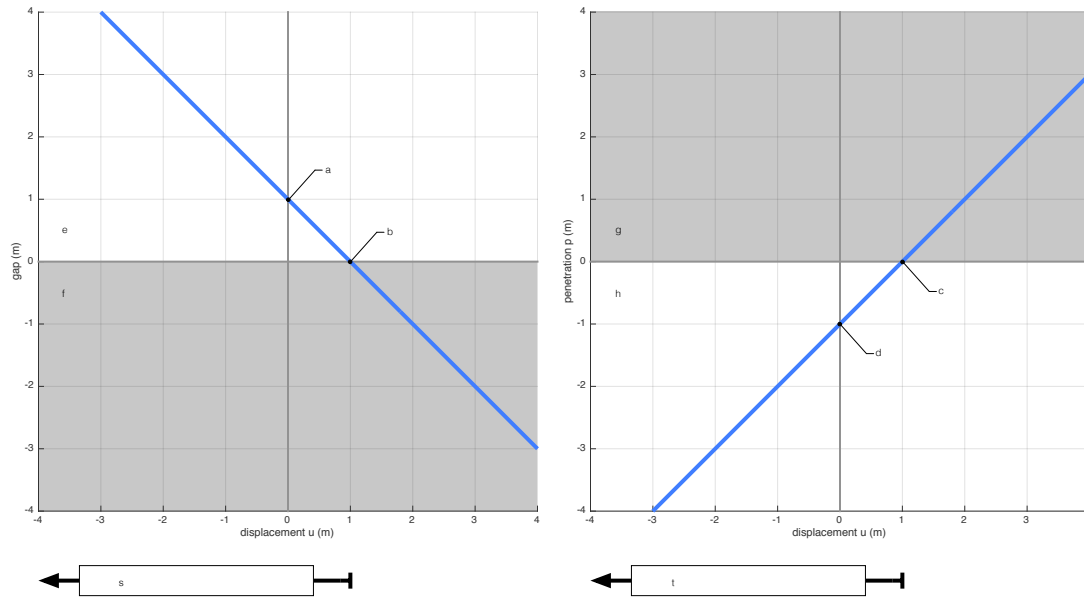


Figure 17.2: Caption to come.

this example does not yet have contact, it sets the context of energy minimization as method used to find static equilibrium.

For concreteness, let $k = 2 \text{ N/m}$, $m = 0.2 \text{ kg}$, and $g = 10 \text{ m/s}^2$, (for simplicity, rounded up from 9.81 m/s^2). Let displacement u of the mass, initially at 0, be measured in the $\hat{\mathbf{f}}_2$ axis. The internal energy (IE) of the spring as a function of u is given by $\Pi_k(u) = \frac{1}{2}ku^2$. The potential energy (PE) of the mass as a function of u is given by $\Pi_m(u) = mgu$. Minimization of the potential energy gives

$$\delta\Pi = \delta(\Pi_k(u) + \Pi_m(u)) = \delta\left(\frac{1}{2}ku^2 + mgu\right) = (ku + mg)\delta u \stackrel{\text{set}}{=} 0, \quad (17.9)$$

$$\implies u = \frac{-mg}{k} = \frac{-0.2 \cdot 10}{2} = -1. \quad (17.10)$$

Fig. 17.4 shows the spring internal energy, $\Pi_k(u)$, in dashed blue, which shows a minimum at $u = 0$, and a symmetric, parabolic response for nonzero displacements, as expected. The potential energy due to the mass, $\Pi_m(u)$, is shown in dot-dash green, which shows a linear relationship in u with intercept 0 and slope $mg = 0.2 \cdot 10 = 2 \text{ N}$.

The addition of the two aforementioned potentials, $\Pi(u) = \Pi_k(u) + \Pi_m(u)$, is shown in solid red. The system potential, $\Pi(u)$, has a minimum at $u = -1 \text{ m}$, as expected. Next, we used this concept of minimizing potential energy in the context of contact.

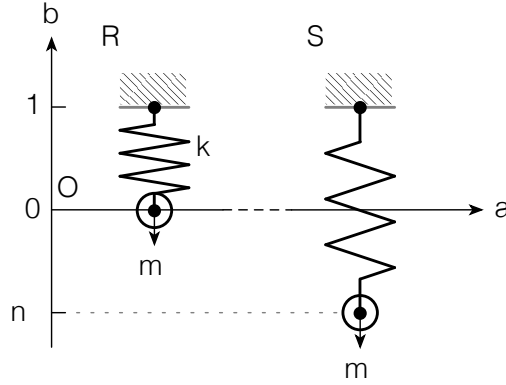


Figure 17.3: Caption to come.

17.2 Example with Spring and Rigid Wall

Consider a spring, shown in Fig. 17.5, with elastic constant $k = 2 \text{ N/m}$, undeformed length $L_0 = 2 \text{ m}$ subjected to a prescribed displacement in the amount of $\bar{u} = 2 \text{ m}$ on the left-hand-side.¹ A rigid wall is $g_0 = X^{PQ} = 1 \text{ m}$ from the right-hand-side of the spring at $t = 0$. At $t = t_1$, the spring has undergone its prescribed displacement and penetrates the wall by amount p . In this configuration, internal energy Π for the spring for the as a function of $u = u^P$ is given by

$$\Pi(u) = \frac{1}{2} k \delta^2 = \frac{1}{2} 2 (u - \bar{u})^2 = (u - 2)^2 \text{ (Nm)}. \quad (17.11)$$

17.2.1 Principle of Minimum Potential Energy

17.2.2 Lagrange Multiplier Enforcement

¹This example follows Laursen TA (2013), *Computational contact and impact mechanics: fundamentals of modeling interfacial phenomena in nonlinear finite element analysis*. Springer Science & Business Media, page 87.

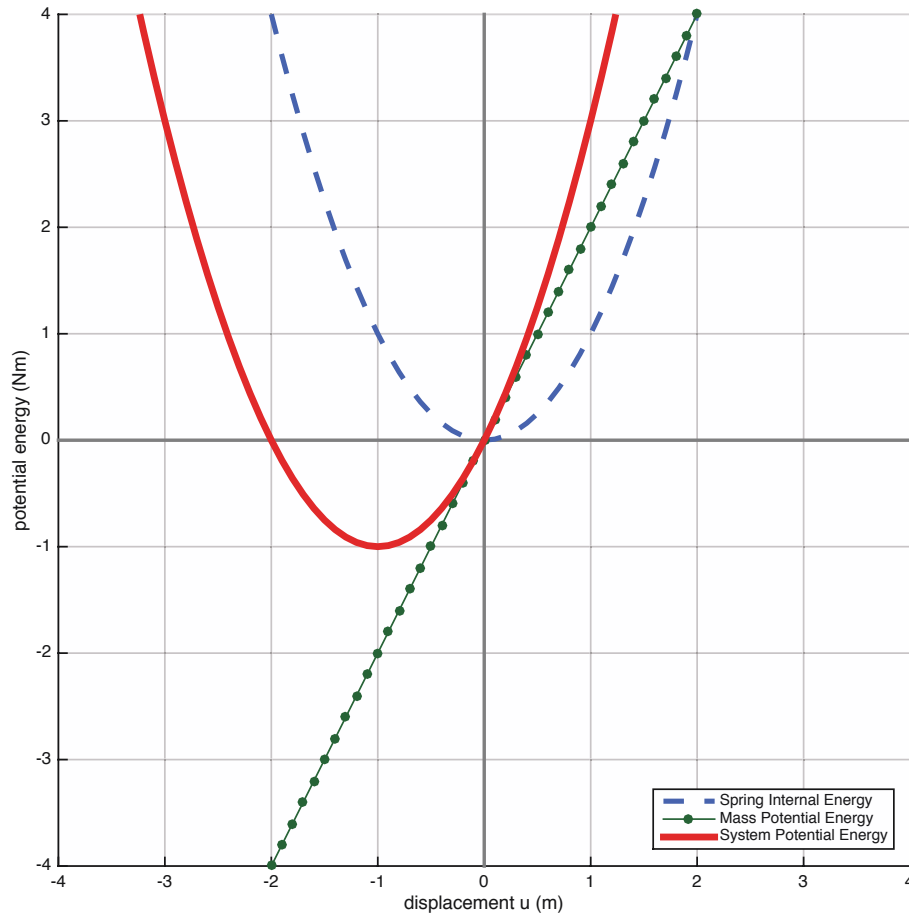


Figure 17.4: Caption to come.

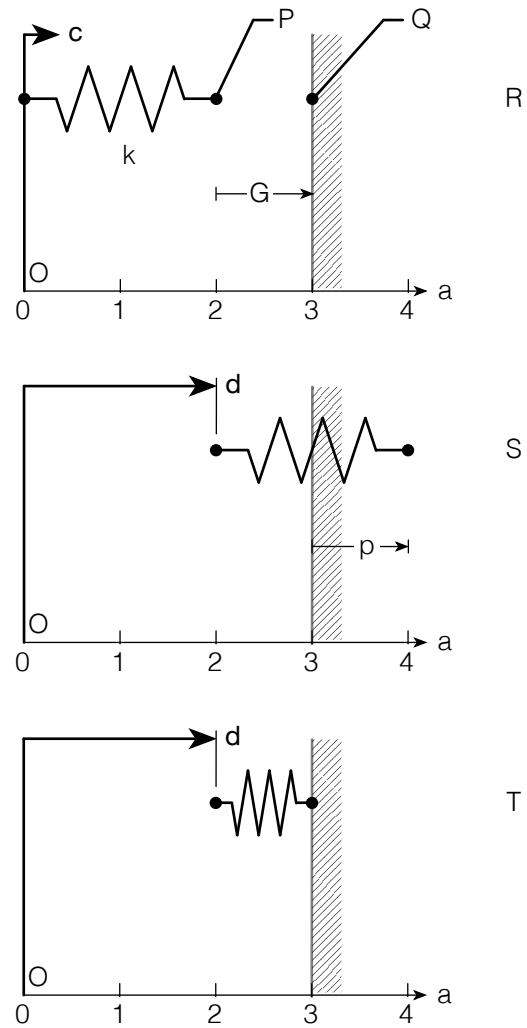


Figure 17.5: Caption to come.

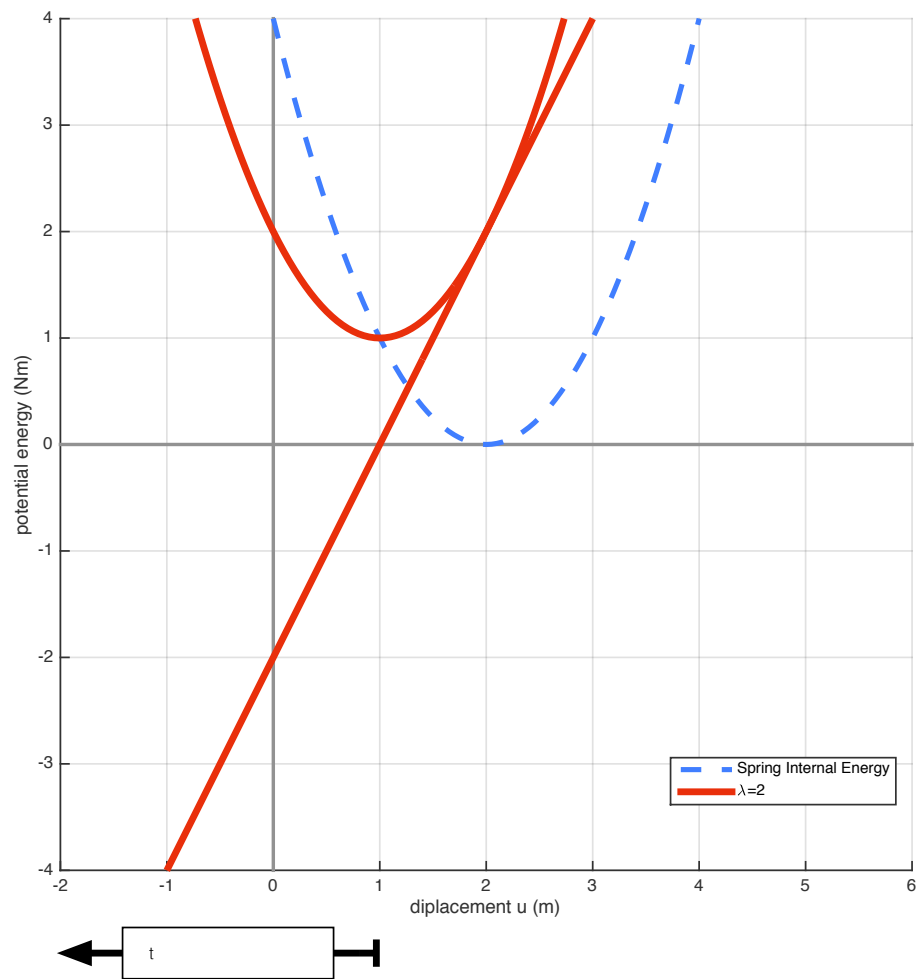


Figure 17.6: Caption to come.

$$\Pi(u) = \Pi_k(u) + \Pi_c(u). \tag{17.12}$$

$$= \frac{1}{2}k(u - \bar{u})^2 + \lambda p(u). \tag{17.13}$$

$$= (u - \bar{u})^2 + \lambda(u + p_0). \tag{17.14}$$

$$\text{where } \bar{u} = 2 \text{ m, and } p_0 = -1 \text{ m.} \tag{17.15}$$

17.2.3 Penalty Enforcement

17.2.4 Augmented Lagrange Multiplier Enforcement

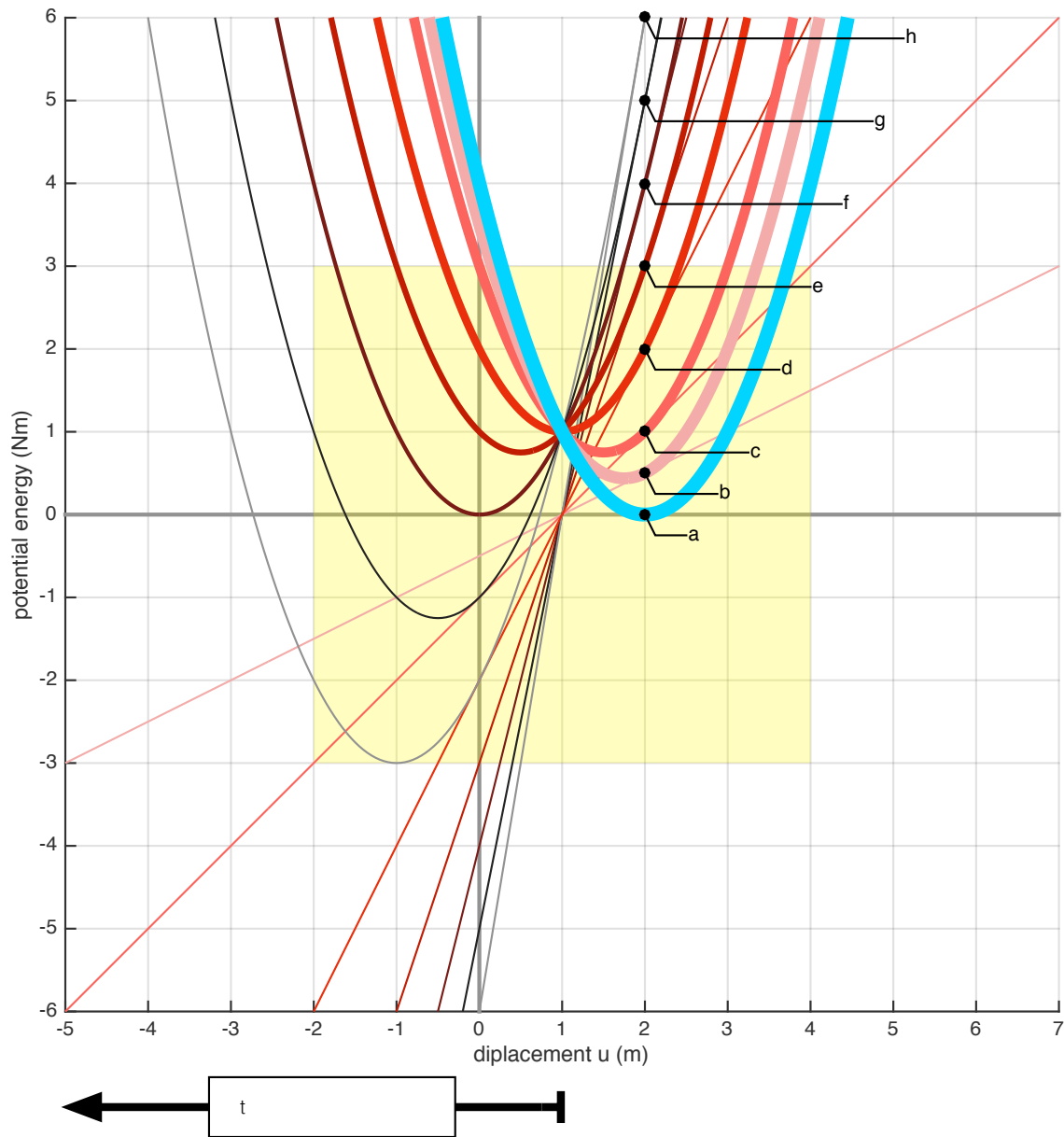


Figure 17.7: Caption to come.

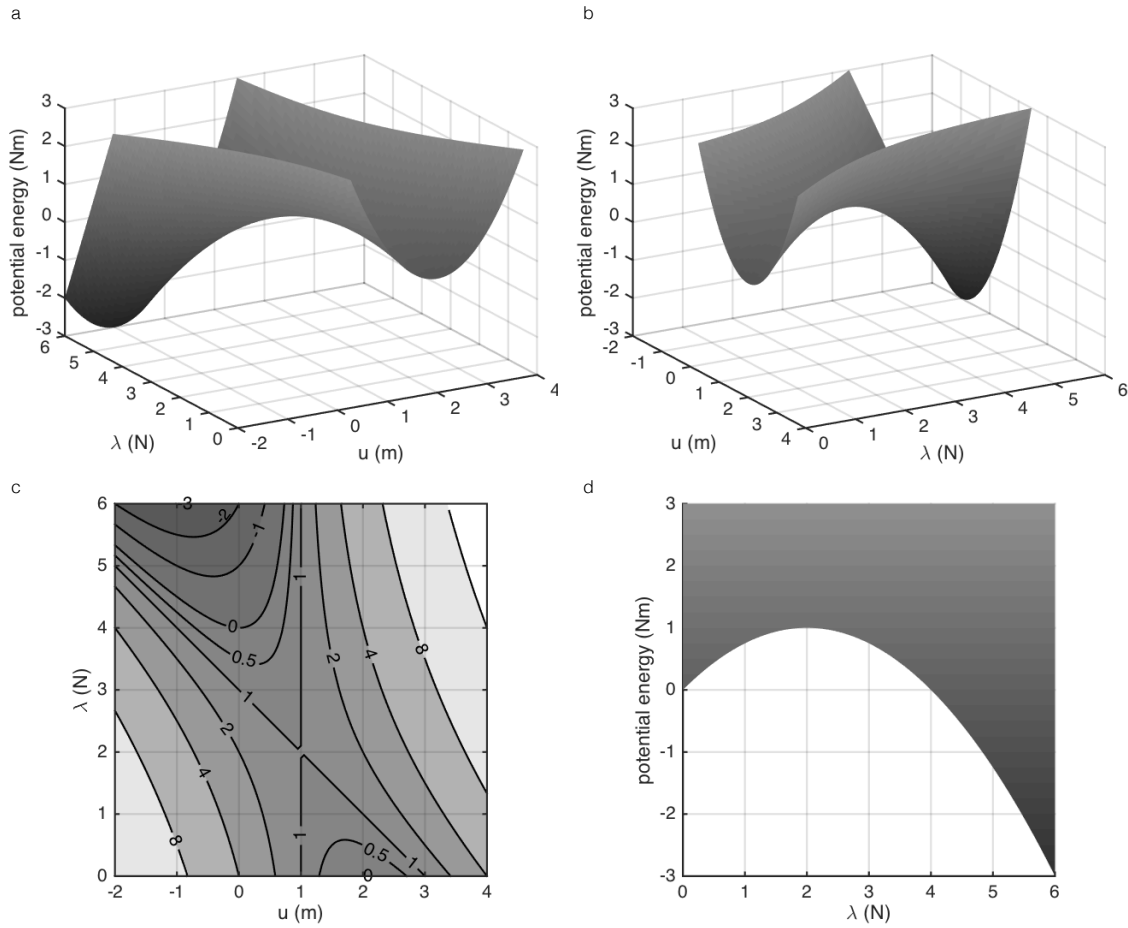


Figure 17.8: Caption to come.

Chapter 18

Shock

18.1 Introduction

18.2 An Ideal Fluid

18.3 Definition of a Shock

18.4 Conservation Laws for a Single Shock

18.4.1 Eulerian (Laboratory Fixed) Coordinate Frame

The derivation that follows an Eulerian description, which assumes a **laboratory-fixed reference frame**,¹ as shown in Fig. 18.1. Consider a rigid tube containing a rigid piston and an ideal fluid at ambient density ρ_0 and ambient pressure P_0 at time $t = t_0$.

Assume at time $t = (t_0 + \Delta t)$, the piston moves left-to-right along the x -axis with speed u_P , measured in the fixed laboratory reference frame. Assume the piston speed u_P causes a shock to propagate through the material with speed U_S . We require the following:

$$0 < u_P < +\infty \quad (18.1)$$

$$0 < U_S < +\infty \quad (18.2)$$

$$u_P < U_S \quad (18.3)$$

From these considerations, we conclude that the material to the left of the shock wave in Fig. 18.1 moves to the right at speed u_P . The material to the right of the shock wave remains at rest.²

Since the piston compresses the material in the tube, we know

¹A shock-fixed reference frame derivation is developed in a subsequent section.

²Furthermore, we assume the material throughout the tube and for all time is inviscid, non-diffusive, and non-reacting. We assume an adiabatic boundary, thus no heat conduction occurs through the boundary of

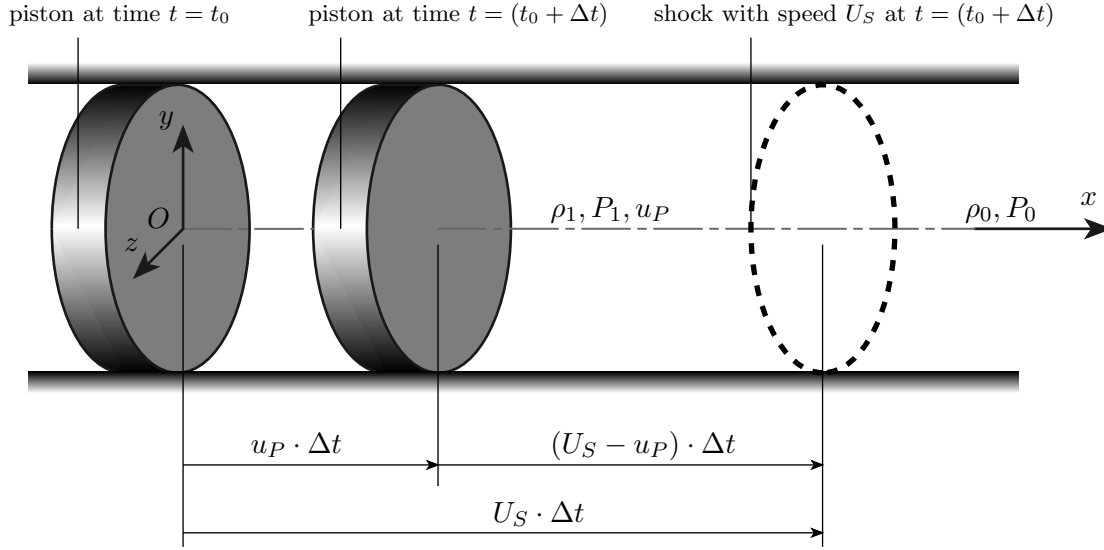


Figure 18.1: Piston moving left-to-right along the x -axis, with piston speed u_P , causing shock with shock speed U_S .

- $\rho_1 > \rho_0$,

where ρ_1 is the density of the material to the left of the shock.

Allow some time interval Δt to elapse, as illustrated in Fig. 18.1. Consider the distances along the x -axis, written as a multiplication of speeds and elapsed time. In the time interval Δt , the piston moves along the x -axis a distance of $u_P \cdot \Delta t$. In the same time interval, the shock wave moves along the x -axis a distance of $U_S \cdot \Delta t$. Thus, the control volume at $t = (t_0 + \Delta t)$ has an axial length of $(U_S - u_P) \cdot \Delta t$.

From these axial lengths and piston cross-sectional area A , we can write the following relationships:

- volume

$$V_0 = V|_{t=t_0} = A \cdot U_S \cdot \Delta t,$$

$$V_1 = V|_{t=(t_0+\Delta t)} = A \cdot (U_S - u_P) \cdot \Delta t,$$

- density

$$\rho_0 = \rho|_{t=t_0},$$

$$\rho_1 = \rho|_{t=(t_0+\Delta t)},$$

the tube walls or piston. We assume no heat transfer from radiation occurs across the shock. Finally, we assume body forces from gravity and electromagnetism are negligible.

- mass

$$\begin{aligned}\rho_0 V_0 &= \rho_0 \cdot A \cdot U_S \cdot \Delta t, \\ \rho_1 V_1 &= \rho_1 \cdot A \cdot (U_S - u_P) \cdot \Delta t.\end{aligned}$$

18.4.2 Conservation of Mass

Conservation of mass within the control volume requires that mass at time $t = (t_0 + \Delta t)$ be unchanged from mass at time $t = t_0$. Thus

$$\rho_0 \cdot \mathcal{A} \cdot U_S \cdot \Delta t = \rho_1 \cdot \mathcal{A} \cdot (U_S - u_P) \cdot \Delta t, \quad (18.4)$$

or

$$\boxed{\rho_0 U_S = \rho_1 (U_S - u_P), \quad \text{(Conservation of Mass).}} \quad (18.5)$$

Using specific volume (12.1), conservation of mass be rewritten as

$$\mathcal{V}_1 = \mathcal{V}_0 \left(1 - \frac{u_P}{U_S} \right), \quad (18.6)$$

or

$$\boxed{\mathcal{V}_1 - \mathcal{V}_0 = \frac{-v_0 u_P}{U_S}, \quad \text{(Conservation of Mass, Volume Form).}} \quad (18.7)$$

From (18.1)–(18.3), we know $\mathcal{V}_1 < \mathcal{V}_0$, which makes sense since we are compressing the material and generally have the understanding that increased pressures causes decreased volumes.

18.4.3 Conservation of Momentum

The net driving force from the piston on the material is $(P_1 - P_0) \cdot A$. Conservation of momentum requires $F = d(ma)/dt$. Written over the time interval Δt and for constant mass, conservation of momentum can be written

$$F \cdot \Delta t = m \cdot \Delta U. \quad (18.8)$$

Rewriting this expression in terms of pressures and densities, we have

$$(P_1 - P_0) \cdot \mathcal{A} \cdot \Delta t = (\rho_0 \cdot \mathcal{A} \cdot U_S \cdot \Delta t) (u_P - 0) \quad (18.9)$$

which gives

$$\boxed{P_1 - P_0 = \rho_0 U_S u_P, \quad \text{(Conservation of Momentum).}} \quad (18.10)$$

Using (12.1), (18.10) can be rewritten as

$$\boxed{P_1 - P_0 = \frac{U_S u_P}{\mathcal{V}_0}, \quad \text{(Conservation of Momentum, Volume Form).}} \quad (18.11)$$

18.4.4 Conservation of Energy

The work done on the control volume by the piston moving left-to-right is given by

$$W_P = \underbrace{P_1 \cdot A}_{\text{force}} \cdot \underbrace{u_P \cdot \Delta t}_{\text{disp.}} \quad (18.12)$$

Note that the right boundary of the control volume, located at the shock location, does not move, so the work across this boundary is identically zero.

The kinetic energy increase of the control volume is simply the one-half times the mass times the square of the change in velocity, thus

$$KE = \frac{1}{2} \underbrace{(\rho_0 \cdot A \cdot U_S \cdot \Delta t)}_{\text{mass}} \underbrace{(u_P - 0)^2}_{\text{velocity}} \quad (18.13)$$

Finally, the increase in internal energy is just the internal energy **per unit mass**, $\mathcal{E}$, multiplied by the mass,

$$IE = (\mathcal{E}_1 - \mathcal{E}_0) \underbrace{(\rho_0 \cdot A \cdot U_S \cdot \Delta t)}_{\text{mass}} \quad (18.14)$$

Equating these quantities gives

$$P_1 \cdot \mathcal{A} \cdot u_P \cdot \Delta t = \frac{1}{2} \rho_0 \cdot \mathcal{A} \cdot U_S \cdot \Delta t \cdot u_P^2 + (\mathcal{E}_1 - \mathcal{E}_0) \rho_0 \cdot \mathcal{A} \cdot U_S \cdot \Delta t \quad (18.15)$$

or simply

$$P_1 u_P = \rho_0 U_S \left(\frac{u_P^2}{2} + \mathcal{E}_1 - \mathcal{E}_0 \right), \quad \textbf{(Conservation of Energy).}$$

(18.16)

The foregoing equation can be written without kinematic variables U_S and u_P . Next, note that conservation of mass can be rewritten as

$$\frac{u_P}{U_S} = 1 - \frac{\rho_0}{\rho_1}. \quad (18.17)$$

Then, with (18.16),

$$P_1 u_P \stackrel{(18.10)}{=} (P_1 - P_0) \frac{u_P}{2} + \rho_0 U_S (\mathcal{E}_1 - \mathcal{E}_0), \quad (18.18)$$

$$= -P_0 u_P + 2 \rho_0 U_S (\mathcal{E}_1 - \mathcal{E}_0), \quad (18.19)$$

$$(\mathcal{E}_1 - \mathcal{E}_0) = \frac{u_P (P_1 + P_0)}{2 \rho_0 U_S}, \quad (18.20)$$

$$\stackrel{(18.17)}{=} \left(\frac{P_1 + P_0}{2 \rho_0} \right) \left(1 - \frac{\rho_0}{\rho_1} \right), \quad (18.21)$$

$$\stackrel{(12.1)}{=} \left(\frac{P_1 + P_0}{2 \mathcal{V}_0} \right) \left(1 - \frac{\mathcal{V}_1}{\mathcal{V}_0} \right), \quad (18.22)$$

which simplifies to

$$\mathcal{E}_1 - \mathcal{E}_0 = \frac{1}{2} (P_1 + P_0) (\mathcal{V}_0 - \mathcal{V}_1), \quad \textbf{(Rankine-Hugoniot curve)}. \quad (18.23)$$

The Rankine-Hugoniot equation is the Conservation of Energy jump condition with state variables alone; there are no kinematic variables in the equation. To plot the Rankine-Hugoniot relationship in the $P\mathcal{V}$ plane with $\mathcal{E}_1 > \mathcal{E}_0$, the initial state $(\mathcal{V}_0, P_0)$ must be known.

18.4.5 Simplified Forms

To create shock waves in solids, the shock pressure P_1 is typically far greater than the initial pressure P_0 . Often the initial pressure P_0 is ambient atmospheric pressure, which is small enough compared to P_1 that P_0 can be ignored.

Also, the internal energy per unit mass $\mathcal{E}_0$ can be set to zero because it is a thermodynamic variable, and can be set relative to any datum.

With $P_0 =$ and $\mathcal{E}_0 = 0$, the foregoing conservation equations simplify to

$$P_1 = \rho_0 U_S u_p \quad \textbf{(Conservation of Momentum } P_0 = \mathcal{E}_0 = 0) \quad (18.24)$$

$$\mathcal{E}_1 = \frac{P_1}{2} (\mathcal{V}_0 - \mathcal{V}_1) \quad \textbf{(Conservation of Energy } P_0 = \mathcal{E}_0 = 0) \quad (18.25)$$

18.4.6 Speed-Volume-Pressure Relationships

The alternative forms for mass conservation (18.7) and momentum conservation (18.11), when multiplied together or divided by one another, readily allow for the shock speed U_S and particle speed u_P to be written in terms of volume and pressure. These relationship are found to be

$$U_S = \mathcal{V}_0 \sqrt{\frac{P_1 - P_0}{-(\mathcal{V}_1 - \mathcal{V}_0)}} = \mathcal{V}_0 \sqrt{\frac{\Delta P}{-\Delta \mathcal{V}}}, \quad (18.26)$$

and

$$u_P = \sqrt{-(P_1 - P_0) (\mathcal{V}_1 - \mathcal{V}_0)} = \sqrt{-\Delta P \Delta \mathcal{V}}. \quad (18.27)$$

18.4.7 Alternative Formulations

Now we develop the shock-fixed reference frame equations for mass, momentum, and energy conservation. (To come.)

18.5 PV Diagram

The PV diagram shows pressure P as a function of specific volume $\mathcal{V}$, as shown in Figure 18.2 (compare to Figure 13.1). Thus, the state variables P and $\mathcal{V}$ are considered in isolation from the other state and kinematic variables. The initial laboratory state is denoted $(\mathcal{V}_0, P_0)$. A large compressive pressure P_1 is applied to the material, which causes the volume to shrink to $\mathcal{V}_1$. If this applied pressure P_1 causes a shock, we record the pressure and volume state, and then repeat the experiment at a different pressure, e.g., P_2 . Several compression experiments are run, and the PV states that created a shock in the material are recorded in the PV diagram.

For concreteness, we show four experiments with applied pressures $P_1 \dots P_4$ and resultant volumes $\mathcal{V}_1 \dots \mathcal{V}_4$, shown in Fig. 18.2. Note that all experiments start from the same initial pressure P_0 and initial specific volume $\mathcal{V}_0$.

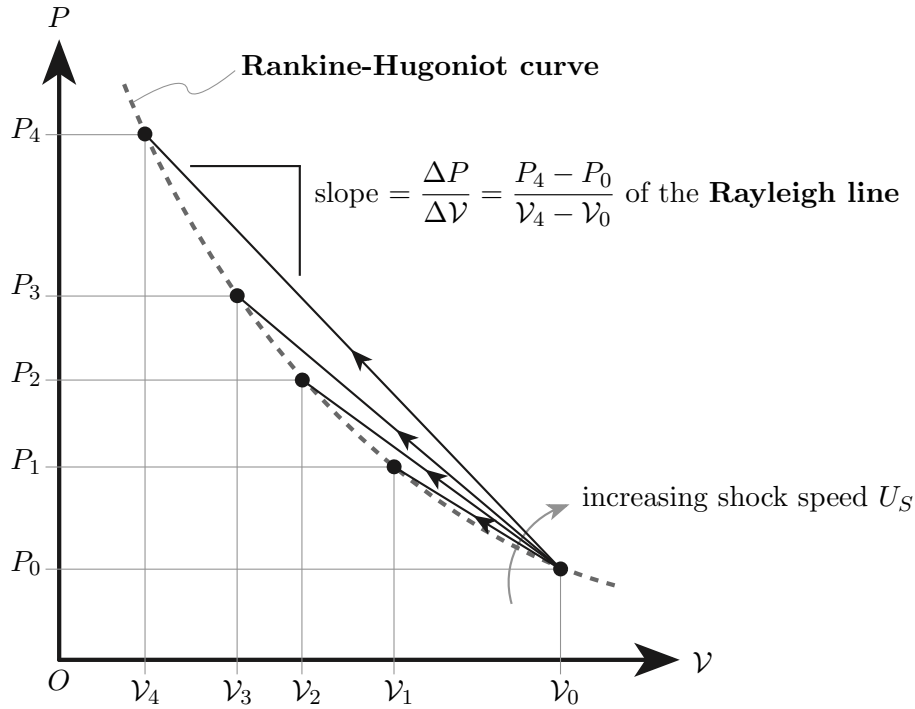


Figure 18.2: Pressure as a function of volume for the shock Hugoniot locus of thermodynamic equilibrium states.

These four thermodynamic states lie on a locus of all possible thermodynamic states that can exist in the material behind a shock. This locus is referred to as the **shock**

Hugoniot. The curve connecting the locus is the **Rankine-Hugoniot curve**. The line connecting two end states is the **Rayleigh line**.

It is important to emphasize that the Rankine-Hugoniot curve **is not** the path taken by the material between two thermodynamic states.

Note that (18.26) describes the mathematical increase in shock speed with increasing slope of the Hugoniot, indicated in Fig. 18.2. Specifically, the slope of the Hugoniot goes as the negative squared value of the shock speed U_S per initial specific volume $\mathcal{V}_0$, or

$$\frac{\Delta P}{\Delta \mathcal{V}} = - \left(\frac{U_S}{\mathcal{V}_0} \right)^2. \quad (18.28)$$

18.6 PU_P Diagram